

OUTAGES IN SOVEREIGN BOND MARKETS

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October 2025

Abstract

We exploit outages in sovereign bond markets as natural experiments. When the euro area futures market goes down, trading volumes on the cash market decline, liquidity evaporates, and prices deviate from fundamental values. Micro-level evidence reveals two mechanisms: the loss of a hedging instrument reduces dealers' intermediation capacity, and the missing benchmark price widens information asymmetries for clients. Outages on the cash market, in contrast, barely affect the futures market, implying one-way price formation and liquidity provision. Our findings highlight the trade-offs of market centralization, support cross-asset learning over symmetric arbitrage models, and demonstrate how intermediaries impose limits to arbitrage.

Keywords: Yield curve, market microstructure, natural experiment.

JEL classification: D47, G12, G14, G23

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1 Introduction

Recent stress episodes have shown that sovereign bond markets can become fragile and may need reform.¹ A key feature of sovereign bond markets is that they consist of two parallel segments: the cash market, where trading is decentralized and intermediated by dealers, and the futures market, where trading is fully centralized on a single exchange.

This paper exploits technical disruptions on either of these two segments as natural experiments. These outages were unanticipated and exogenous to market conditions and hence provide a rare glimpse into the role each market segment – and the investors active on them – play in liquidity provision and price discovery in sovereign bond markets.

Our first key contribution is to highlight the role of bond futures. We focus on two days in 2020, 14 April and 1 July, when technical glitches abruptly halted trading on the euro area futures exchange Eurex for several hours. To study the effects on the cash market for sovereign bonds, we combine non-anonymous regulatory transaction-level data with proprietary data from major trading platforms, interdealer brokers, and indicative quote providers. Thanks to the sharp discontinuities caused by the outages, we can identify causal effects using a simple event-study by comparing outages with matched control days. We document dramatic macro-level effects of Eurex outages and trace these back to the underlying micro-level mechanisms.

When the futures market goes down, trading volumes and liquidity on the cash market drop sharply for euro area government bonds. Indicative quotes, which usually provide a guidepost to negotiate trades in the over-the-counter market, become stale as soon as bond futures become unavailable. Finally, futures market outages cause a large spike in mispricing on the cash bond market. Focusing on German bonds, the euro area’s risk-free benchmark, we show that the remaining cash market transactions during outages occur at prices far from fundamental values. Under normal conditions, arbitrage forces keep observed market yields closely aligned with a smooth fitted yield curve. When Eurex goes down, however, pricing errors – the difference between observed and fitted yields – increase sharply and are many times higher than normal. The root mean squared pricing error – [Hu, Pan, and Wang \(2013\)](#)’s noise measure – matches or exceeds levels observed during the peak of the Covid-19 market turmoil in March 2020.

Taken together, these macro-level results suggest that bond futures are vital for the euro area fixed-income market to function smoothly. As our second major contribution, we exploit the granularity of our transaction dataset and identify two micro-level mechanisms causing the cash market to break down: outages remove a hedging instrument, reducing the intermediation capacity of dealers, and outages remove a benchmark price, exacerbating information asymmetries for clients. We provide extensive evidence for these mechanisms. Outages reduce trading

¹Recent bond market stress episodes include the Covid-19 turmoil in March 2020 (see e.g. [Haddad, Moreira, and Muir, 2021](#); [Eisenbach and Phelan, 2022](#); [Duffie, Fleming, Keane, Nelson, Shachar, and Van Tassel, 2023](#); [Duffie, 2025](#)), the UK Gilt crisis in late 2022 ([Pinter, Siriwardane, and Walker, 2024](#); [Czech and Monroe, 2025](#)), and the US tariff shock in April 2025 ([Jiang, Krishnamurthy, Lustig, Richmond, and Xu, 2025](#); [Pinter, Uslu, and Smets, 2025](#)). Ongoing bond market policy reforms include the U.S. Securities and Exchange Commission’s initiative to increase central clearing and all-to-all trading ([press release](#)), the Federal Reserve’s proposal to increase incentives for dealer banks to intermediate in U.S. Treasuries ([draft notice](#)) and the EU Commission’s work towards a consolidated tape ([press release](#)).

volumes most for dealer-to-client trades and least for client-to-client trades, consistent with dealers cutting back intermediation. Mispricing, in contrast, spikes most for client-to-client trades and does not increase for dealer-to-dealer trades. In fact, dealers trade more profitably during outages, consistent with clients relying on bond futures as benchmarks.

To corroborate the hedging mechanism, we exploit the dealer-level heterogeneity in hedging strategies. Dealers with the most intense use of bond futures retract most from the cash market during Eurex outages, forcing clients to trade directly with each other, usually at prices far from fundamental values. Dealers with a more moderate use of bond futures, in contrast, retain some intermediation capacity, but they demand higher markups when trading with clients, particularly for large trades in long-term bonds. To corroborate the benchmark mechanism, we show that less sophisticated types of clients fare worse during outages (e.g. retail investors compared to hedge funds), and that private signals and relationships become more important without the public benchmark. Better connected clients cause less mispricing, pay less markup, and trade more profitably while Eurex is down. The same is true for strong dealer-client relationships. Dealers extract less profits from loyal clients during outages. Further evidence shows that outages cause severe mispricing also in interest rate swaps, despite swaps not being directly linked to Eurex-traded derivatives. This suggests that Bund futures provide the pricing benchmark not only for the bond but also the swap market.

Our third and final major contribution is to document how asymmetric the outage effects are, using five outages on four different cash trading platforms as natural experiments. All outages reduce the aggregate trading activity on the cash market, showing that trading does not seamlessly migrate to alternative platforms in case of individual outages. Yet, cash venue outages barely spill over to the futures market. Three outages are particularly informative. First, on 12 January 2010 a technical glitch delayed the market opening on MTS by roughly two and a half hours. This outage has almost no discernible impact on bond futures traded on Eurex, in stark contrast to the widespread breakdown of MTS during Eurex outages. Second, on 17 April 2015, Bloomberg terminals worldwide were inaccessible for about two hours. This outage significantly reduces trading volumes on the futures market, but barely affects market functioning. Standard illiquidity measures rise by less than a tenth of a standard deviation. The third major cash market outage affected the Brokertec platform for about two hours on 11 January 2019. Again, this outage reduces trading volumes but barely affects the liquidity on the futures market. Importantly, this null effect also holds for US Treasury futures, despite the fact that Brokertec is the dominant electronic trading platform for US Treasuries. Taken together, the asymmetric effects of outages suggest that price discovery and liquidity provision seem to be more of a one-way street from the futures to the cash market than previously thought.

Literature Our quasi-experimental evidence is informative for three major strands in the literature. It highlights the trade-offs between a (de)centralized market structure, distinguishes competing theories about the link between liquidity and arbitrage and lastly, it shows how financial intermediaries create limits to arbitrage.

Market Structure (On vs. Off-Exchange Trading) A key issue in market microstructure is the prevalence and desirability of trading outside of exchanges. Positive network effects push trading towards a single central exchange, but information asymmetries pull in the opposite direction. In particular, because more informed and faster traders impose adverse-selection costs on liquidity suppliers, these liquidity suppliers have an incentive to trade with uninformed traders off-exchange, potentially at a discount.² Lee and Wang (2025) formalize this intuition. In their model, less informed traders optimally choose the OTC market. Nonetheless, closing the OTC market raises welfare, particularly for assets traded mostly OTC. Allen and Wittwer (2023a) use transaction-level data to estimate a structural model of the Canadian government bond market. They find that shifting trades to a centralized platform could decrease welfare, unless competition among dealers is sufficiently strong.³

Our results provide an important qualifier. On the decentralized cash market, investors – particularly clients – free-ride on the price discovery provided by other investors on the centralized futures market. Bond futures serve as benchmarks, reducing information gaps and increasing liquidity on the cash market (Glosten and Milgrom, 1985; Duffie, Dworczak, and Zhu, 2017). Welfare calculations should take these positive externalities into account. Centralizing the OTC market would have additional benefits, insofar as it reduces the dependency on the futures market and leads to self-sustained price discovery on the cash market. This is in line with Kutai, Nathan, and Wittwer (2025), who document that the Israeli bond market, the only major government bond market operating on an exchange, performed better during the Covid-19 crisis than most other markets operating OTC. We find similar results across euro area countries: the liquidity in Italian bonds drops least during futures market outages, in line with the fact that most cash trading is centralized in a single limit order book. At the same time, our results show that full market centralization comes at the cost of introducing systemic risks.

Arbitrage across Time and across Markets Our natural experiments distinguish two margins of market efficiency: arbitrage *across time* (Grossman and Miller, 1988) and *across markets* (Gromb and Vayanos, 2010, 2018). The Eurex outages show that the intertemporal margin alone does not ensure market functioning. When the futures market goes dark, cash market prices dislocate, because arbitrageurs fail to fully absorb the shock by temporarily warehousing bonds. Instead, market efficiency hinges on the cross-market linkage between Bunds and Bund futures.

Cross-market arbitrage models, however, typically assume symmetry. They assume that arbitrageurs provide liquidity on one market conditional on the other being active. We test

²de Roure, Moench, Pelizzon, and Schneider (2025) provide empirical evidence in line with the price discrimination channel. They document that for German government bonds, transaction prices in the OTC market are favorable compared to the centralized MTS exchange. This ‘OTC discount’ is in line with the fact that most German bonds are traded off-exchange.

³Dugast, Üslü, and Weill (2022) provide a model for on vs. off-exchange trading and show that, depending on their trading capacity, some market participants benefit from a decentralized OTC market. Biais and Green (2019) provide historical context and document that up until World War II, most US bond trading occurred on-exchange.

and reject this symmetry.⁴ Outages on cash market venues barely affect the futures market, suggesting that price formation and liquidity provision is a one-way street from the futures to the cash market.⁵ This asymmetry aligns with *cross-asset learning* models (see e.g. [Admati, 1985](#); [Veldkamp, 2006](#); [Cespa and Foucault, 2014](#); [Asriyan, Fuchs, and Green, 2017](#)). Those models posit that liquidity providers use some particularly informative prices – in our case bond future prices – when trading related instruments – in our case sovereign bonds and interest rate swaps. An inherently linked source of the asymmetry is the higher liquidity of the futures market. Arbitrageurs can hedge cash market exposures more easily and cheaply through futures than vice versa. Structural features of the futures market, such as the above-mentioned centralized trading, central clearing and margin-based leverage, contribute to this liquidity advantage. Yet, disentangling liquidity from information is challenging because they reinforce one another: liquidity concentrates where prices are informative, and informativeness rises with liquidity, potentially causing endogenous concentration on a single venue ([Pagano, 1989b,a](#); [Vayanos and Wang, 2007](#); [Vayanos and Weill, 2008](#)).

Intermediary Asset Pricing and Limits to Arbitrage A closely related literature studies the role of intermediaries in financial markets and the limits to arbitrage they impose (see [Long, Shleifer, Summers, and Waldmann, 1990](#); [Shleifer and Vishny, 1997](#); [Gromb and Vayanos, 2002](#); [Mitchell, Pedersen, and Pulvino, 2007](#); [Brunnermeier and Pedersen, 2008](#); [He and Krishnamurthy, 2013](#); [Allen and Wittwer, 2023b](#); [Bräuning and Stein, 2024](#)). In a recent paper on the U.S. Treasury market, e.g., [Duffie et al. \(2023\)](#) show that expected volatility explains most of the time variation in Treasury market liquidity, but when dealers exhaust their balance sheet capacity – as observed in March 2020 – liquidity drops abruptly and market functionality gets impaired. The outages we study provide a clear natural experiment that corroborates this functional chain. The Eurex outages diminish dealers’ capacity to intermediate bond trades on the cash market, since they are unable to hedge their inventory risk on the futures market. Crucially, our evidence is not subject to the usual endogeneity concerns between expected volatility, market liquidity, and dealer capacity.

2 European Government Bond Market Structure

This section briefly outlines the market microstructure for euro area government bonds (EGBs) on the cash and futures market. See [Appendix A](#) for further details and [Bessembinder, Spatt,](#)

⁴[Harding and Ma \(2010\)](#) show that outages on the US Treasury futures exchange lead to a dramatic fall in liquidity on an electronic cash market trading platform. This is similar to our finding that liquidity on MTS evaporates when Eurex is down. [Harding and Ma \(2010\)](#), however, cannot test for asymmetry, because there had been no suitable outages of cash trading venues.

⁵Vector error correction models confirm that bond futures ‘lead’ price discovery, reflecting new information faster than cash bonds (see e.g. [Mizrach and Neely, 2008](#); [Dobrev and Schaumburg, 2023](#); [Upper and Werner, 2007](#); [Jappelli, Lucke, and Pelizzon, 2022](#)). We find an even more pivotal role: bond futures are essential for smooth fixed-income market functioning. This aligns with the equity literature on dominant vs. satellite markets, where price discovery occurs mainly on the primary venue (see e.g. [Hasbrouck, 1995](#); [Guillaumie, Loiacono, Winkler, and Kern, 2020](#); [Hagströmer and Menkveld, 2023](#); [Degryse, Hagströmer, and Landsberg, 2025](#)).

and Venkataraman (2020) for a recent survey on fixed-income market structure. Our analysis covers the four largest euro area member states: Germany, France, Italy and Spain. They are also the only countries with corresponding bond futures.

The futures market for EGBs is highly homogeneous and centralized, as these futures and their associated options are traded exclusively on Eurex’s central limit order book (CLOB). There are only a handful of bond futures covering selected maturity segments. A 10-year bond future exists for all four countries. For Italy, also a 2-year bond future is actively traded while for Germany, bond futures are also available for the 2-, 5-, and 30-year segment. Together, the trading volume in German bonds is roughly ten times larger on the futures than the cash market in terms of underlying value. For France and Italy, the two markets are of comparable size, while for Spain, where a future was introduced last, trading volumes in futures are only about 1% of the cash market volume.

In terms of market participants, the futures market is rather concentrated, with relatively few high frequency trading firms dominating trading activity (Hautsch, Noé, and Zhang, 2017; Kirilenko, Kyle, Samadi, and Tuzun, 2017). The cash market, in contrast, has the typical dealer-client and core-periphery structure (see e.g. Di Maggio, Kermani, and Song, 2017; Li and Schürhoff, 2019; Bessembinder et al., 2020; Hendershott, Li, Livdan, and Schürhoff, 2020; Allen and Wittwer, 2023a).

The cash market for EGBs is also much more fragmented, segmented, and opaque compared to the futures market. There are hundreds of individual bonds outstanding at any point in time and these can be traded on very different venues. These venues differ along multiple dimensions, most importantly the trading mechanism they employ and the types of investors they grant access to. Consider concrete examples from two polar opposites. On the one side, bonds can be traded anonymously and immediately in a CLOB, just like bond futures. MTS is the dominant platform in this regard, but it is only accessible to dealers, i.e. large banks. Bonds can also be traded in a CLOB on regular stock exchanges, which are also open to small retail traders. But out of the multitude of exchanges, no single exchange captures a significant market share and the order book for most bonds is correspondingly thin. Some exchanges offer incentive programs for designated market makers to improve this poor liquidity. Importantly, on-exchange trading is ‘lit’, i.e. all market participants can observe quotes and transaction prices and volumes.

On the other side, bonds can be traded over-the-counter (OTC). Bilateral OTC trades are neither anonymous nor immediate. Such trades are typically negotiated by voice or chat. To preserve some anonymity and to reduce search costs, market participants can also trade OTC via a broker. In this case, the initiating party communicates its trade request to a broker, who then tries to find a suitable counterparty on a ‘matched principal’ basis, see de Roure et al. (2025) for a detailed description of this market segment. This way, the two counterparties do not have to reveal their identity to each other. Yet another venue on the OTC market are systematic internalisers, also called single-dealer platforms. On these platforms, large dealer banks act as a central counterparty for trades initiated by their clients. Compared to on-exchange trades, all three types of OTC trades are comparatively ‘dark’, i.e. there is little pre- and post-trade

transparency. In between these two extremes, another increasingly important market segment for EGB trading are electronic trading platforms, which dominate the dealer-to-client segment. Examples of such platforms are Tradeweb and Bloomberg, which use a request-for-quote (RFQ) mechanism, i.e. clients request quotes for a certain bond from multiple dealers.

Moreover, to facilitate the matching process between buyers and sellers, various companies – e.g. trading platforms themselves – provide indicative quotes for European government bonds. These quotes are often available only for specific bonds, e.g. ‘benchmark’ bonds of selected maturities, and the exact calculation methods behind those quotes are not disclosed.

Lastly, the market structure of EGBs differs substantially across countries. Italian bonds, e.g., are predominantly traded ‘on-exchange’ (on MTS), whereas German bonds are traded more on electronic trading platforms and OTC.⁶ To capture this heterogeneous bond market as much as possible, we combine various data sources, as explained in the following subsections.

2.1 Cash Market Data

For EGB transactions on the cash market, we start with the non-anonymous MiFIR dataset, which contains information on bond transactions since 2018 and which is collected under the MiFID II regulation. For each transaction, this dataset contains the ISIN, price, volume, time, the involved counterparties, as well as the venue on which the trade was executed. This dataset has been made available relatively recently and to the best of our knowledge, we are the first to use it for academic research. While relatively new, the dataset can be seen as an extended version of the Bafin dataset used in prior work (see e.g. [de Roure et al., 2025](#)), which was collected under the MiFID I regulation during 2008-2017, see [Bundesbank website](#) and [German Securities Trading Act](#).

An important caveat for our purposes is that the regulatory data only covers trades in which a German security is traded or where at least one of the involved parties – the buyer or seller, the trading venue, or a central counterparty (CCP) – is domiciled in Germany. This is why trades in German bonds are overrepresented.

For the six most important days in our sample – the two Eurex outage days and the respective control days – we address this limitation by complementing the regulatory dataset with transactions data sourced directly from trading platforms (MTS, MTS BondVision, Tradeweb) and interdealer brokers (TPICAP, BGC, GFI, Aurel). This way we are able to capture also a large share of transactions in French, Italian and Spanish government bonds. In contrast to the regulatory data, however, these datasets are anonymous, i.e. they do not contain information about the involved counterparties. Appendix [A.5](#) describes our transaction-level data in detail.

Besides the transaction data, we also use quote data. In particular, we have executable quotes and volumes from MTS’s CLOB and indicative quotes from Bloomberg, Refinitiv and TPICAP.

⁶Our distinction of trading venue types (on-exchange, electronic platforms, OTC) differs somewhat from the MiFID II regulation, see Appendix [A.3](#) for details.

2.2 Futures Market Data

For bond futures, we exploit three different datasets. First, we have the full history of transaction prices and volumes at the millisecond frequency going back to 2002. Second, we have the full intraday order book data, i.e. bid and ask quotes and volumes for all order book levels, going back to April 2019 from Deutsche Boerse’s A7 platform. Third and lastly, we have non-anonymous investor-level data on bond future transactions since 2019 from the European Market Infrastructure Regulation (EMIR) dataset.

2.3 Investor-Level Data

The two regulatory datasets mentioned above (MiFIR and EMIR) are non-anonymous, they contain a unique legal identifier (LEI) for the counterparties involved in each transaction. To classify investors into different sectors, we can thus use supervisory lists and the Register of Institutions and Affiliates Database (RIAD, see [Perrella and Catz, 2020](#)). We differentiate between banks, non-bank financial institutions (NBFI), investment funds, hedge funds, insurance companies and pension funds (ICPF), non-financial corporations (NFC), an official sector (including central banks), and households (retail investors). We further identify whether an investor is a dealer and whether the investor is usually active also on the bond futures market or only the cash market. Appendix [A.6](#) provides details.

3 Macro-Level Effects of Futures Market Outages

On 14 April 2020 and 1 July 2020, technical glitches caused outages on Eurex, the leading futures market exchange in the euro area. The first outage lasted approximately four and a half hours, from 9:25 a.m. to 2:00 p.m. while the second outage lasted less than three hours, from 8:49 a.m. to 11:31 a.m. (all in local time). In both cases, Eurex blamed ‘faulty third-party software’ as the root cause.⁷ We use these outages as natural experiments to study the role of the futures market for the broader fixed-income market in the euro area.

We will show how these outages affect the *cash* market for EGBs, namely in terms of trading activity ([Section 3.1](#)), market liquidity ([Section 3.2](#)), and pricing ([Section 3.3](#)).

3.1 Trading Activity

To investigate the effect of the outages on cash market trading activity, we estimate the following dummy regression:

$$\log(1 + Volume_{cmt}) = \alpha + \beta \times D_t + \gamma \times FE + \epsilon_t \quad (1)$$

⁷Deutsche Börse commented: ‘*The disruption in the T7 system in April and today’s failure had the same origin. They were due to faulty third-party software that is part of the trading system. [...] External causes can be ruled out.*’ Besides Eurex, the T7 outage also took down several stock exchanges, most notably XETRA but also local exchanges in Eastern Europe. These simultaneous outages on the spot market do not contaminate our results because they did not affect the bond market. [Appendix B](#) contains further narrative evidence on the outages.

where $Volume_{cmt}$ is the total trading volume in bonds of country c and maturity bucket m in the 30-minute time interval t . D_t is a dummy that equals one during the Eurex outages and is zero otherwise, and FE captures fixed effects. We include six days (the two outage days plus the preceding and subsequent week) and 15 intraday observations per day (from 08:30 a.m. to 4:00 p.m.). We use $\log(1 + Volume_t)$ to account for periods with zero trading volume.⁸

Table 1 reports the results.⁹ Column (1) confirms that trading volumes on the cash market decrease dramatically during the futures market outages. Column (2) shows that trading volumes drop particularly for bonds above 2.5 years of maturity, while column (3) shows that all four euro area countries are affected similarly by the outage.

	(1) Average	(2) Maturities	(3) Countries
Outage	-3.10*** [0.56]		
Outage $\times$ <2.5y		-1.07** [0.35]	
Outage $\times$ 2.5-5.5y		-3.69*** [0.47]	
Outage $\times$ 5.5-10.5y		-3.81*** [0.64]	
Outage $\times$ >10.5y		-3.83** [1.01]	
Outage $\times$ DE			-2.75*** [0.36]
Outage $\times$ FR			-3.36** [1.03]
Outage $\times$ IT			-3.24*** [0.16]
Outage $\times$ ES			-3.05** [0.88]
FE Day	✓	✓	✓
FE Time	✓	✓	✓
FE Country	✓	✓	
FE Maturity Bucket	✓		✓
Observations	1440	1440	1440
Adjusted R^2	0.324	0.335	0.323

Table 1: Effect of Eurex Outages on Cash Trading Volume. Each column shows results of a different regression, see Equation 1. The dependent variable is the log of the transaction volume in bonds of a given country and maturity bucket, in 30-minute intervals. Column (2) adds an interaction term between the outage dummy and the maturity buckets, column (3) does the same for countries.

3.2 Market Liquidity

3.2.1 Executable Quotes

MTS is the only major trading venue for euro area sovereign bonds with a central limit order book, i.e. immediately executable quotes. In this regard, MTS is the closest alternative to Eurex to trade fixed-income securities. So does trading transition to MTS when Eurex goes down?

⁸Chen and Roth (2023) show that the regression coefficients in this setting should not be interpreted as approximate percentage effects, since they depend arbitrarily on the units of the outcome variable. We mitigate the problem by aggregating volumes to relatively wide time buckets, minimizing periods with no trading. The maturity buckets c cover bonds of less than 2.5 years to maturity, 2.5 to 5.5 years, 5.5 to 10.5 years, and more than 10.5 years, see Appendix A.4 for details.

⁹Throughout this paper, *, **, *** indicates statistical significance at the 10%, 5% and 1% level, respectively and standard errors (shown in brackets) are clustered at the daily level.

For simplicity and maximum comparability, we first look at a single bond per country, namely the cheapest-to-deliver (CTD) bond underlying the 10-year bond future.¹⁰

Figure 1 shows the order book depth on MTS, i.e. the quantity of bonds investors are willing to buy or sell at any price. To put the events into perspective, we compare outage days with the same day in the previous and subsequent week. Liquidity evaporates during the Eurex blackout. Roughly three minutes after Eurex goes down on 14 April 2020, virtually all quotes vanish from the MTS platform, resulting in an order book depth of zero. The first quotes reappear only at 14:06, six minutes after trading on Eurex resumed. The same is true for the second outage. While trading usually starts before 9:00 a.m. on MTS, most quotes appear only at 11:43 a.m. on 1 July 2020, 12 minutes after Eurex was back online. These results suggest that the MTS cash market platform functions properly only if the futures market is active. One might think this is true only for CTD bonds, due to their close connection to the futures traded on Eurex, but we will show that the breakdown on MTS extends far beyond these specific bonds.

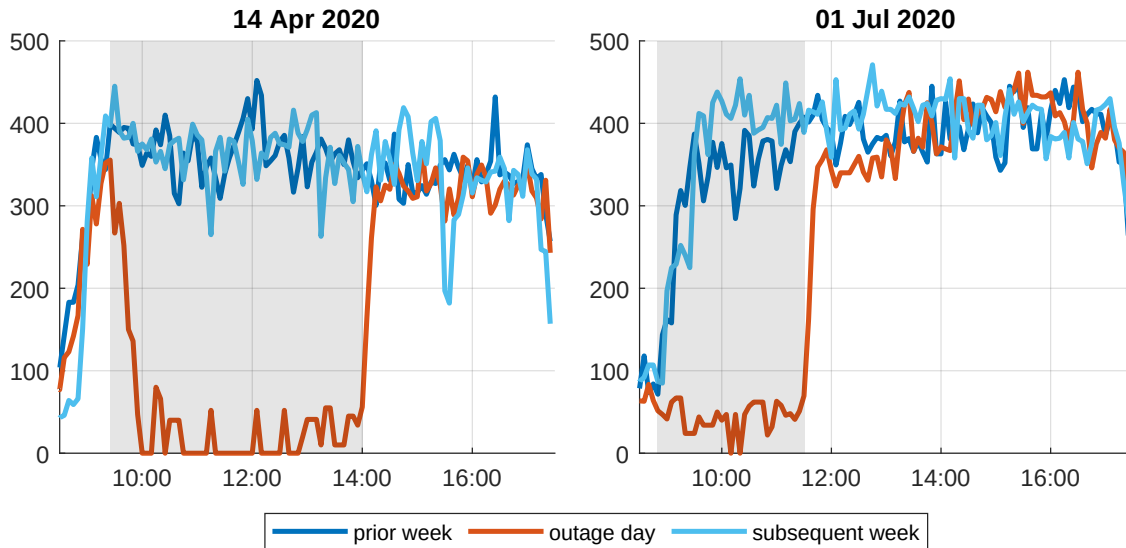


Figure 1: Order Book Depth on MTS. This figure shows the total quoted volume for German, French, Italian and Spanish 10-year CTD bonds (in million €) across all three levels and both sides of the order book, at 5-minute snapshots.

Before delving into further details, recall that [Harding and Ma \(2010\)](#) report broadly similar results to what we find here: outages of the main US futures exchange (CBOT, Chicago Board of Trade) lead to a dramatic fall in trading and quoting activity on a major electronic cash market trading platform (Espeed). They attribute this finding to high-frequency trading firms that are only active on the cash market if the futures market is online and vice versa. However, we can rule out this explanation in our setting, because only dealers are allowed to trade on MTS.¹¹ The fact that liquidity on MTS evaporates nonetheless suggests that the forces at work are more general. We will show that the more likely explanation is simply that price discovery

¹⁰See Appendix A.4 for the ISIN of each CTD bond. Appendix D.5 confirms that the results presented here are not confined to these particular bonds.

¹¹The current list of members is available on the [MTS website](#).

and liquidity provision crucially depend on an active futures market. Without bond futures as a hedging instrument and pricing signal, market functioning on the cash market deteriorates dramatically.

To study the dependency of MTS on Eurex in more detail, we estimate dummy regressions of the following form:

$$\log(1 + OBdepth_{cmt}) = \alpha + \beta \times D_t + \gamma \times FE + \epsilon_t \quad (2)$$

where $OBdepth_{cmt}$ is the order book depth (in €) of all bonds of country c and maturity bucket m at time t , measured at 5-minute snapshots. D_t is a dummy that equals one during Eurex outages and zero otherwise, and FE captures fixed effects. We include six days (the two outage days plus the preceding and subsequent week) and 91 intraday observations per day (5-minute intervals from 08:30 a.m. to 4:00 p.m.). We use $\log(1 + OBdepth_{cmt})$ to account for periods with empty order books.

Table 2 shows the results. For most countries and maturity buckets, the order book depth on MTS drops dramatically when Eurex goes down, see column (1). Column (2) shows that liquidity drops more for bonds with longer maturity, in line with the trading volume results above.

	(1) Average	(2) Maturities	(3) Countries
Outage	-10.86*** [0.42]		
Outage $\times$ <2.5y		-4.95*** [0.21]	
Outage $\times$ 2.5-5.5y		-13.45*** [0.32]	
Outage $\times$ 5.5-10.5y		-12.68*** [0.62]	
Outage $\times$ >10.5y		-12.35*** [0.71]	
Outage $\times$ DE			-18.03*** [0.43]
Outage $\times$ ES			-9.46*** [2.04]
Outage $\times$ FR			-14.15*** [0.44]
Outage $\times$ IT			-1.79** [0.52]
FE Day	✓	✓	✓
FE Time	✓	✓	✓
FE Country	✓	✓	
FE Maturity Bucket	✓		✓
Observations	8736	8736	8736
Adjusted R^2	0.518	0.558	0.644

Table 2: Effect of Eurex Outages on MTS Order Book Depth. Each column shows results of a different regression, see Equation 2. Throughout, the dependent variable is the log of the quoted bid and ask volume of all bonds of a given country and maturity bucket, at 5-minute snapshots. All explanatory variables are dummies, either for the maturity bucket or for different countries.

Before examining the differential effect across countries, recall that MTS is the main trading platform for Italian bonds and aggregate trading volumes in Italian bonds have a similar magnitude on the cash and futures market. This is in stark contrast to Germany e.g., where

trading volumes on the futures market are roughly ten times larger than on the cash market and where MTS has only a negligible cash market share (see Appendix A.3 for details). Hence, we would expect that Italian bonds are less affected by the Eurex blackout, which column (3) confirms. Still, our results suggest that market functioning on MTS is severely impaired during Eurex blackouts even for Italian bonds.

3.2.2 Indicative Quotes

Apart from the executable quotes on MTS, there are several providers of indicative quotes for EGBs. We obtained data from three different providers, namely Bloomberg, Refinitiv and TPICAP. Bloomberg terminals and Refinitiv’s Eikon application are the two most widely used sources for real-time financial information on the D2C segment, whereas TPICAP operates as an interdealer broker. All three provide indicative quotes to facilitate trading, but the methodologies behind these quotes are proprietary and not publicly disclosed.¹²

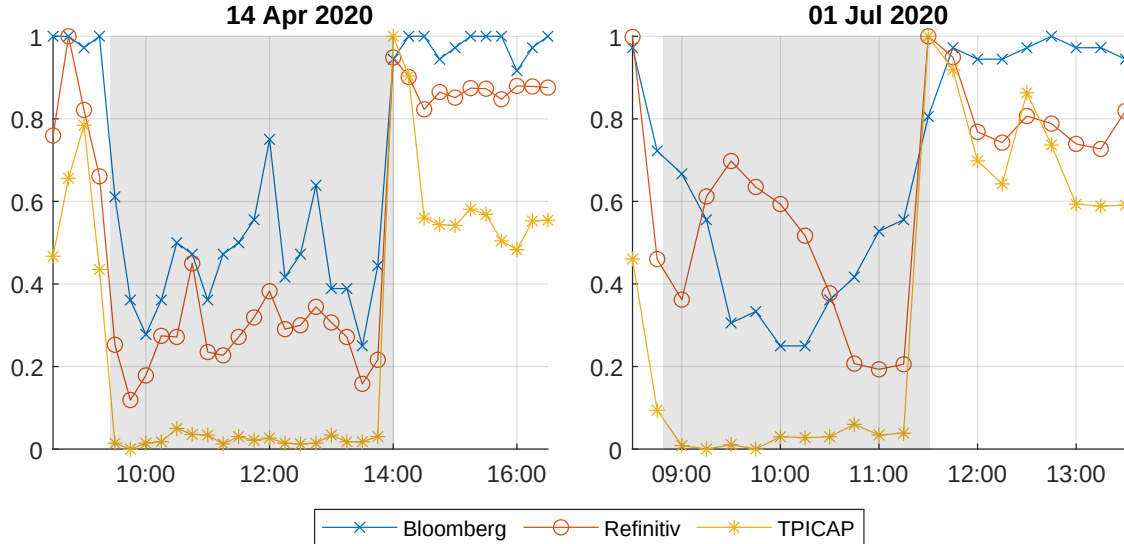


Figure 2: Quote Update Frequency for 10-Year Government Bonds on Different Platforms. Bloomberg and TPICAP data refer to quote updates in the cheapest-to-deliver bond, Refinitiv data to quote updates in the on-the-run (‘benchmark’) bond. Results for TPICAP and Refinitiv are based on the exact number of new quotes per bond. For Bloomberg, we approximate the number of new quotes as the number of tick-by-tick price changes. To show all series on a single scale, we aggregate the number of quote updates in 15-minute windows and normalize them to a 0-1 range for each data provider.

Figure 2 shows that across all three data providers, indicative quotes become stale when Eurex suffers an outage. The number of quote updates drops dramatically the moment Eurex goes down and quickly recovers once trading resumes, consistent with our claim that price discovery on the euro area fixed-income market hinges on bond futures.¹³

¹²Schestag, Schuster, and Uhrig-Homburg (2016) describe Bloomberg’s quotes as ‘a weighted average of quotes from participating dealers’. Similarly, Refinitiv’s ‘Tick History’ database contains the real-time feed updates shown on Eikon. For German and French bonds, quotes are from multiple ‘pricing contributors’, all of which are major European banks. For Italian and Spanish bonds, only a ‘composite price’, computed by Refinitiv, is available.

¹³Appendix D.7 confirms that these results hold across countries.

3.3 Pricing

We have documented that futures market outages lead to lower trading volumes and lower liquidity on the cash market for euro area government bonds. What we are ultimately interested in, however, is whether the price discovery process for EGBs is actually impaired due to the unavailability of bond futures.

Some *prima facie* evidence points in this direction: on 14 April 2020, the Dutch State Treasury Agency cancelled three bond auctions planned for that day, citing the Eurex outage as the reason.¹⁴ The auctions were postponed to the next day, when Eurex was back online. Dutch authorities apparently feared that the bonds might not be properly priced by market participants while the futures market is offline.

Was this fear justified? To find out, we study yield curve fitting errors, a popular measure for how well the bond market functions. [Hu et al. \(2013\)](#) e.g. argue that arbitrage forces usually keep the yield curve smooth. Hence, a low dispersion in bond yields along the yield curve is a sign that bond prices are in line with fundamental values. So were the prices of bond transactions that *did* occur during the Eurex outages ‘fair’? We focus on German bonds with a maturity between one and ten years to answer this question, which constitute the benchmark risk-free yield curve for the euro area. We convert transaction prices observed in the market into yields and then fit a term structure model to these observed yields. We do this separately for all transactions that occurred while Eurex was offline, and for all transactions that occurred during the same intraday window but in the previous or subsequent week. [Figure 3](#) shows the results.

The observed market yields usually lie on a smooth yield curve, but not when Eurex is offline. During the outages, many transactions deviate strongly from model-implied ‘fair’ yields. In fact, the dispersion in bond yields is so large that the exact method to compute ‘fair’ yields is secondary. We fit yield curves based on [Svensson \(1994\)](#), but we would obtain virtually the same increase in ‘pricing errors’ using the [Nelson and Siegel \(1987\)](#) or spline-based methods. During the outages, the noise measure proposed by [Hu et al. \(2013\)](#), defined as the root mean squared pricing error, is roughly three to six times higher than during the same intraday window one week before or after. Some market participants apparently struggle to price risk-free German sovereign bonds without bond futures as a guidepost. [Section 4](#) will show who these investors are.

To put the Eurex outages into perspective, we repeat the above exercise for all trading days between 1 March 2020 and 8 July 2020. [Figure 4](#) shows that the mispricing of German government bonds during the two Eurex outages was severe, comparable to or even higher than at the peak of the Covid-19 market turmoil in March 2020. For reference, the figure also shows the yield curve noise measure for US Treasuries published by [Hu et al. \(2013\)](#). Due to our use of intraday transaction prices, rather than end-of-day prices, our measure of noise is itself more noisy. This is because within our intraday windows, surprising macro news might hit the market, moving fair yields and hence mechanically increasing pricing errors. Nonetheless, the mispricing

¹⁴‘Decided to postpone today’s [...] auctions [...], due to technical issues at Eurex’ ([DSTA press release](#)).

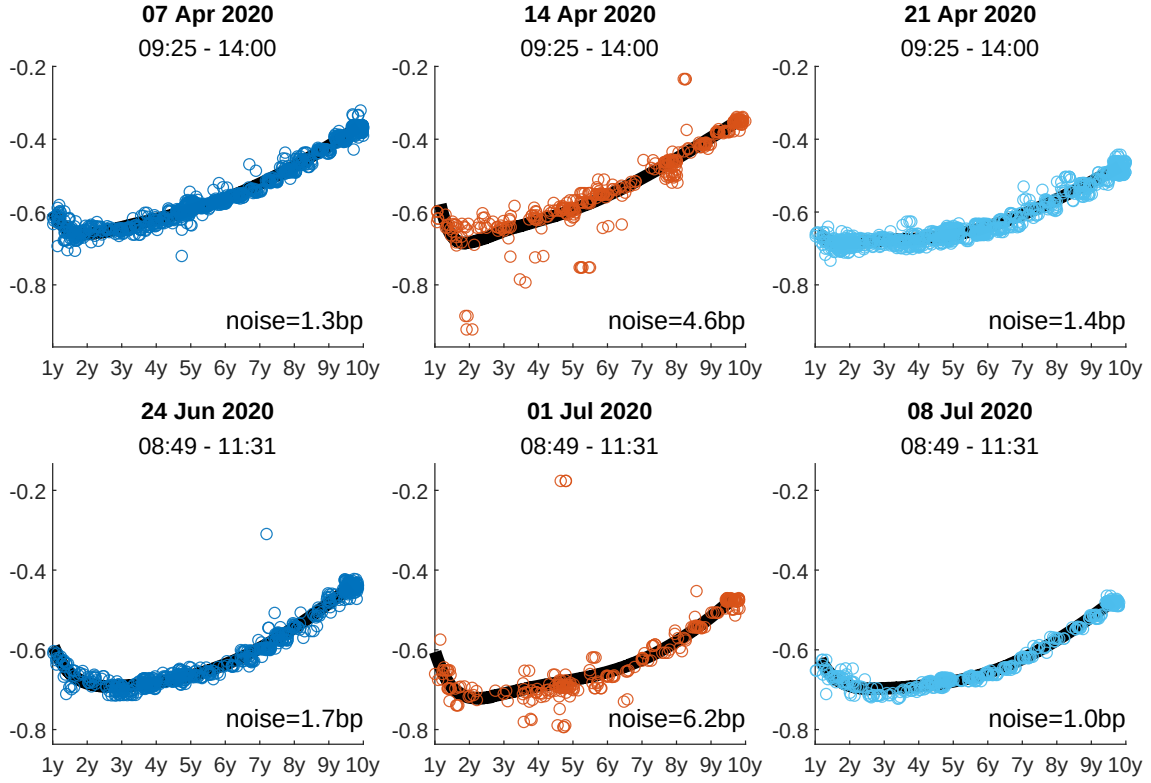


Figure 3: German Yields and Fitted Yield Curves. Each circle shows the implied yield of a transaction in German bonds during the intraday windows stated at the top of each panel. Red circles refer to the two Eurex outage days (middle column), dark and light blue circles refer to the same intraday window in the previous and subsequent week (left and right column, respectively). The vertical axis refers to the yield in percentage points, the horizontal axis to the remaining maturity in years. The black lines are fitted yield curves based on [Svensson \(1994\)](#). The value in the bottom right corner of each panel is the noise measure proposed by [Hu et al. \(2013\)](#), defined as the root mean squared pricing error (difference between actual and fitted yields).

we compute for German bonds is remarkably close to US Treasuries. In both countries, the yield curve noise hovers around 1.5 basis points during normal times, but shoots up to more than 4 basis points during the Covid-19 market turmoil. On the two Eurex outage days, lastly, we observe a spike in mispricing only for German bonds.¹⁵

[Table 3](#) confirms that cash bond transactions become significantly more mispriced. By sequentially adding fixed effects, we show that the mispricing spike is not due to different bonds being traded during the outages (column 2). Instead, a sizable share of the pricing errors is caused by a shift in trading activity toward investors and venues with systematically higher mispricing. The outage coefficient drops by 14% when we add venue-level fixed effects (column 3), and by another 33% with investor-level fixed effects (column 4). Nevertheless, the spike in mispricing remains economically large and highly significant even after controlling for all these compositional effects.

To summarize our macro-level findings, the euro area sovereign bond market becomes dysfunctional when the futures market goes down. Next, we use granular transaction-level data to

¹⁵Appendix [D.2](#) zooms in on the two outage days and confirms that the mispricing in German bonds spikes the moment Eurex goes down and quickly recovers once trading resumes.

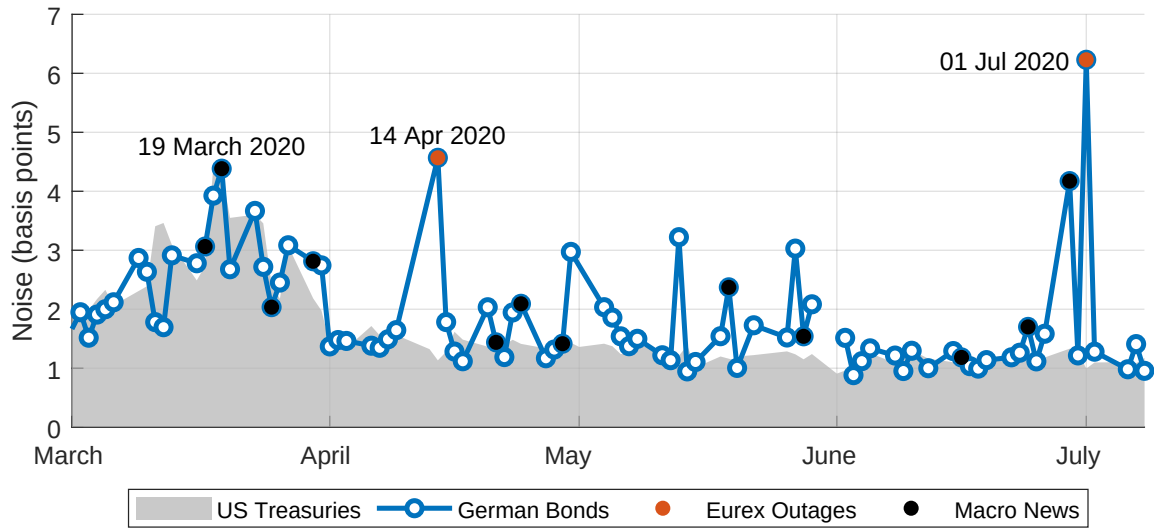


Figure 4: Yield Curve Noise during 2020. The grey area is the US Treasury noise measure published by [Hu et al. \(2013\)](#). It refers to the root mean squared pricing error, based on the cross-section of end-of-day bond prices from the CRSP database and using the [Svensson \(1994\)](#) method to fit yield curves. The blue line applies the same methodology to German government bonds, as in [Figure 3](#). Before (after) 1 June 2020, the measure is based on all transactions between 9:25 a.m. and 2:00 p.m. (8:49 a.m. and 11:39 a.m.) each day, which corresponds to the intraday times of the first (second) Eurex outage. Red dots mark the Eurex outage windows. Black dots mark windows with major macroeconomic data releases (ifo survey, ZEW survey, German CPI, or US mortgage applications). 19 March 2020 indicates the peak of the Covid-19 market turmoil. The ECB announced its €750 billion Pandemic Emergency Purchase Programme shortly before midnight on 18 March.

	(1)	(2)	(3)	(4)
Outage	1.27*** [0.11]	1.25*** [0.12]	1.07*** [0.14]	0.72*** [0.18]
Constant	0.98*** [0.07]	0.98*** [0.08]	0.99*** [0.08]	1.00*** [0.07]
FE Minute	✓	✓	✓	✓
FE ISIN		✓	✓	✓
FE Venue			✓	✓
FE Buyer/seller ID				✓
Observations	3352	3352	3055	2404
Adjusted R^2	0.072	0.115	0.127	0.374

Table 3: Effect of Eurex Outages on Mispricing. Each column shows results of a different regression. Throughout, the dependent variable is the absolute pricing error in basis points, i.e. the difference between the observed and fitted yield based on [Svensson \(1994\)](#). The sample spans all trades shown in [Figure 3](#), i.e. all trades in one to ten year German bonds during the Eurex outages and during the same intraday window on the same day in the previous and subsequent week. We sequentially add fixed effects: (1) for the time-of-day, using 15-minute buckets, (2) for the bond ISIN, (3) for the venue type the transaction is executed on, (4) for the buyer and seller ID.

identify the mechanisms behind this breakdown.

4 Micro-Level Mechanisms of Futures Market Outages

This section provides micro-level evidence on how outages in the futures market affect the cash market for Bunds. The data point to two main mechanisms. First, by removing a hedging instrument, outages reduce the intermediation capacity of dealers. Second, by removing a benchmark price, outages exacerbate information asymmetries for clients. We present two stylized facts for these mechanisms before analyzing each one in more detail in [Section 4.1](#) and [Section 4.2](#).

First, [Table 4](#) shows that outages reduce volumes most for dealer-to-client trades and least for client-to-client trades, consistent with dealers cutting back intermediation when they cannot hedge. Mispricing, in contrast, rises sharply for client-to-client trades and does not increase for dealer-to-dealer trades, consistent with clients relying on futures prices as a benchmark.

	(1) Volume	(2) Mispricing
Outage $\times$ C2C	-1.10*** [0.12]	3.30*** [0.18]
Outage $\times$ D2C	-2.54*** [0.24]	0.72*** [0.12]
Outage $\times$ D2D	-1.45*** [0.11]	0.36 [0.37]
FE Time	✓	✓
FE ISIN		✓
Observations	252	3231
Adjusted R^2	0.656	0.163

Table 4: Effect of Eurex Outages on Trading Volumes and Mispricing across Investor Segments. Column (1) refers to a balanced panel regression using the log of the trading volume in Bunds in Euro plus one, in 15-minute intervals, as the dependent variable. Column (2) refers to a transaction-level regression with absolute pricing errors as the dependent variable. Throughout, the sample covers all trades in 1-10y Bunds during the two outages and the same intraday windows on the same day in the previous and subsequent week. We differentiate trades across three investor segments: client-to-client (C2C), dealer-to-client (D2C) and dealer-to-dealer (D2D). For brevity, the table shows results only for the outage dummy interaction terms.

Second, [Table 5](#) documents that dealers’ trading profits increase during outages, both in relative and absolute terms, at the expense of clients. This is exactly in line with [Duffie et al. \(2017\)](#): *introducing* a benchmark reduces the informational advantage of dealers relative to clients, and hence lowers dealers’ profit margins. The outages are natural experiments of this effect in reverse: *removing* the benchmark exacerbates information asymmetries and hence raises dealers’ profit margins.

4.1 Bond Futures as Hedging Instruments

This section provides direct evidence that the reduced trading volumes of dealers is indeed driven by their reduced intermediation capacity. To do so, we first establish that dealers usually use bond futures to hedge their temporary inventory risk that arises from intermediating trades on the cash market, as already shown by [Naik and Yadav \(2003\)](#) for UK Gilts and [Fleming, Nguyen, and Rosenberg \(2024\)](#) for US Treasuries. We confirm this for German Bunds by running the

	Profit (€ million)		Profit (% of volume)		% profitable trades	
	No Outage	Outage	No Outage	Outage	No Outage	Outage
Client	-27.79	-48.85	-0.18	-3.73	48.97	39.38
Dealer	28.40	48.85	0.19	3.73	51.19	60.62

Table 5: Distribution of Daily Trading Profits across Dealers and Clients. To obtain a profit measure, we follow [Allen and Wittwer \(2023a\)](#) and compute markups as the deviation of a trade’s yield from the fair yield based on a fitted curve ([Figure 3](#)), appropriately signed for buys and sells such that markups are positive when favorable for the investor. We then multiply markups by trade volume and maturity to measure the euro value of gains for one counterparty and losses for the other, reflecting the zero-sum nature of trading (equivalently, we could multiply volumes with the deviation between actual and fair *prices*). The sample covers all dealer-to-client trades during the outages and non-outage control windows. To obtain a daily measure, we scale the profits from 7.28 hours (the combined duration of both outages) to 10 hours (the typical trading hours for Bunds), and divide the sum during control windows by two (since we use two control windows for each outage). Profits do not necessarily sum to zero due to a few transactions where only one counterparty can be identified.

following regression:

$$\Delta Duration_{it}^f = \alpha - \beta \times \Delta Duration_{it}^c \times Dealer_i + \epsilon_{it} \quad (3)$$

where $\Delta Duration^f$ is the duration-weighted net trading volume (buy minus sell) of investor i on day t in German government bond futures, based on EMIR data ([Section 2.2](#)). For each of the four futures, we multiply the number of traded contracts with the contract value and the modified duration and conversion factor of the underlying cheapest-to-deliver bond on day t . $\Delta Duration^c$ analogously is the duration-weighted net trading volume in German bonds on the cash market. $Dealer_i$ is a dummy variable equal to one if investor i is a dealer. We are interested in the β coefficient, the contemporaneous hedge ratio.

[Table 6](#) shows a strong and highly significant negative relationship between the cash and bond future trading volumes for dealers. On average, dealers offset 8% of the duration risk arising from their cash market trades by taking opposite positions on the futures market on the same day (column 1). This hedge ratio increases to 15% when we exclude days where we expect the trading behavior to be dominated by exogenous factors, namely auctions, quarter ends, and futures expiries (column 2). Restricting the sample to days and investors with non-zero inventory changes in both markets makes no difference (column 3).¹⁶ Equally importantly, we find a precisely estimated null effect for clients across all three specifications, confirming that clients do not use bond futures for hedging purposes. For many clients, this is obvious because they are not active on the futures market at all (see [Table A11](#) in [Appendix A.6](#)). But our results confirm that it is also true for the subset of clients that is active on both the cash and futures market.

The average hedging coefficient β of roughly 15% for the dealer sector masks substantial

¹⁶Our estimated hedge ratios are slightly smaller than those reported by [Naik and Yadav \(2003\)](#) for UK Gilts and by [Fleming et al. \(2024\)](#) for US Treasuries. Beyond differences in data and methodology, a likely reason is that Bund futures serve as the natural hedging instrument for many other euro area government bonds that do not have a dedicated futures market, particularly bonds of low default risk countries like the Netherlands, Finland, and Austria.

	(1) all days	(2) excl. auctions, EoQ, fut.exp.	(3) excl. zeros
Client	0.19 [0.41]	0.13 [0.38]	0.18 [0.41]
Dealer	7.81*** [2.42]	14.96*** [2.99]	15.30*** [3.07]
FE Day	✓	✓	✓
FE LEI	✓	✓	✓
Observations	45805	36861	16425
Adjusted R^2	-0.005	0.003	-0.020

Table 6: Duration-weighted Net Trading Volumes in German Bunds on the Cash and Futures Market. This table shows regression results of Equation 3, predicting daily changes in the duration exposure on the futures market with those on the cash market, at the investor-level, distinguishing clients from dealers. In column (2), we drop roughly 50 trading days related to primary auctions, quarter ends, and days around the expiration of bond futures. In column (3), we further drop all observations with zero inventory change at the day-LEI level on either the futures or cash market. Coefficients are multiplied by 100 such that they refer to percentage effects.

variation *across* dealers. To see this, we re-estimate Equation 3 only for the dealer sector and allow the β coefficient to vary across individual dealers. Table 7 shows the results. The β coefficients range from 0% to 92.6%, meaning that some dealers do not hedge their cash market trades at all while others hedge their entire exposure.¹⁷ This heterogeneity lets outages bite different dealers with different strength. We exploit this heterogeneity by grouping dealers into a ‘no hedging’ group if the dealer hedges less than 5% of the daily cash market exposure via futures, an ‘intense hedging’ group if they hedge more than 25%, and a ‘moderate hedging’ group with the remaining dealers.

	min	p50	mean	max
no hedging (<5%)	-0.0	0.0	0.9	3.2
moderate hedging	5.2	11.8	12.8	23.8
intense hedging (>25%)	29.4	39.2	49.4	92.6
All Dealers	-0.0	11.8	16.7	92.6

Table 7: Dealer-Heterogeneity in Bond Futures Use for Hedging. This table refers to the cross-section of β coefficients when estimating Equation 3 at the dealer-level. The estimation sample is identical to Table 6 column (3), excluding clients. β measures the percent of daily cash duration exposure changes offset via bond futures. For the two dealers that are not active on the futures market at all (see Appendix A.6), we set $\beta = 0$. The final classification includes five dealers in the ‘no hedging’, five in the ‘intense hedging’, and 22 dealers in the ‘moderate hedging’ group.

Table 8 shows how the different hedging strategies of dealers relate to their D2C trading behavior during Eurex outages. Besides trading volumes and absolute mispricing, we also look at markups (in basis points, following Allen and Wittwer, 2023a) and trading profits (in Euros, used in signed-log form in the regressions; see Table 5 for definitions).

The ‘no hedging’ group shows no obvious reaction. These dealers keep trading with their clients as usual, both in terms of volumes and prices. That means they neither increase their

¹⁷Naik and Yadav (2003) document similarly large differences in hedging ratios across UK Gilt dealers. In Appendix D.3, we show that dealers’ hedging intensity is barely related to other characteristics like connectedness, centrality and size.

	(1) volume	(2) mispricing	(3) markup	(4) profit
Outage $\times$ Dealer not hedging	0.55 [1.02]	0.35 [0.30]	-0.27 [0.60]	-0.41 [4.34]
Outage $\times$ Dealer moderately hedging	-2.35** [0.59]	0.83*** [0.12]	0.30** [0.10]	3.31* [1.31]
Outage $\times$ Dealer intensely hedging	-5.35** [1.71]	0.35* [0.15]	0.22 [0.38]	1.44 [2.49]
FE Minute	✓	✓	✓	✓
FE ISIN		✓	✓	✓
FE Client Sector		✓	✓	✓
Observations	162	1915	1915	1915
Adjusted R^2	0.365	0.069	0.028	0.031

Table 8: Effect of Eurex Outages on D2C Trades across Dealer Groups. The sample includes all D2C transactions during the two outage and four non-outage control windows. In column (1), we construct a balanced panel of trading volumes (in logs plus 1) at the 30-minute frequency, to avoid compositional effects. Columns (2)-(4) are at the transaction-level. The dependent variables are calculated from the dealer perspective. Positive coefficients in (3) and (4) hence imply a favorable increase in markups or profits for a dealer group. For brevity, the table shows results only for the outage dummy interaction terms.

activity to compensate for the withdrawal of other dealers nor do they exploit the situation by increasing markups. D2C volumes drop significantly for the ‘moderate hedging’ group and even more for the ‘intense hedging’ group, reflecting these dealers’ progressively greater reliance on the futures market. Regarding prices, we see that both groups cause mispricing but markups and profits only rise significantly for the ‘moderate hedging’ group.

One interpretation of these results is that for dealers in the ‘intense hedging’ group, intermediation capacity constraints become binding when Eurex goes down, forcing them to forego even highly profitable trades. Dealers in the ‘moderate hedging’ group, in contrast, retain some intermediation capacity, allowing them to selectively deploy capacity when it is profitable.¹⁸

Overall, this section provides quasi-experimental evidence that dealers are important for market functionality (see e.g. [Duffie et al., 2023](#)). Outages on the futures market reduce dealers’ capacity to intermediate bond trades on the cash market, as they are unable to hedge additional inventory risk. Consequently, clients either accept unfavorable quotes by dealers, or they have to trade directly with other clients, causing large mispricing.

4.2 Bond Futures As Benchmark Prices

Outages cause severe mispricing in client-to-client trades and reallocate profits toward dealers ([Table 4](#); [Table 5](#)). This suggests that dealers can price Bunds without Bund futures, while clients rely on them as benchmarks (in line with [Hortaçsu and Kastl, 2012](#); [Paiardini, 2015](#)). This section tests two further predictions of the benchmark channel.

First, less sophisticated clients should fare worse when futures go dark. We test this by separating clients into different types: non-dealer banks, non-bank financial institutions (NBFIs), investment funds, hedge funds, insurance companies and pensions funds (ICPF), non-financial cor-

¹⁸Appendix [D.3](#) confirms that these results hold when controlling for the additional effect of dealer inventory: dealers with larger pre-outage inventories raise markups and extract profits less during the outage.

porations (NFC), the official sector (including central banks) and households (retail investors).

Table 9 column (1) shows that absolute pricing errors rise broadly across client types, with the largest increase for households and hedge funds. The increase in mispricing is significant for all but three client types (ICPFs, NFCs and the official sector), probably because they execute few trades during the outages (see Appendix A.6). Looking at markups in column (2), the largest effect is on households, whose markups decrease by 4 basis points while Eurex is down, i.e. they buy Bunds at excessively low yields (high prices) and sell at excessively high yields (low prices). Hedge funds, in contrast, increase their markups significantly. Column (3), lastly, confirms that dealer banks' trades become significantly more profitable when the futures market is down, despite the non-significant increase in markups. This is because their markups are concentrated in long-term bonds and in larger trades. The opposite is true for households, which mitigates the negative effect of markups on their profits. Similarly, the significant spike in markups for hedge funds and the drop in markups for investment funds turns insignificant once we take into account the traded volume and maturity.

	(1) mispricing	(2) markup	(3) profit
Outage $\times$ Bank Dealer	0.62*** [0.13]	0.25 [0.13]	2.27** [0.62]
Outage $\times$ Bank Non-Dealer	1.86*** [0.13]	0.66 [0.48]	-1.22 [0.61]
Outage $\times$ NBFI	1.25** [0.38]	0.48** [0.13]	0.95 [0.53]
Outage $\times$ Investment Fund	0.44** [0.14]	-0.67*** [0.09]	-2.94 [1.46]
Outage $\times$ Hedge Fund	2.63*** [0.31]	2.75*** [0.45]	1.57 [1.80]
Outage $\times$ ICPF	0.27 [0.41]	-0.51** [0.15]	-6.18 [3.21]
Outage $\times$ NFC	1.78 [1.64]	-1.28 [2.18]	-1.14 [2.55]
Outage $\times$ Official	0.20 [0.13]	-0.31 [0.27]	-3.10 [1.91]
Outage $\times$ Household	4.30*** [0.69]	-4.04*** [0.34]	-2.45* [1.16]
FE Minute	✓	✓	✓
FE ISIN	✓	✓	✓
Observations	6117	4795	4795
Adjusted R^2	0.165	0.054	-0.009

Table 9: Effect of Eurex Outages on Absolute Pricing Errors, Markups, and Profits across Investor Types. The dependent variable is the transaction-level pricing error, in basis points of yield, in absolute value (column 1), in net terms (i.e. as markups, column 2), and volume- and duration-weighted (i.e. as profits, column 3). In columns (2)-(3), we drop trades between investors of the same type and the variables are calculated from the perspective of the investor type listed in the first column. Positive coefficients hence imply a favorable increase in markups or profits for that investor type. For brevity, the table shows results only for the outage dummy interaction terms.

To sum up, the client cross-section broadly fits the information asymmetry channel. Households, typically thought of as the least sophisticated client type, fare worst during outages, particularly compared to sophisticated clients like hedge funds.

A second prediction of the benchmark channel is that when the public benchmark disappears, private signals and relationships should become more important. This prediction moves us from sectors to networks. We study three established network proxies, namely the number of clients' connections (Kondor and Pintér, 2022), the intensity of client-dealer relationships (Di Maggio

et al., 2017), and dealer centrality (Li and Schürhoff, 2019).¹⁹

In Table 10, we use all three measures to explain the dysfunction on the dealer-to-client segment. Note that this analysis does not cover the most mispriced trades. That is because we cannot compute network measures for trades by households (since individuals remain anonymous in our dataset), and for trades on exchanges (as these occur anonymously in a CLOB). Nonetheless, we find that better connected clients fare better during outages. They cause less mispricing, they pay less markup, and they incur less trading losses. This is exactly in line with client’s connections measuring informational advantages in decentralized markets. Equally important is the dealer-client relationship. The stronger the relationship, the less trades become mispriced, and the less dealers extract profits from their clients. The centrality of the dealer seems less important. Trades with more central dealers are less mispriced, but the effects on markups and profits are insignificant. This could indicate that during outages, more central dealers are better informed as they rely less on the futures market as a pricing signal, but they do not act as liquidity providers of last resort and are therefore not compensated as such.

	(1) mispricing	(2) markup	(3) profit
Outage	-1.14*** [0.18]	-2.47*** [0.26]	-7.58 [9.72]
Outage $\times$ # Connections	-0.02** [0.01]	0.04*** [0.00]	0.32*** [0.06]
Outage $\times$ Relationship w/ Dealer	-1.70** [0.58]	0.44 [0.78]	5.08** [1.38]
Outage $\times$ ln(Dealer Centrality)	-0.35*** [0.03]	-0.28 [0.16]	-0.00 [0.97]
Outage $\times$ Dealer moderately hedging	0.97*** [0.14]	0.23 [0.13]	-3.01 [4.76]
Outage $\times$ Dealer intensely hedging	1.14*** [0.23]	0.45* [0.22]	-1.48 [5.24]
FE Minute	✓	✓	✓
FE ISIN	✓	✓	✓
FE Sector	✓	✓	✓
Observations	1700	1700	1700
Adjusted R^2	0.128	0.036	0.032

Table 10: Role of Dealer-Client Network Characteristics during Outages. The dependent variables are calculated from the client perspective. # Connections refers to the number of different dealers a client has traded with. Dealer Relationship measures the fraction of buying/selling volume of each client-dealer pair as a share of the total volume of each client. Dealer Centrality is the volume-weighted eigenvector centrality of the dealer involved in the transaction. The last two regressors are dummy variables equal to one if the dealer hedges between 5-25% or more than 25% of its cash market duration risk via bond futures (see Table 7). The sample includes all D2C transactions during the two outage and four non-outage control windows. We drop trades by households, since the various network measures cannot be computed, and trades on exchanges, as these occur anonymously in a CLOB. This leads to a slightly smaller sample compared to Table 8.

Importantly, these results hold even though we control for the differential effect of Eurex outages on dealer’s intermediation capacity. Just like in Table 8, we find that D2C trades become more mispriced the more the dealer involved in the transaction relies on bond futures to

¹⁹Following Kondor and Pintér (2022) we measure the number of different dealers a client has traded with. Following Di Maggio et al. (2017), we measure the intensity of each client-dealer pair as the fraction of sales by client j to dealer k , normalized by j ’s total selling volume, and we compute the analogous measure for buys. Following Li and Schürhoff (2019), lastly, we capture each dealer’s centrality as the volume-weighted eigenvector centrality. All measures are based on the universe of Bund trades in 2020.

hedge inventory risk. The differential effects on markups and profits, in contrast, mostly turn insignificant in [Table 10](#).

To conclude, the dysfunction observed on the D2C segment during Eurex outages is not entirely explicable by the reduced intermediation capacity of dealers. Instead, outages also exacerbate information asymmetries and increase the importance of client-dealer relationships.²⁰ Who you trade with matters more when the common price benchmark is gone.

4.3 Further Evidence

Swap Market If futures only hedged inventory risk, the swap market could anchor price discovery for the risk-free yield curve. Instead, [Figure 5](#) shows that interest rate swaps are severely mispriced during Eurex outages, just as we observed for Bunds on the cash market (see [Figure 3](#)).²¹

Why do Eurex outages spill over to the swap market, even though swaps are not directly linked to any Eurex-traded derivative and the outages do not mechanically remove a hedging instrument from this market? An unsatisfactory explanation is that just like on the cash bond market, dealers on the swap market — which are mostly the same large banks — hedge their trades with bond futures. Hence, when the futures market suffers an outage, these dealers stop intermediating on the swap market, causing the illiquidity and mispricing we observe.

In our view, a more satisfactory answer is that bond future prices are the most informative asset price on the fixed-income market. Bund futures provide the benchmark that anchors pricing not only in the bond but also the swap market. When that benchmark disappears, price discovery in both markets collapses, in line with cross-asset learning models ([Cespa and Foucault, 2014](#)).²²

Venue-Level [Table 11](#) shows that outages affect trading venues very differently. On electronic platforms, volumes collapse and mispricing rises, consistent with dealers pulling back intermediation when they cannot hedge. Trading on systematic internaliser platforms also declines but prices remain fair, consistent with dealers favoring loyal clients. On regular exchanges, volumes barely move, yet mispricing spikes, consistent with unsophisticated retail investors continuing

²⁰In [Appendix D.3](#), we run the same regression using a larger set of investor characteristics. We find that clients that are usually active on Eurex fare better during outages, probably because they are more likely to be aware of the Eurex outages, and adjust their trading on the cash market accordingly. Trades that are the first between a particular dealer-client pairs are also more profitable for the client. This is driven by trades on electronic trading platforms: during outages, clients are more likely to accept advantageous quotes by new dealers.

²¹[Appendix D.4](#) provides further evidence of a swap market breakdown. Indicative quotes become stale the moment bond futures become unavailable, particularly for longer-dated swaps. On Bloomberg, bid-ask spreads spike during outages, if quotes are available at all. Lastly, on one particular interdealer broker platform (for which we were able to obtain the full order book of executable quotes), Eurex outages cause a complete evaporation of liquidity, which is reversed within half an hour after Eurex goes back online.

²²An ideal test of the impaired price formation hypothesis would be to check if important fundamental news is properly incorporated into bond prices while Eurex is offline. Unfortunately, there are no such news. According to the extensive intraday news database of [Kerssenfischer and Schmeling \(2024\)](#), the most significant news during either outage was the Swedish central bank’s announcement to extend its QE program on 1 July 2020 at 9:00 a.m., see [Riksbank press release](#). However, this decision was largely anticipated and caused only minor market reactions, even in Sweden, see [Riksbank staff memo](#).

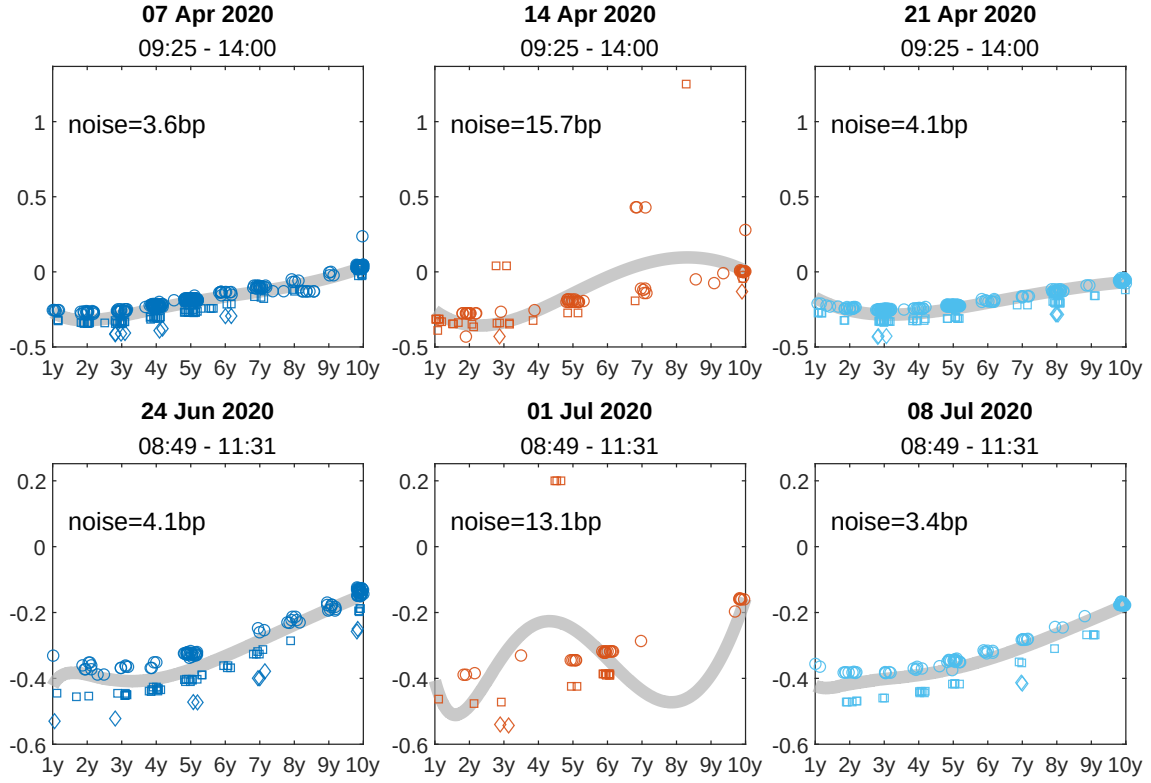


Figure 5: Euribor Swap Rates and Fitted Yield Curves. Each marker refers to an interest rate swap transaction during an outage (middle column) or the same intraday window in the previous and subsequent week (left and right column, respectively). The vertical axis refers to the fixed-leg interest rate (in percentage points), the horizontal axis to the contract horizon (in years). The floating rate is the Euro Interbank Offered Rate (Euribor) with a maturity of one, three, or six months, marked as diamonds, squares and circles, respectively. Grey lines are fitted yield curves based on Svensson (1994). The value in the top left corner of each panel is the noise measure proposed by Hu et al. (2013), defined as the root mean squared deviation of actual rates from the fitted curve. Based on EMIR data, see Section 2.2.

to trade without a benchmark. These venue-level discrepancies reinforce both mechanisms via which Eurex outages operate: they impair dealer capacity and exacerbate information asymmetries.

Transaction Sizes Table 12 shows that outages also reshape the size of trades. On the dealer-to-dealer segment, trade sizes remain unchanged, consistent with dealers postponing rather than scaling down their own trades. In contrast, dealer-to-client trades shrink by $\exp(-1.05) - 1 \approx 65\%$, pointing to reduced willingness of dealers to take on inventory when they cannot hedge. Client-to-client trades also contract, though less strongly. Across venues, the sharpest decline occurs on electronic platforms, the main D2C venue. On systematic internaliser platforms, by contrast, trade sizes hold up, consistent with dealers filling full orders for loyal clients. On regular exchanges, trade sizes also remain stable, consistent with unsophisticated retail investors continuing to execute small orders, likely unaware of the Eurex outage.

The consequences are visible in pricing. Table 13 shows that small trades become particularly mispriced during outages. Controlling for transaction size absorbs about half of the differen-

	(1) Volume	(2) Mispricing
Outage $\times$ OTC bilateral	-1.51** [0.40]	1.97*** [0.40]
Outage $\times$ OTC via IDB	-1.74*** [0.34]	0.14 [0.28]
Outage $\times$ OTC via SI	-1.64** [0.46]	0.17 [0.25]
Outage $\times$ electronic platforms	-2.30*** [0.12]	0.91*** [0.17]
Outage $\times$ MTS	-0.39** [0.11]	0.00 [.]
Outage $\times$ regular exchange	-0.23 [0.13]	4.46*** [0.97]
FE Time	✓	✓
FE ISIN		✓
Observations	504	3055
Adjusted R^2	0.639	0.166

Table 11: Effect of Eurex Outages on Trading Volumes and Mispricing across Venues. See Table 4 for details. In column (2), the differential effect for MTS cannot be estimated, since there was not a single trade in 1-10y German bonds on MTS during the outages.

	(1) Segment	(2) Venue
Outage $\times$ C2C	-0.86* [0.37]	
Outage $\times$ D2C	-1.05*** [0.15]	
Outage $\times$ D2D	0.34 [0.18]	
Outage $\times$ OTC bilateral		-0.13 [0.58]
Outage $\times$ OTC via IDB		-0.69** [0.18]
Outage $\times$ OTC via SI		-0.31 [0.35]
Outage $\times$ electronic platforms		-1.02*** [0.20]
Outage $\times$ regular exchange		-0.22 [0.25]
FE Time	✓	✓
FE ISIN		✓
Observations	3231	3033
Adjusted R^2	0.264	0.310

Table 12: Effect of Eurex Outages on Transaction Sizes. See Table 4 for details.

tial mispricing across investor segments and venues. In other words, outages impair pricing partly by forcing clients to slice trades into smaller pieces, which then execute at worse prices. This reduced-trade-size channel reflects both constrained dealer capacity as well as amplified information asymmetries in the smaller trades that remain.

4.4 Ruling out alternative explanations

Urgency or Liquidity Demand The venue-level trading volume results mentioned above are closely aligned with Menkveld, Yueshen, and Zhu (2017)’s ‘pecking order hypothesis’. According to this hypothesis, investors rank low-cost-low-immediacy venues (like the OTC segment) first and high-cost-high-immediacy venues (like exchanges) last. But when investors’ trading needs become more urgent, they reverse this ranking. This is exactly what we find for cash bonds

	(1) Segment	(2) + vol.-interact.	(3) Venue	(4) + vol.-interact.
Outage $\times$ C2C	3.30*** [0.18]	1.54*** [0.20]		
Outage $\times$ D2C	0.72*** [0.12]	0.37* [0.16]		
Outage $\times$ D2D	0.36 [0.37]	0.68 [0.41]		
Outage $\times$ OTC bilateral			1.97*** [0.40]	1.36*** [0.17]
Outage $\times$ OTC via IDB			0.14 [0.28]	0.27 [0.27]
Outage $\times$ OTC via SI			0.17 [0.25]	0.29 [0.22]
Outage $\times$ electronic platforms			0.91*** [0.17]	0.50* [0.24]
Outage $\times$ regular exchange			4.46*** [0.97]	2.06 [1.37]
Outage $\times$ log(Volume)		-0.68*** [0.01]		-0.59*** [0.07]
FE Minute	✓	✓	✓	✓
FE ISIN	✓	✓	✓	✓
Observations	3231	3231	3055	3033
Adjusted R^2	0.163	0.228	0.166	0.210

Table 13: Reduced-trade-size channel of Eurex Outages on Mispricings. This table replicates [Table 4](#) column (2) and [Table 11](#) column (2) and compares the results to a regression that includes an interaction term between the outage dummy and transaction volume.

during Eurex outages. So one might think that a higher urgency to trade, or equivalently a higher demand for liquidity, is driving the results. However, in [Menkveld et al. \(2017\)](#)’s framework, a pure ‘urgency shock’ would push more flow to high-immediacy venues and raise total volume. We see the venue shift but not the volume rise. Outages merely change the composition of investors. Those with a higher urgency to trade withdraw less from the market.

Another way to rule out that higher liquidity demand plays a major role is to look at the comovement of mispricing and dealer inventory. As [Goldberg \(2020\)](#) and [Goldberg and Nozawa \(2021\)](#) explain, higher liquidity demand increases both [Hu et al. \(2013\)](#)’s noise measure as well as dealer’s inventory. Lower liquidity supply, in contrast, increases noise but reduces dealer’s inventory. This latter channel is clearly more in line with our evidence that dealers withdraw from the market, particularly those dealers that usually use bond futures as a hedging instrument.

Market panic Is the dysfunction we observe on the cash bond market simply the result of futures market outages causing a market panic? To rule out this possibility, we first look at other asset classes. [Figure 6](#) shows that stock prices do not crash and the USD/EUR exchange rate barely moves during the outage, at odds with a market-wide panic.

Next, we rule out the possibility that outages cause a ‘flight to safety’ or ‘dash for cash’. In a ‘flight to safety’ event, Bunds of all maturities should appreciate in value, i.e. the entire yield curve should shift downwards, given that Bunds are deemed to have zero default risk (see e.g. [Baele, Bekaert, Inghelbrecht, and Wei, 2020](#)). A ‘dash for cash’ scenario predicts the exact opposite: as investors try to sell Bunds in order to get cash, bond prices should fall and the

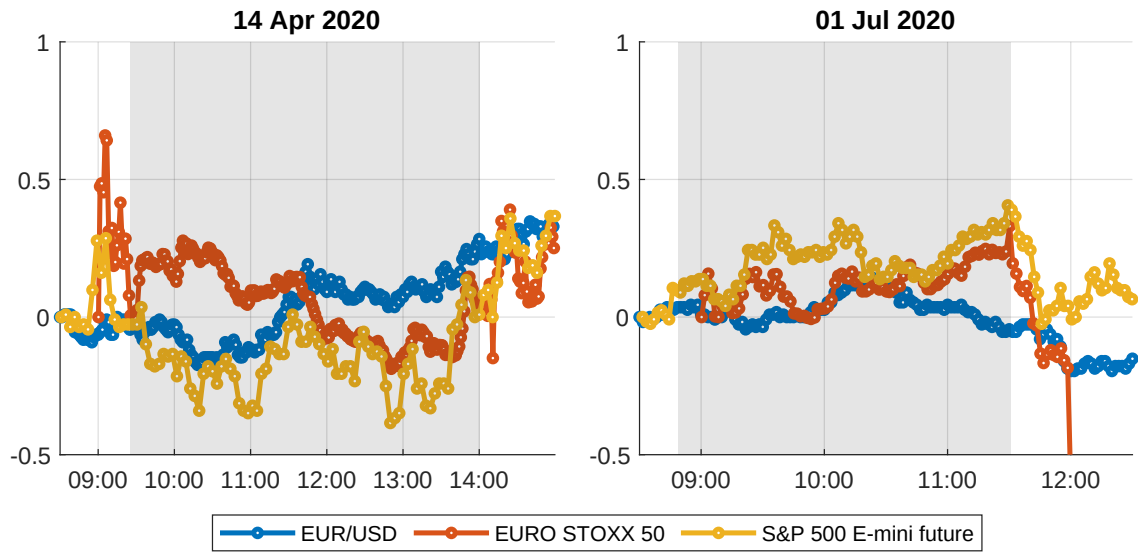


Figure 6: EUR/USD Exchange Rate and Stock Prices During Eurex Outages. All price changes are in percentage points since 08:30 a.m., the S&P 500 future refers to the front-end contract. Data is from Bloomberg. The grey area refers to the outage period on Eurex.

yield curve should shift upward.²³

While the dispersion of transaction yields *around* the curve spikes during outages, Figure 7 shows that the yield curve *itself* barely changes. This null effect is at odds with both a ‘flight to safety’ and ‘dash for cash’ explanation.

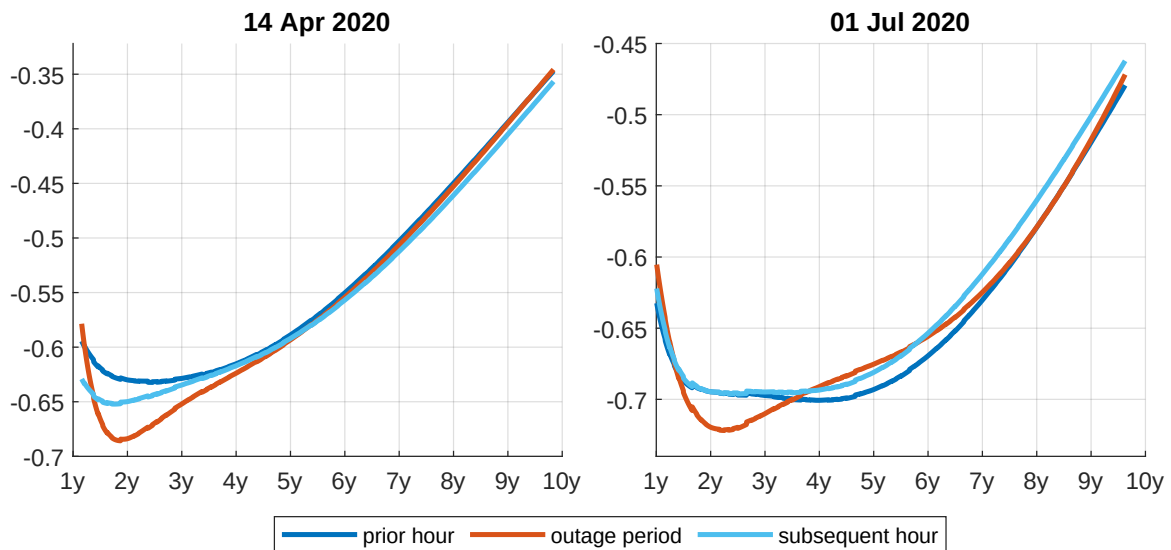


Figure 7: German Yield Curve before, during, and after Eurex Outages. This figure shows yield curves based on the Svensson (1994) method, using all transactions in German sovereign bonds with 1-10y maturity one hour before the outage, during the outage, and one hour after the outage. The fitted yield curves during the outages are identical to the ones shown in Figure 3

²³Eisenbach and Phelan (2022) show in a model how this ‘dash for cash’ effect can outweigh the ‘flight to safety’ effect, rationalizing the temporary spike of Treasury yields in mid-March 2020.

Dealer Quotes and New Trade Pairs Table 4 shows that mispricing spikes most strongly on the client-to-client market, where dealers are not directly involved. This points to an impaired price formation process. One caveat, however, is that some C2C trades might occur only because dealers stop providing quotes during outages, forcing clients to search for alternative counterparties. While we cannot observe individual quote provision directly, Table 14 provides suggestive evidence: Eurex outages lead to a significant increase in newly formed buyer-seller pairs on the cash market. The probability that a trade is the first between a given pair rises by 2 percentage points in the C2C segment and 9 percentage points in the D2C segment. These numbers might appear small, but when compared to the low baseline probabilities, they imply a 60% and 160% increase in new trade pair formations, respectively. These new connections arise mostly on electronic trading platforms, where the likelihood of a trade being the first between two investors more than doubles during outages. The increase on the OTC market is smaller, unsurprising given higher search frictions and the short duration of the disruptions.

Do these new pairings explain the dislocations we observe? Column (3) suggests not. Mispricing rises sharply during outages regardless of whether trades occur between established partners or first-time counterparties. In other words, while outages push clients into new trading relationships, the broader dysfunction in the cash market is not driven by these first-time trades.

	New Buyer-Seller Pair		Mispricing
	(1) Venue	(2) Segment	(3) Venue
Outage $\times$ C2C	0.02* [0.01]		3.40*** [0.20]
Outage $\times$ D2C	0.09*** [0.02]		0.66*** [0.16]
Outage $\times$ OTC bilateral		0.06* [0.03]	
Outage $\times$ OTC via SI		0.03* [0.01]	
Outage $\times$ electronic platforms		0.10*** [0.01]	
Outage $\times$ C2C $\times$ New Pair			-2.20 [1.34]
Outage $\times$ D2C $\times$ New Pair			-0.17 [0.98]
FE Minute	✓	✓	✓
FE ISIN	✓	✓	✓
Observations	2516	2248	2516
Adjusted R^2	0.024	0.033	0.180

Table 14: Effect of Eurex Outages on the Formation of New Buyer-Seller Pairs and their Mispricing. The dependent variable in columns (1)-(2) equals one if a trade is the first between a given investor pair and is zero otherwise (using all Bund trades in 2020). In column (3), the dependent variable is the absolute pricing errors (in basis points of yield), and we add an interaction term between the explanatory variables and the aforementioned dummy indicator. For brevity, the table shows results only for the outage dummy interaction terms. The sample covers all Bund trades, except D2D trades (due to the lack of new buyer-seller pairs) and trades on regular exchanges, as these occur anonymously in a CLOB.

5 Evidence from Cash Venue Outages

The previous sections show that when the centralized futures market suffers an outage, the decentralized cash market for EGBs is severely impaired. What about the other way around?

An ideal experiment to answer this question would require a random incident that halts trading on the entire cash market, i.e. on all exchanges, electronic trading platforms, as well as the over-the-counter segment. Fortunately for the public at large, but unfortunately for us researchers, such a comprehensive and uniform shock has never occurred. Hence, the best we can do is to study outages on individual cash trading platforms. We found five such outages affecting four different platforms:²⁴ MTS, Bloomberg, the Frankfurt stock exchange (Frankfurter Wertpapier Börse), and Brokertec.

To study the effect of these cash venue outages on the aggregate cash and futures market, we run dummy regressions like below:

$$Y_{it} = \alpha + \beta \times D_t + \gamma \times FE + \epsilon_{it} \quad (4)$$

where D_t is a dummy variable equal to one during the outage. We include the outage period and the same intraday window during the preceding and subsequent week (as controls).

For the cash market, Y_{it} measures the transaction volume in € or the number of transactions (both in logs plus one), separately for German, French, Italian and Spanish government bonds, and t refers to a 5-minute frequency. For the futures market, Y_{it} also covers the trading volume (in number of contracts or the number of trades, both in logs plus one), separately for each of the eight bond futures traded on Eurex, and t refers to a minutely frequency. Since we have the entire history of transactions on Eurex, we also look at more direct measures of market functioning. In particular, Y_{it} also covers volatility (the sum of absolute log returns), an intraday [Amihud \(2002\)](#) measure (ratio of volatility over volume), and the [Roll \(1984\)](#) bid-ask spread estimate. For simplicity, these last three measures are normalized to mean zero and unit variance for each future.²⁵

[Table 15](#) contains results. The top panel shows that all cash venue outages reduce the aggregate trading activity on the cash market. This is at odds with the common claim that bond trading seamlessly moves to alternative platforms in case of individual outages, see the narrative evidence in [Appendix F.1.1](#). So given that individual cash venue outages affect the aggregate cash market, do they also affect the futures market?

Yes, but mainly in terms of trading activity, not market functioning, as the lower panel in [Table 15](#) shows. The trading volume on Eurex drops significantly during cash market outages, but not nearly as much as in the other direction: the pooled estimate is roughly one-tenth as

²⁴[Appendix B](#) confirms all outages through newspaper reports, except the most recent one on MTS. We could not confirm this outage independently, but the episode we identified has the same characteristics as an earlier confirmed outage: no intraday transactions or quotes for any bond for a prolonged period during trading hours and in the absence of any major news.

²⁵We focus on average effects here, see [Appendix E](#) for more granular country-level results. We do not attempt to link the differential effects across outages and countries to the different characteristics or market shares of each platform. We leave this to future research.

large as the equivalent estimate in Table 1. More direct measures of market functioning show even less of an impact. Bond future prices do not become more volatile and the illiquidity measure of Amihud (2002) increases only during two of the five outages, as does the bid-ask spread estimate of Roll (1984). Pooling all cash venue outages together, the two measures imply that the liquidity on Eurex drops by only .05 and .09 standard deviations, respectively. These effects pale in comparison to the complete evaporation of liquidity on the cash market during Eurex outages (see Section 3.2).

	MTS 12 Jan 2010 8:00-10:35	Bloomberg 17 Apr 2015 9:20-10:10	FWB 27 May 2015 8:00-11:00	Brokertec 11 Jan 2019 19:43-21:35	MTS 26 Jul 2019 12:30-13:20	Pooled
<i>Cash market</i>						
Volume	-1.33*** [0.11]	-4.45*** [0.13]	-0.43* [0.10]		-2.35* [0.69]	-1.46*** [0.44]
#Trades	-0.17*** [0.00]	-0.38 [0.17]	-0.08 [0.08]		-0.28*** [0.01]	-0.17*** [0.05]
<i>Futures market</i>						
Volume	0.02 [0.03]	-0.25* [0.06]	-0.59 [0.50]	-0.23*** [0.01]	-0.55** [0.12]	-0.31* [0.17]
#Trades	0.03 [0.05]	-0.15** [0.02]	-0.29 [0.26]	-0.16** [0.03]	-0.43* [0.14]	-0.17* [0.09]
Volatility	0.09 [0.10]	-0.02 [0.03]	-0.44 [0.41]	-0.04 [0.02]	-0.11 [0.09]	-0.20 [0.19]
Amihud	0.02** [0.00]	0.03*** [0.00]	0.09 [0.05]	-0.09 [0.15]	0.07 [0.02]	0.05* [0.03]
Roll	0.02 [0.05]	0.08* [0.02]	0.15* [0.04]	-0.13 [0.14]	0.05 [0.02]	0.09** [0.03]

Table 15: Effect of Cash Venue Outages. Each column refers to a different outage, each row to a different dependent variable. Each sample includes the outage period and the same intraday window on the preceding and subsequent week (as controls). The last column refers to a pooled regression of all outages. For the cash market, variables are at the country-level and five minute frequency and Volume refers to the transaction volume in €. For the futures market, variables are at the future-level and one minute frequency and Volume refers to the number of traded contracts. Volume and #Trades are in logs plus one, Volatility is the sum of absolute log returns, Amihud (2002) is the ratio Volatility/Volume, and Roll (1984) is a bid-ask spread estimate. The last three measures are normalized to mean zero and unit variance. Since these three measures are undefined when there are zero trades in a given period, estimates may suffer from selection bias. Replacing missing values with zeros renders all pooled estimates statistically insignificant. All regressions include fixed effects for the minute within a day and for each country or bond future.

Why do cash and futures market outages have so asymmetric effects? Recall that futures market outages cause dysfunction on the cash market via two mechanisms: they remove a

hedging instrument and a benchmark price (see [Section 4](#)). In contrast, [Table 15](#) suggests that cash market outages only operate via the hedging channel. In particular, trading volumes drop on the futures market mainly because investors need to hedge fewer trades on the cash market. The liquidity on the futures market is barely affected, however, since cash trading venues are negligible for price discovery.

Let us now elaborate on the three most informative cash market outages below.

MTS Outage We know from [Section 3.2](#) that the MTS cash market platform is dependent on the futures market: trading and quotation activity stops when Eurex goes down. Is there a similar dependency of Eurex on MTS? January 12, 2010, provides a natural experiment to test this hypothesis. On this day, MTS suffered a roughly two and a half hour long outage at the beginning of the trading day: instead of 8:00 a.m., the first quotes for any bond appeared at 10:35 a.m in the order book. [Table 15](#) suggests that this MTS outage barely affects the futures market. Similarly, the unconfirmed but suspected outage on MTS in 2019, when trading and quoting activity stopped entirely for almost an hour, reduces the trading activity on the futures market somewhat but has no significant impact on any of the market functioning proxies. Since this latter outage occurred quite recently, [Appendix E](#) provides further evidence. Neither the country-level order book liquidity on the futures market, nor indicative bond quotes on Bloomberg show any obvious reaction to the MTS outage. That these null effects also hold for Italy is worth stressing. MTS is the dominant cash trading venue for Italian bonds but still, outages on MTS seem to have little macro-level market impact even for Italy.

Bloomberg Outage On Friday, 17 April 2015, Bloomberg suffered an outage from 8:20 a.m. till at least 9:10 a.m. London time. Some newspaper reports suggest a longer duration, probably because Bloomberg terminals came back online gradually. In any case, traders worldwide were not able to access their terminals temporarily. [Table 15](#) shows that this cash venue outage has the biggest impact on the futures market for European government bonds. Trading activity in bond futures slows down and the illiquidity measures of [Amihud \(2002\)](#) and [Roll \(1984\)](#) increase significantly. The magnitude of these effects is small, however, corresponding to an increase of .03 and .08 standard deviations, respectively. [Appendix E](#) provides further evidence for a limited impact of the Bloomberg outage.

Brokertec Outage On 11 January 2019, another Friday, Brokertec, the dominant electronic trading platform for US Treasuries (see [Fleming, Nguyen, and Ruela, 2024](#)), suffered a roughly two-hour-long outage, from 1:43 till 3:35 p.m. US Eastern Time. [Table 15](#) shows that this outage reduces trading activity in euro area bond futures, but does not affect actual market functioning.²⁶ Since this outage occurred quite recently, we can again look at actual liquidity on the futures market, both in the euro area as well as the US. [Figure 8](#) shows that the order book depth in German Bund futures seems unaffected by the outage on Brokertec. For US Treasury

²⁶The European *cash* market was unaffected by the Brokertec outage because it was effectively idle during this time of day (7:43-9:35 p.m. local time) and because Brokertec has a minor market share in EGBs.

futures, liquidity slightly *improves* after the outage and falls back to normal once Brokertec goes back online. This seemingly counterintuitive effect suggests that outages reduce latency arbitrage between Treasuries and Treasury futures, similar to the effect of adverse weather between NYC and Chicago on the equity market. In that setting, (Shkilko and Sokolov, 2020) show that adverse weather interrupts the early access to futures price movements in Chicago for some high-frequency trading firms, which improves liquidity for cash equities in NYC. The overall benign effects of the Brokertec outage suggest that one of our main findings – that price formation and liquidity provision is more of a one-way street from the futures to the cash market than previously thought – also applies to the US.

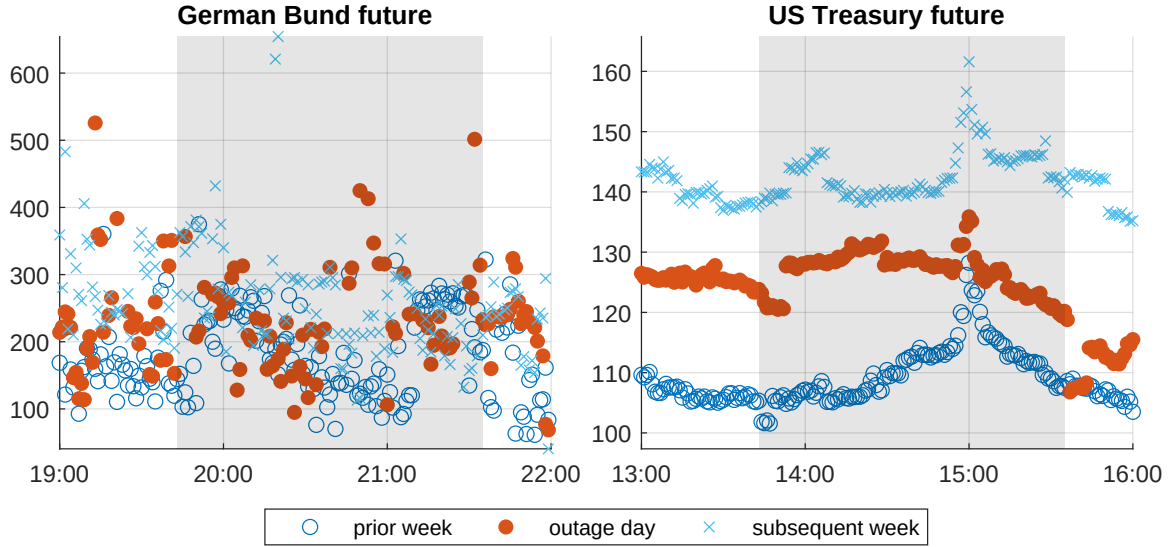


Figure 8: Order Book Depth of Bond Futures during Brokertec outage on 11 January 2019. This figure shows the number of quoted contracts in 10-year German Bund futures on Eurex (left panel; covering both sides of the first three levels of the order book; data comes from Bloomberg, since our Eurex order book data starts in April 2019) and 10-year US Treasury futures on CME (right panel; in thousands; covering both sides of the first 15 levels of the order book). The grey area refers to the outage period on the Brokertec cash trading platform. Red markers refer to the outage day, dark and light blue markers refer to the previous and subsequent week. Timestamps are in local time.

Overall, we conclude that outages on individual cash trading platforms affect the fixed-income market not nearly as much as outages on the centralized futures market. Next, we will investigate whether outages on the European and US futures market affect each other.

6 Conclusion

Sovereign bond markets build the core of the financial system. We show that in the euro area this market becomes dysfunctional without bond futures. When the Eurex futures exchange suffers an outage, trading activity on the cash market declines, liquidity dries up, and the remaining transactions occur at prices far from fundamental values.

Our non-anonymous transaction-level dataset reveals two mechanisms behind this breakdown: outages remove a hedging instrument, limiting dealers’ capacity to intermediate, and

they remove a benchmark price, amplifying information asymmetries among clients. Dealers that rely heavily on futures retreat most from the cash market, while dealers with a moderate use deploy their remaining intermediation capacity selectively. In fact, dealers' profit margins rise during the outage, as the lack of a public benchmark price widens the informational advantage of dealers over clients (in line with [Duffie et al., 2017](#)). With private signals and relationships becoming more important, clients fare very differently: sophisticated clients and those with strong dealer relationships are least affected by the outage, whereas households and less connected clients suffer most.

Lastly, we show that outages on individual cash trading platforms barely affect bond futures, suggesting that price formation and liquidity provision is more of a one-way street from the futures to the cash market than previously thought. This asymmetry suggests that for sovereign bonds, cross-asset learning models (e.g. [Cespa and Foucault, 2014](#)) offer a more realistic framework than symmetric arbitrage models (e.g. [Gromb and Vayanos, 2010](#)).

Our results have important implications for a number of ongoing policy debates, e.g. regarding the pros and cons of centralization (see e.g. [Duffie, 2025](#)). Notably, the U.S. Securities and Exchange Commission's mandate to centrally clear U.S. Treasury cash transactions, currently scheduled to take effect in 2027 ([SEC press release](#)), promotes all-to-all trading. This would make the cash market, which is currently fragmented across multiple trading platforms, many of which lack central clearing, more similar to the futures market, which is fully centralized. We show that this centralization makes the futures market highly liquid and clearly dominant in terms of price discovery. The downside is that an outage of the futures market becomes a systemic risk. In contrast, the decentralized cash market is less liquid, provides little price discovery, heavily relies on dealers as financial intermediaries, but it is robust to an outage of any individual cash platform.

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INTERNET APPENDIX

for

OUTAGES IN SOVEREIGN BOND MARKETS

by MARK KERSSENFISCHER and CASPAR HELMUS

(not for publication)

[Appendix A](#) provides details on our data, especially our cash transaction dataset.

[Appendix B](#) lists the outages we study and confirms them through newspaper reports.

[Appendix C](#) uses prior outages on Eurex as a robustness check of our main results.

[Appendix D](#) provides further evidence on the two most recent Eurex outages.

[Appendix E](#) contains further results regarding outages on other trading venues.

[Appendix F](#) presents narrative evidence, namely quotes from private market participants about their views on euro area bond market functioning.

Appendix A Data Details

This section contains background information on our data. Appendix A.1 provides an overview of our datasets and Appendix A.2 explains how to get access to the data. Appendix A.3 outlines the market structure on the European cash and futures market for bonds. Appendix A.4 describes the universe of cash bonds we study, Appendix A.5 provides details on our transaction-level data, and Appendix A.6 explains how we classify investors into different types and segments.

A.1 Overview of Data Sources

Dataset / Source	Content	Anony- mous	Sample period
<i>Cash Market</i>			
MiFID I ('Bafin')	transactions	no	2008-2017
MiFID II ('Mifir')	transactions	no	since 2018
MTS	transactions and order book updates	yes	since 2008
MTS BondVision, Tradeweb, BGC, TPICAP, GFI, Aurel	transactions	yes	six days in 2020 (two Eurex outage days and four control days)
Bloomberg, Refinitiv, TPICAP	indicative quotes	yes	
<i>Futures Market</i>			
EMIR	transactions	no	since 2008 (prior to 2018, data comes from the MiFID I dataset)
Eurex	transactions	yes	since 2002
Eurex (A7)	order book updates	yes	since April 2019 (data since January 2019 comes from Bloomberg)
CME (A7)	order book updates	yes	since February 2019

Table A1: Overview of Data Sources Used in This Paper.

Table A1 lists all datasets we use. We focus most of our analysis on the two most recent Eurex outages in 2020 for data availability and quality reasons. In particular, the MiFID I dataset captures less than half as much daily trading volume as the MiFID II dataset and covers particularly few trades by clients such as investment funds. MiFID II has better data coverage and much more granular information on trades, e.g. a flag identifying trades by retail investors. See Appendix A.5 for a detailed description. However, MiFID II data quality improved gradually over time and remains relatively poor for 2018 and 2019. Moreover, we obtained transaction-level data from various trading venues only for the six relevant days in 2020. For these reasons, we compute yield curve fitting errors and run venue- and investor-level regressions only for the outages in 2020.

A.2 Access and Availability

Access to the non-anonymous regulatory MiFID I, MiFID II and EMIR datasets can be requested via the [Research Data and Service Centre \(RDSC\)](#) at Bundesbank and is subject to strict requirements, see [rules for visiting researchers at the RDSC](#).

The [application process](#) requires researchers to write a short proposal outlining the research question and its policy relevance. The data request is then evaluated by the RDSC and economists in the statistics department providing the data. Once the evaluation has been successfully completed, researchers need to sign a contract with the RDSC, before being granted secure on-site access to the data.

All other data is available directly from data vendors.

A.3 Future and Cash Market Structure

This section provides further details on the market structure for European government bonds briefly outlined in [Section 2](#) of the main text.

[Table A2](#) gives an overview of the available bond futures and their trading volumes in 2022. Because of their benchmark status in the euro area fixed income markets, futures on German government bonds dominate trading with roughly 80% of the total in 2022. Italian (10%), French (8%) and Spanish futures (<1%) are much less relevant. Importantly, the futures market is fully centralized on Eurex. For each future, three contracts with different expiration horizons can be traded, one for each of the three nearest months in the March, June, September and December cycle. We focus on the nearest-to-maturity contracts at any point in time. These account for over 90% of trading volume, as most investors ‘roll over’ to the next-nearest contract one or two days before expiration. That means for the Eurex outage (and control days) in April 2020, we use future contracts that expire in June 2020. For the July outage, we use contracts that expire in September 2020.

[Table A3](#) gives a rough comparison of the relative size of the future and cash markets. For Germany, the trading volume on the futures market is roughly ten times as large as on the cash market. For France and Italy, the two markets are of comparable size. For Spain, where a future was introduced last, namely in October 2015, trading volumes in futures are only 1% of the cash market volume. Within the cash market, the trading volume in German and Italian bonds is similar, whereas the volume is only 75% as large for French and 30% as large for Spanish bonds.

[Figure A1](#) visualizes EGB trading volumes, not only across but also within the futures and cash market. The figure highlights that trading volumes on the futures market are concentrated into a handful of bond futures whereas on the cash market, the trading volume is dispersed across hundreds of individual bonds, many with tiny trading volumes. For Germany, France and Italy, not even the most-traded cash bond comes close to any of the bond futures in terms of trading volume. For Spain, in contrast, the trading volume in the bond future is roughly equal to the median volume across cash bonds.

[Table A4](#) gives a stylized overview of the different cash market venues for EGBs, differentiating between exchanges, electronic platforms, and the over-the-counter segment. We classify venues based on the employed trading protocol, which leads to some discrepancies with the MiFID II regulation. MTS, e.g., is regulated as a multilateral trading facility, whereas we classify it as an exchange, since MTS employs a central limit order book just like stock exchanges.²⁷

²⁷MiFID II Art. 4(1)(20)-(24) differentiates between regulated markets (exchanges in our terminology) and multilateral and organised trading facilities (MTF and OTF; mostly electronic platform in our terminology). Single-dealer platforms are called ‘systematic internalisers’ (SI) and are defined as ‘an investment firm which, on an organised, frequent systematic and substantial basis, deals on own account when executing client orders outside a regulated market, an MTF or an OTF without operating a multilateral system’. We follow the MiFID II regulation and consider these trades a particular form of OTC trading. In practice, systematic internalisers are usually large banks and the ‘SI’ status comes with extra regulatory responsibilities, e.g. an obligation to report

The market shares of different EGB trading venues are notoriously difficult to measure, due to the poor post-trade transparency and the lack of a consolidated tape. The last row in [Table A4](#) provides a rough estimate based on public sources, which suggest that roughly half of the trading volume occurs OTC, with the remainder split 2:1 between electronic platforms and exchanges (mostly MTS).²⁸

In fact, [Figure A2](#) roughly confirms these aggregate market shares based on our transaction-level dataset, which we present in detail in [Appendix A.5](#). The figure also documents substantial country-level differences. German bonds, e.g., are traded predominantly OTC or on electronic platforms and rarely on-exchange.²⁹ For Italian bonds, in contrast, on-exchange trading on MTS accounts for more than 40% of volume. Note, however, that the venue market shares for France, Italy and Spain should be interpreted with caution, since they rely solely on the anonymous *trade* – not *transaction* – reports mentioned in [Appendix A.5](#). This is because proper identification of trade chains (affiliate trades) works best in German bonds and also the reason why we focus on German bonds in [Section 3.3](#) and [Section 4](#) in the main text when studying the venue- and investor-level market microstructure.

Name	Code	Country	Maturity (years)	Contracts (million)	%	Volume (billion €)	%
Bund	FGBL	DE	8.5-10.5	216	33%	32,835	37%
Bobl	FGBM	DE	4.5-5.5	158	24%	19,873	23%
Schatz	FGBS	DE	1.75-2.25	142	21%	15,543	18%
OAT	FOAT	FR	8.5-10.5	54	8%	7,745	9%
BTP	FBTP	IT	8.5-11	42	6%	5,390	6%
Short Term BTP	FBTS	IT	2-3.25	27	4%	2,977	3%
Buxl	FGBX	DE	24-35	22	3%	3,793	4%
Bono	FBON	ES	8.5-10.5	0	0%	21	0%
*Mid-Term BTP	FBTM	IT	4.5-6	0		0	
*Mid-Term OAT	FOAM	FR	4.5-5.5	0		0	
Sum				662		88,177	

Table A2: Euro Area Government Bond Futures on Eurex. Trading volumes refer to 2022 ([source: Eurex](#)). The maturity column refers to the remaining maturity a cash bond must have in order to be deliverable into the respective futures contract. *Since their trading volume is virtually zero, we exclude the FBTM and FOAM futures from our analysis.

trades through an Approved Publication Arrangement (APA). A recent [ESMA Consultation Paper](#) provides a concise overview of European trading venues and the [ESMA Register](#) contains a list of regulated markets, MTFs, OTFs, SIs, and APAs. Lastly, a [2021 ICMA Directory](#) catalogues cash trading venues including eligible participants and trading protocols.

²⁸A 2021 [European Commission Report](#) about the proposed merger between the London Stock Exchange Group (owner of the MTS platform at that time) and Refinitiv (owner of the Tradeweb platform) provides a detailed overview of the European government bond market and the different trading platforms. The merger was approved only after the London Stock Exchange Group sold off its MTS platform to Euronext (see [European Commission press release](#)).

²⁹[de Roure et al. \(2025\)](#) study dealer-to-dealer trades in German bonds (using the ‘Bafin’ dataset mentioned in [Section 2.1](#)) where they can identify both counterparties. In this subsample, they find that interdealer brokers account for roughly 80% of trading volume. Our data puts the IDB share at roughly 25% of the aggregate trading volume.

Country	Future Volume	Cash Volume	Ratio Future/Cash
Germany	72,044	7,404	9.7
France	7,745	5,435	1.4
Italy	8,367	7,197	1.2
Spain	21	2,002	0.01
Sum	88,177	22,038	4.0

Table A3: Comparison of Euro Area Government Bond Trading Volumes on the Futures and Cash Market in 2022 in billion €. The bond future trading volumes correspond to the aggregate volume of all futures of a given country, see [Table A2](#). The cash market trading volumes are based on the [European Secondary Bond Market Data Report](#) by the International Capital Market Association (ICMA). These numbers are roughly in line with data from the [Government Bond Data Report](#) from the Association for Financial Markets in Europe (AFME), whose numbers for Germany are based on a survey of the debt management agency among dealer banks. This survey puts the cash trading volume in German bonds in 2022 roughly 10% lower at 6,636 billion €, see [Finanzagentur website](#).

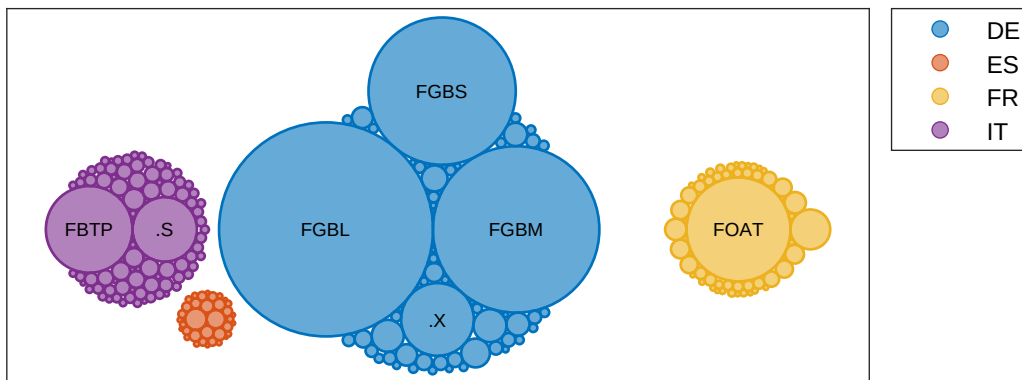


Figure A1: EGB Trading Volumes at the Instrument-Level. Each circle refers to an individual bond or bond future. Bond futures are labelled, e.g. ‘FGBL’ for the 10-year Bund future. The size of each circle is proportional to the trading volume in that instrument. In particular, the relative trading volume across countries and across the futures and cash market matches [Table A3](#). Within the futures market, the relative volumes match [Table A2](#). For the cash market, the number of bonds per country and the distribution of trading volumes across those bonds are based on our transaction dataset for a particular day (7 April 2020), see [Appendix A.5](#). Since this dataset is restricted to plain vanilla bonds, see [Appendix A.4](#), the above figure slightly understates the true number of bonds per country. Lastly, note that the Spanish bond future is not labelled due to its low trading volume (roughly equal to the median volume across Spanish cash bonds).

	On-Exchange	Electronic Platforms	Over-the-Counter
Trading Protocol	Central Limit Order Book	Request for Quote	Voice, Chat
Immediacy, Centralization, Pre-Trade Transparency	High	Moderate	Low
Post-Trade Transparency	High	Low	Low
Anonymity	High	Low	Low if bilateral, moderate if via broker
Examples	MTS, stock exchanges	Tradeweb, Bloomberg	interdealer brokers: TPICAP, BGC, ...
Market Share*	12.5%	37.5%	50%

Table A4: Stylized Differences between Cash Trading Venues for European Government Bonds. *The market shares are rough estimates based on public trading volume data from different sources covering different time periods. In particular, the above-mentioned [2022 ICMA report](#) puts the OTC market share at roughly 50% (labelled ‘systematic internaliser’ trades, which are a major segment of the OTC market, see also [Appendix A.5](#)). To decompose the remaining “on-venue” volume into exchanges and electronic platforms, we use the [2019 EC Report](#) cited in the main text. This report estimates that roughly 25% of “on-venue” volume occurs “on-exchange” (0-5% on regular stock exchanges and 15-30% on MTS, see Table 3 on p.168 of the report).

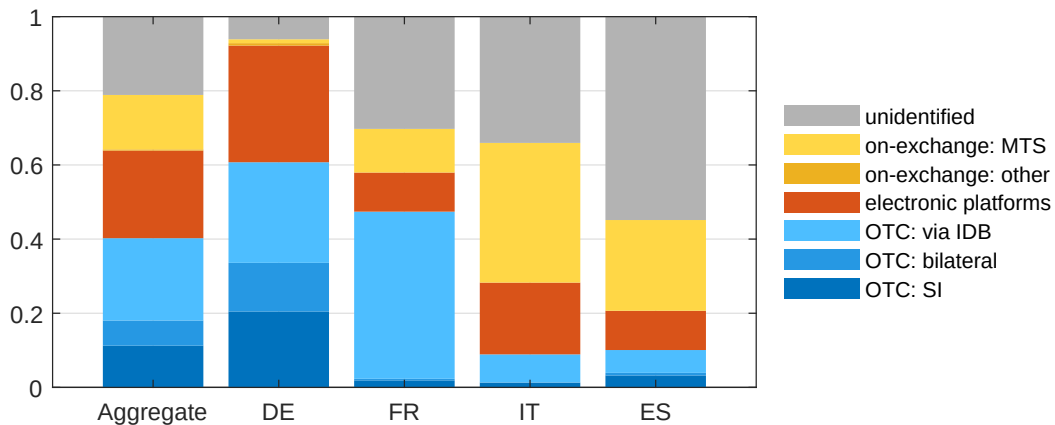


Figure A2: Market Shares of Cash Trading Venues. This figure breaks down the aggregate and country-level trading volume in EGBs into different cash trading venues. Results are based on our transaction dataset for all four non-outage days (i.e. April 7 and 21, June 30 and July 8, 2020), see [Appendix A.5](#). ‘IDB’ refers to interdealer brokers, ‘SI’ refers to systematic internalisers and ‘other exchanges’ refer to regular stock exchanges.

A.4 Cash Bond Universe

To make life easier, we restrict our analysis to ‘plain vanilla’ European government bonds whenever we study the cash market. That means we ignore inflation-indexed bonds, bonds with variable coupon, ‘strips’, ‘green’ bonds, and any other exotic bonds. For each bond ISIN, we obtain a number of bond characteristics from Bloomberg, including the issuance and maturity date, the coupon rate, coupon frequency and day-count basis. [Table A5](#) provides summary statistics regarding the time to maturity, time since issuance, original maturity, and the coupon rate of the bonds we study. [Table A6](#) provides a breakdown of our bond universe by country and maturity bucket.

	min	p50	mean	max
Years to Maturity	0.0	2.9	6.3	46.9
Years since Issuance	0.0	2.5	4.7	28.4
Original Maturity in Years	0.2	10.0	11.0	50.4
Coupon Rate	0.0	0.9	1.7	9.0

Table A5: Bond Characteristics.

Maturity Bucket	DE	ES	FR	IT	Total
< 2.5 years	37	24	46	50	157
2.5 – 5.5 years	14	12	11	25	62
5.5 – 10.5 years	15	16	14	18	63
> 10.5 years	11	12	16	21	60
Total	77	64	87	114	342

Table A6: Number of Bonds by Country and Maturity Bucket.

We use maturity buckets of less than 2.5 years to maturity, 2.5 to 5.5 years, 5.5 to 10.5 years, and more than 10.5 years. These thresholds ensure that we have an ‘on-the-run’ (OTR) bond in each of the first three buckets and a cheapest-to-deliver bond in all buckets (if a corresponding future exists, see [Table A2](#)).

[Table A7](#) shows the ISINs of the bonds that were cheapest-to-deliver into futures contracts at the time of the Eurex outages in 2020. Similarly, [Table A8](#) shows the ISINs of on-the-run bonds, i.e. the most recently issued bond with approximately two, five or ten year original maturity. For France and Spain, we identify OTR bonds for the 1-year rather than the 2-year maturity, since the latter maturity is uncommon in these countries.

Maturity	Country	Future	April 2020	July 2020
2y	Germany	FGBS	DE0001104792	DE0001104800
	Italy	FBTS	IT0005367492	IT0005282527
5y	Germany	FGBM	DE0001102374	DE0001141810
10y	Germany	FGBL	DE0001102465	DE0001102473
	France	FOAT	FR0013407236	FR0013407236
	Italy	FBTP	IT0005365165	IT0005365165
	Spain	FBON	ES0000012E51	ES0000012E51
30y	Germany	FGBX	DE0001135481	DE0001102341

Table A7: Overview of Future Contracts and Cheapest-to-Deliver Bonds.

Country	Time	1y/2y	5y	10y
Germany	April 2020	DE0001104792 [#]	DE0001141810	DE0001102499
	July 2020	DE0001104800 [#]	DE0001141810 [#]	DE0001102507
Italy	April 2020	IT0005388928	IT0005344335	IT0005403396
	July 2020	IT0005412348	IT0005408502	IT0005413171
France	April 2020	FR0125848699	FR0013157096	FR0013451507
	July 2020	FR0126001801	FR0013157096	FR0013516549
Spain	April 2020	ES0L02103056 [*]	ES0000012F92	ES0000012F76
	July 2020	ES0L02106117	ES0000012F92	ES0000012F76

Table A8: Overview of On-the-Run Bonds. For Germany and Italy, we identify 2-year OTR bonds, for France and Spain we identify 1-year OTR bonds. [#]For Germany, some bonds were OTR and CTD bonds at the same time, see [Table A7](#). ^{*}For Spain, the 1-year OTR bond changed between the first Eurex outage on 14 April 2020 and the subsequent control day on 21 April (namely to ES0L02104161 on 17 April 2020).

A.5 Cash Market Transaction Dataset

As explained in [Section 2.1](#), we exploit multiple datasets to capture as many cash transactions in EGBs as possible. For the entire sample since 2008, we use data from the MiFID I/II datasets and from MTS. For the six most important days – the two Eurex outage days and the respective control days in 2020 – we exploit further sources, as explained below.

Data Sources

In a nutshell, we start with a non-anonymous regulatory dataset and add non-duplicate anonymous transactions sourced directly from trading venues.

To give more context, note that the MiFID II regulation differentiates between *trade* and *transaction* reporting. Trade reports are anonymous, need to be sent in near real-time, and must be made publicly available, e.g. via an Approved Publication Arrangement (APA).³⁰ These reports usually contain the intraday timestamp, ISIN, price and volume of a traded bond, with some exceptions.³¹ Transaction reports, in contrast, contain additional information, most importantly about the counterparties involved in each trade.³² These reports need to be sent on a $T + 1$ basis, either to the National Competent Authority (NCA) or ESMA, possibly via an Approved Reporting Mechanism (ARM).

What we refer to as ‘MiFIR’ or ‘MiFID II’ data refers to the *transaction* reports sent to Bafin, Germany’s NCA. This data only covers trades in which a German security is traded or where at least one of the involved parties – the buyer/seller, the trading venue, or a central counterparty – is domiciled in Germany. We augment this data with non-duplicate *trade* reports we obtained directly from trading platforms (MTS, MTS Bondvision, Tradeweb) and interdealer brokers (TPICAP, BGC, GFI, Aurel).³³ See below for further details on how we define duplicates and clean the data.

³⁰However, the regulation does allow to ‘charge fees for the use and redistribution of historic data’, see [ESMA Q&A on MiFID II and MiFIR transparency topics](#). Hence, in practice, trade reports are only available for the current and previous business day, see e.g. [Tradeweb website](#), [Bloomberg website](#) and [Brokertec website](#). In June 2023, the European Commission has announced to improve this state of affairs by creating a ‘*consolidated tape [that] will bring together the prices and volumes of financial instruments, such as shares and bonds, from hundreds of execution venues across all Member states into a single stream of information, equally accessible for everybody*’, see [EC press release](#).

³¹Under MiFID II, transactions above specific ‘large in scale’ (LIS) or ‘size specific to the instrument’ (SSTI) thresholds or transactions in ‘illiquid’ bonds benefit from deferred publication. The thresholds vary across ISINs and are regularly updated by ESMA, as is the list of bonds deemed liquid. At the time of the two Eurex outages in 2020, the LIS threshold for sovereign bond transactions was €15 million, the SSTI threshold was €6.5 million (the latter only applies to trades where a party is dealing on own account) and ESMA deemed a total of 520 bonds liquid (see [ESMA website](#)). In some cases the deferral period can be indefinite, i.e. these transactions are never published individually. Instead, only a weekly aggregate volume is published at the ISIN-level. Lastly, MiFID II allows ‘volume masking’, i.e. some transactions can be reported without volumes. Regarding our dataset, these reporting exemptions apply only to the Tradeweb dataset, which covers only few trades above €6.5 million, none above €15 million, and roughly 5% of transactions omit the volume information. For details on the post-trade reporting requirements, see this [AFME report](#).

³²For the full list of reported fields, see [EU regulation 2017/590 Annex 1 Table 2](#).

³³For the sake of completeness, note that we unsuccessfully tried to obtain intraday transaction data from further data sources. Tradition, another large interdealer broker for European government bonds, see [Tradition website](#), did not retain transaction data for the period we study. Similarly, Bloomberg provides the transaction data it collects under MiFID II only with a limited history. Lastly, various sources provide data on ISIN-level trading volumes, but only at a daily frequency, which is too coarse for our purposes (e.g. Clearstream and Euroclear, the two principal clearing houses in Europe, MarketAxess, a trading venue, and Markit, an independent data provider).

Final Sample

Table A9 panel (a) shows the final number of transactions we use from each data source. As expected, our MiFIR data has the best coverage for trades in German bonds. In theory, the dataset should cover all trades, and we are close to this ideal in practice. For instance, adding transactions from Tradeweb to the MiFIR dataset gives roughly 3% more observations for Germany, but almost doubles the number of observations for France and Spain. Similarly, adding transactions from MTS and MTS Bondvision is crucial for Italy, but rather negligible for Germany.

Table A9 panel (b) contains average daily trading volumes at the country-level. To put these numbers into perspective, we capture a volume of 24 billion Euro in German government bonds, which is slightly higher than the German debt management agency estimates based on a survey among dealer banks (4,255/250 $\approx$ 17 billion Euro daily trading volume in 2020, see [Finanzagentur website](#)).

(a) Number Of Transactions					
	DE	FR	IT	ES	Total
Mifir	16891	1821	13172	1950	33834
Tradeweb	550	1738	2365	1379	6032
MTS	36	431	5278	461	6206
BondVision	125	206	2711	218	3260
Interdealer Broker	601	660	516	103	1880
Total	18203	4856	24042	4111	51212

(b) Average Daily Trading Volume (in billion €)					
	DE	FR	IT	ES	Total
in billion €	24.0	6.0	13.5	4.0	47.5
in % of total	50.5	12.6	28.5	8.4	100.0

Table A9: Number and Volume of Cash Transactions by Data Source and Country. All figures refer to transactions in ‘plain vanilla’ European government bonds, see Appendix A.4 for details. Panel (a) refers to the total number of transactions on all six days in our sample (the two Eurex outage days and the respective control days), whereas panel (b) refers to the average across the four control days.

MiFIR data cleaning procedure

From the raw MiFIR dataset, we first delete transactions i) with missing or implausible prices or quantities, ii) that occurred outside of trading hours (prior to 8 a.m. or after 7 p.m.), and iii) that occurred more than two days prior to the bond’s issuance (so-called ‘when-issued’ transactions). We round prices to one decimal because we observed that multiple reports of the same transaction sometimes deviate slightly in prices. Since timestamps are vital for our deduplication procedure and the linkage of plausible trade chains, we correct those where a misreporting is highly likely. E.g. differences of exactly one or two hours for trades with identical bond, price, and quantity likely reflect misreported time zones. We convert everything to CET. Timestamp deviations by up to two seconds are also accounted for when everything else points towards a trade chain.

Duplicates in bond, timestamp, price, and quantity are simply deleted for the macro-level analyses in Section 3. However, for our more granular analyses on sectoral or venue-level trading behavior in Section 4, we require a more thorough approach. For a probable trade chain with multiple different reported trade venues, we identify the ultimate trading venue by placing actual venues over systemic internalisers and OTC flags to identify the ultimate trading venue. Our final goal as regards trades’ counterparties is to identify the ultimate buyers and sellers, i.e. those that

ultimately exchange ownership in the bond. First we delete duplicates in timestamp, bond, buyer ID, seller ID, price and quantity but different submitting IDs. Trade chains are then identified by timestamp, bond, price and quantity. In the large majority of cases, netting over all involved buyers and sellers results in only two remaining counterparties of which one is a buyer and the other a seller with identical quantities. Only if this is not the case, we deploy a carefully designed algorithm. To give a basic example, when one buyer but two sellers remain after netting and both of these sellers show identical quantities, i.e. this is not a cumulative trade, we place the selling counterparty less likely to be an intermediary first (e.g. an investment fund) and drop the other seller (e.g. a dealer bank). A set of these measures results in the identification of two (or in the case of cumulative trades three or more) ultimate counterparties for the bulk of identified trade chains. For the remainder, we drop the trade side that remains ambiguous. We also drop intragroup trades, i.e. trades between different LEIs belonging to the same investor, see Appendix A.6

Non-representative sample

One issue with our transaction dataset are the relative shares of the trading volume we capture across countries, reported in the last row in Table A9. German bonds account for 51% of volume in our dataset compared to a ‘true’ cash market share of only 34% in 2022 based on Table A3. The analogous figures are 13% vs. 25% for France, 29% vs. 33% for Italy, and 8% vs. 9% for Spain. That means even after augmenting the regulatory data with data from trading platforms, trades in German bonds are still overrepresented.

These data issues are not too concerning for our purposes, however, since we focus on the differential effect of outages. In particular, we compare outage days with ‘control’ days, usually the same day one week before and after the outage, and we compare the intraday periods just prior and just after the outage. In this setting, data issues such as duplicates and non-representative market shares should have little impact on the estimated treatment effect.

Additional Filters for Yield Curve Fitting

In Section 3.3, where we compute yield curve fitting errors for German government bonds, we apply some additional filters to the ones mentioned above. In particular, we exclude intragroup trades (see next section) and drop trades in bonds with less than one or more than ten years to maturity or that have implausible prices. For the latter filter, we convert each transaction price to an implied yield³⁴ and compute the difference to the daily maturity-matched yield from the Bundesbank’s term structure model (see Bundesbank website). We drop a few transactions where the absolute yield difference is larger than 75 basis points, or larger than 25 basis points and the transaction price is exactly 100. These transaction prices most likely reflect reporting errors. These filters leave us with over 3,000 transactions during the six selected intraday windows. Table A10 provides some descriptive statistics on these transactions.

³⁴We use Matlab’s built-in *bndyield* function, taking into account the price of the bond, its maturity and issuance date, the coupon rate, coupon frequency, first and last coupon date, and assuming $T+2$ settlement.

	min	p50	mean	max
No Outage				
Volume (million Euro)	0.00	1.22	9.96	1292.00
Years to Maturity	1.00	5.36	5.77	9.86
Years since Issuance	0.11	2.27	3.27	26.47
Coupon Rate	0.00	0.00	0.53	6.50
Yield	-0.73	-0.62	-0.59	-0.31
Yield Curve Fitting Error	-12.23	-0.23	-0.18	27.94
Outage				
Volume (million Euro)	0.00	0.90	2.37	200.00
Years to Maturity	1.01	5.84	6.11	9.84
Years since Issuance	0.09	2.25	3.59	26.28
Coupon Rate	0.00	0.25	0.62	6.50
Yield	-0.92	-0.57	-0.56	-0.18
Yield Curve Fitting Error	-23.80	-0.12	-0.20	50.53

Table A10: Descriptive Transaction Statistics for Yield Curve Fitting Sample. These statistics refer to the over 3,000 transactions in German sovereign bonds with 1-10 year maturity shown in [Figure 3](#) and underlying the regression results in [Table 3](#).

A.6 Investor-Level Info

A key feature of our regulatory transaction data is that it contains a unique legal identifier (LEI) for at least one of the two counterparties involved in each cash transaction.³⁵ We can thus use Eurosystem supervisory lists and the Register of Institutions and Affiliates Database (RIAD) database, see [Perrella and Catz, 2020](#)) to group LEIs into unique investors.³⁶ Then, we classify investors into the following categories: banks, non-bank financial institutions (NBFI), investment funds, hedge funds, insurance companies and pensions funds (ICPF), non-financial corporations (NFC), an official sector (including central banks) and households (retail investors).³⁷

[Table A11](#) gives an overview of the number of investors of each type and [Table A12](#) gives an overview the number of transactions across investor segments and types.

	Eurex-Active	Eurex-Inactive	Total
Bank Dealer	30	2	32
Bank Non-Dealer	13	68	81
NBFI	6	39	45
Investment Fund	5	220	225
Hedge Fund	6	61	67
ICPF	1	55	56
NFC	0	20	20
Official	0	16	16
Household	0	1	1
Total	61	482	543

Table A11: Number of Investors by Type. Data refers to investors with Bund trades on the cash market on the six selected days in 2020. Investors are aggregated to group-level. The category ‘household’ comprises an unknown number of retail investors, as these are not uniquely identifiable via a LEI.

For banks, it is important to differentiate between (non)dealers. We identify dealers using the list of primary dealers participating in Bund auctions, see [Finanzagentur website](#). In total, our sample covers cash market transactions by 32 different dealer banks. With this classification at hand, we categorize each transaction into one of three investor segments: client-to-client (C2C), dealer-to-client (D2C), and dealer-to-dealer (D2D).

We also separate investors into those that are usually active on the bond futures market and those that are not, see last column of [Table A11](#). In particular, we exploit the European Market Infrastructure Regulation (EMIR) dataset, see [Section 2.2](#). We deem any investor with more than 20 trades in any German bond future on the six selected days in 2020 (outage and control days) as ‘active on Eurex’. This way we can differentiate between cash market trades in which both, only one, or none of the two counterparties are usually active on Eurex.

For our trade report data, the investor classification is less straightforward, because this data is anonymous. However, the data source itself is informative about the venue type and investor

³⁵This is in contrast to the Canadian transaction data used e.g. by [Allen and Wittwer \(2023a\)](#), which contains a LEI only for dealers, brokers, and the largest investors, whereas other investors remain anonymous. The US transaction data used e.g. by [Duffie et al. \(2023\)](#) only covers trades by registered broker-dealers, identified via Market Participant Identifier (MPID) codes. The UK transaction data used e.g. by [Czech, Huang, Lou, and Wang \(2021\)](#), lastly, contains a Business Identifier Code (BIC, issued by SWIFT) or a Firm Reference Number (FRN, issued by the UK’s Financial Conduct Authority).

³⁶The 32 dealers in our sample, e.g., comprise over 100 different LEIs (similar to [Duffie et al., 2023](#)).

³⁷We start classifying investors using the Eurosystem lists of financial institutions (covering banks, investment funds and ICPFs, see [ECB website](#)) and use RIAD for the remaining investors. In some cases, we manually adjust the investor classification. Some of the largest global banks trade in EGBs via European subsidiaries, for instance, and RIAD identifies these as NBFIs rather than banks, probably because they are no depository institutions in the euro area.

(a) Investor Segments			
	No Outage	Outage	Total
C2C	879	261	1140
D2C	3195	640	3835
D2D	978	164	1142
Total	5052	1065	6117

(b) Investor Types			
	No Outage	Outage	Total
Bank Dealer	2578	484	3062
Bank Non-Dealer	590	180	770
NBFI	474	117	591
Investment Fund	489	99	588
Hedge Fund	256	64	320
ICPF	175	19	194
NFC	30	12	42
Official	323	25	348
Household	137	65	202
Total	5052	1065	6117

Table A12: Number of Cash Transactions by Investor Segment and Type. All figures refer to transactions in German sovereign bonds with 1-10 year maturity during the two Eurex outages in 2020 and the four non-outage control windows. With two sides to each trade, the totals sum to roughly twice the number of observations shown in [Figure 3](#) and underlying the regression results in [Table 3](#) (abstracting from unknown investor types due to anonymous transaction data).

segment and we use this information to the extent possible, see [Table A13](#). For instance, we attribute transactions on MTS to the dealer-to-dealer segment, based on the fact that MTS is restricted to dealers. The classification is equally uncontroversial for data we obtained from interdealer brokers. The MTS Bondvision and the Tradeweb platform, in contrast, are open to both dealers and clients, and hence we leave the investor segment of these trades unclassified. However, besides transactions executed directly on its platforms, Tradeweb also publishes data on OTC trades (for which Tradeweb acted as an APA). Within these trades, systematic internaliser trades are explicitly flagged and we classify the remaining trades as bilateral OTC trades.

Data Source	Venue Type	Venue Subtype	Segment
MTS	On Exchange	MTS	D2D
TPICAP, BGC, GFI, Aurel	OTC	OTC via IDB	D2D
MTS Bondvision	Electronic Platforms		-
Tradeweb MTF*	Electronic Platforms		-
Tradeweb SI [†]	OTC	OTC via SI	-
Tradeweb other [†]	OTC	Bilateral OTC	-

Table A13: Venue and Investor Segment Classification for Anonymous Transaction Data. This table shows how we classify anonymous transactions, sourced directly from trading venues, into venues and investor segments. Empty cells mean we leave the segment unidentified. *Due to Brexit, Tradeweb runs two separate multilateral trading facilities (MTF) in the UK and the EU (MIC Codes ‘TREU’ for London and ‘TWEM’ for Amsterdam). [†]Tradeweb also offers an APA service, reporting trades from the OTC market, explicitly flagging trades by systematic internalisers (SI).

Appendix B Overview of Market Outages

This section provides an overview of the outages we study. Appendix B.1 lists all outages and affected assets, Appendix B.2 recounts newspaper reports on all Eurex outages and Appendix B.3 on all cash venue outages. Many articles are from the [Factiva news archive](#), which is how we identified some of the outages in the first place.

B.1 Outage List

Table A14 contains a list of the market outages we study. Figure A3 shows the exact intraday times of each system-wide Eurex outage. We focus on outages since 2008 because we do not have cash market data prior to that. Figure A4 confirms the two partial outages on Eurex studied in Appendix C.2. The figure shows that only 5-year and 10-year German bond futures were unaffected by the outages on 26 May 2014 and 21 November 2016. On the first of these days, there were two separate outages. Figure A5 confirms the known outage of the MTS platform on 12 January 2010 and shows another suspected outage on 26 July 2019.

Exchange/Platform	Affected fixed-income assets	Dates
Eurex	European Government Bond (EGB) futures	1 July 2020
		14 April 2020
		16 March 2018
		13 December 2017
		22 February 2016
		20 July 2015
		17 February 2015
		26 August 2013
		11 October 2011
		23 December 2009
		18 November 2009
		4 February 2009
	all EGB futures except 5/10-year German Bund futures	21 November 2016 26 May 2014
MTS (unconfirmed)	cash bonds	26 July 2019
Brokertec		11 January 2019
Frankfurt stock exchange (FWB)		27 May 2015
Bloomberg		17 April 2015
MTS		12 January 2010

Table A14: Overview of Market Outages Studied in This Paper.

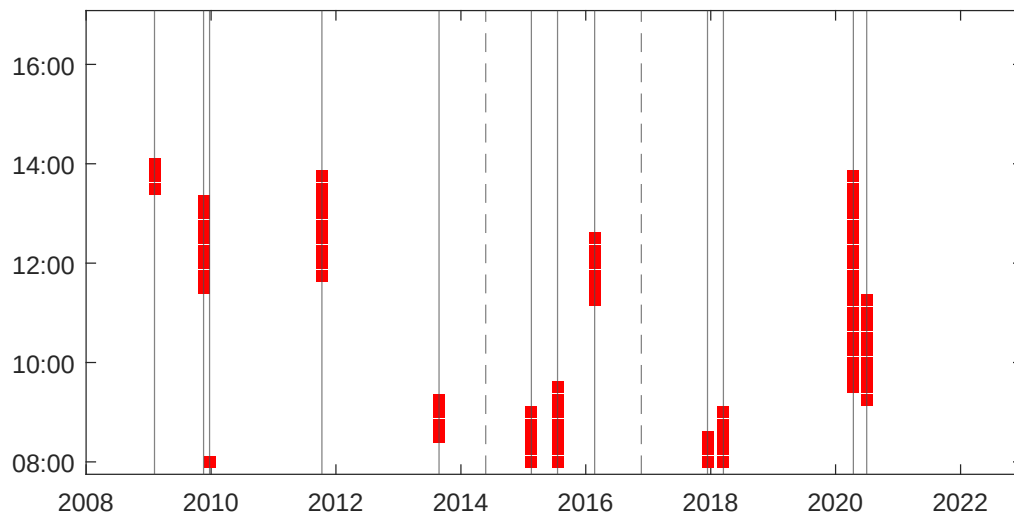


Figure A3: System-Wide Eurex Outages since 2008. For each trading day since 2008, red squares mark 15-minute intraday windows with no trading in the German 10-year bond future (FGBL), which line up perfectly with the twelve known outage days, marked by solid vertical lines. The two dashed lines mark days with partial outages, see [Figure A4](#). The vertical axis ranges from 7:45 a.m. to 5:00 p.m. local time.

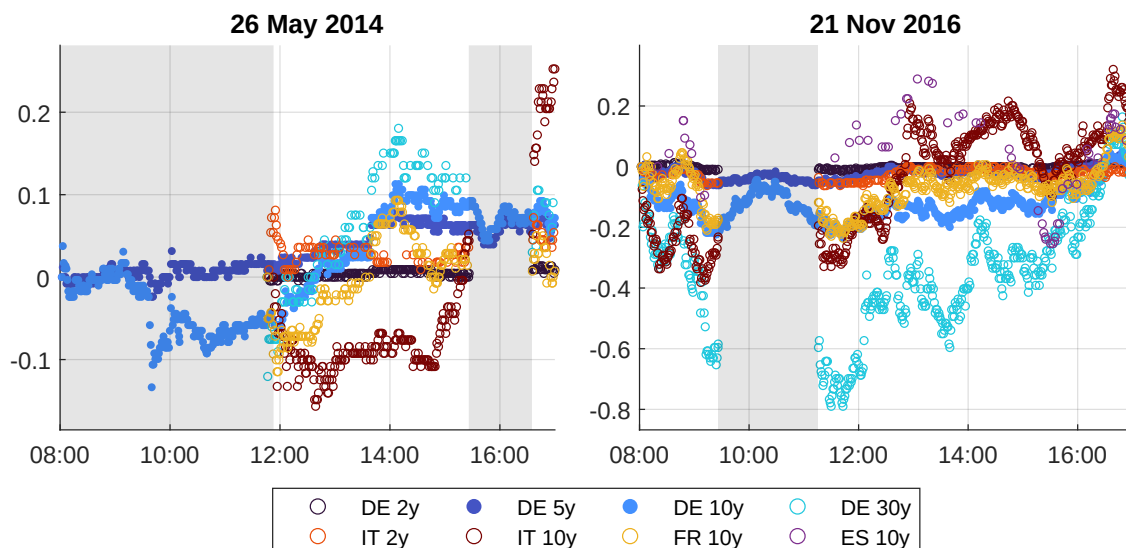


Figure A4: Partial Outages on Eurex. This figure shows intraday transaction prices of all euro area government bond futures (in percentage changes since 8:00 a.m., at minutely frequency). The grey areas mark the outage times that affected all futures except those for 5-year and 10-year German bonds. The Spanish 10-year future was introduced in October 2015, i.e. after the 2014 outage.

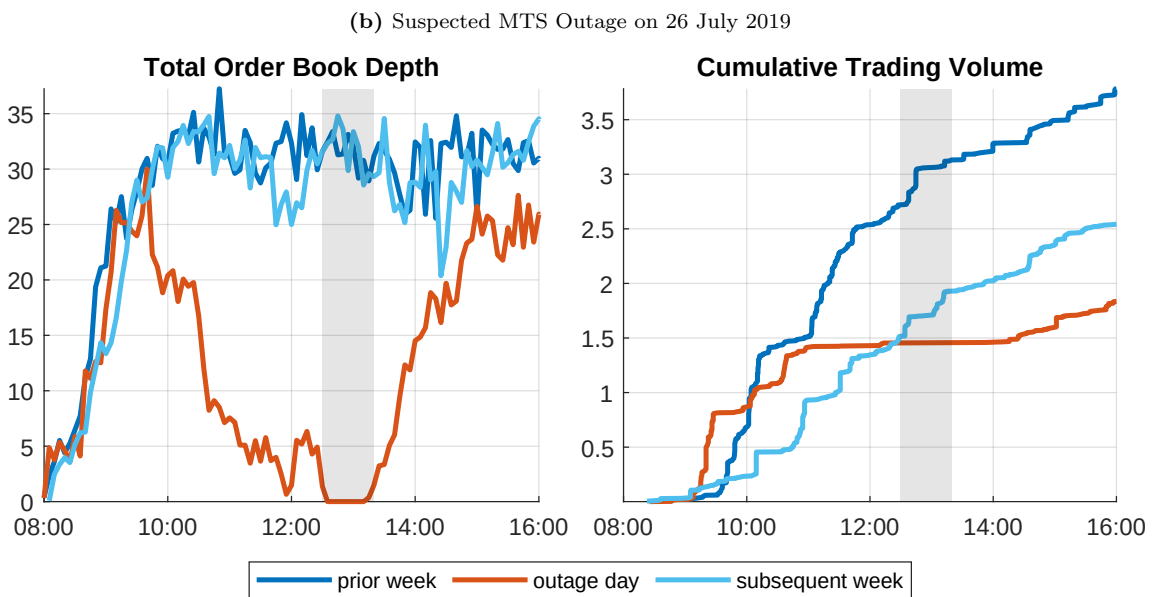
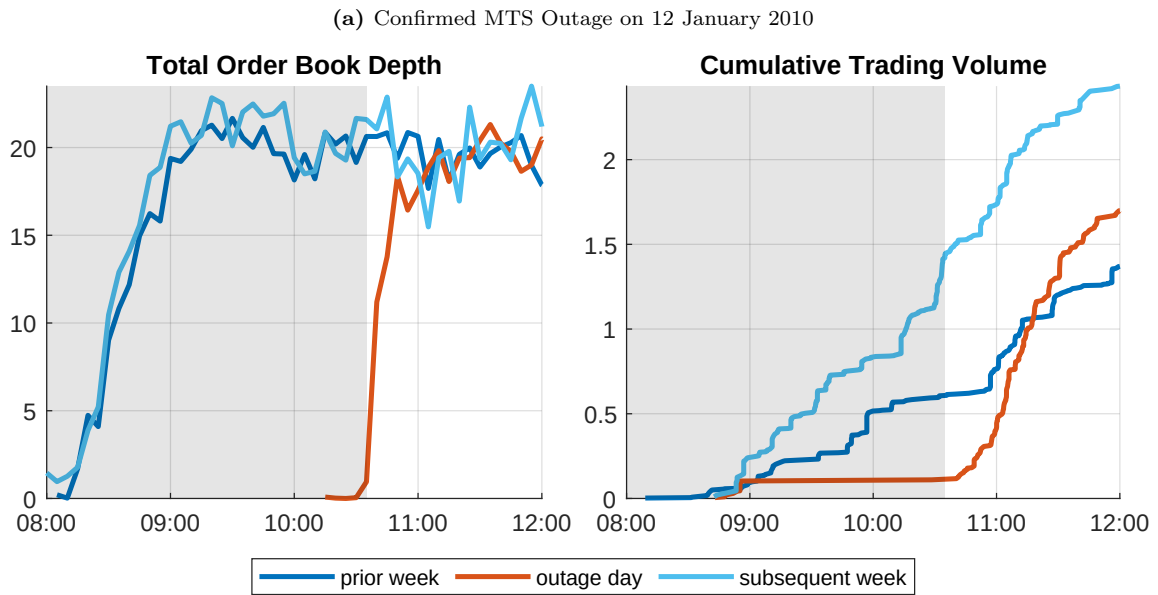


Figure A5: Outages on MTS. The left panels show the total quoted volume (in billion €), across all German, French, Italian and Spanish sovereign bonds and all market segments, in 5-minute intervals. The right panels show the total cumulative trading volume (in billion €). Red lines refer to the (suspected) outage day, dark and light blue lines refer to the previous and subsequent week. Grey areas mark the (suspected) outage times.

B.2 News Reports on Eurex Outages

Table A15 provides a timeline of the two Eurex outages in 2020 and related news.

Table A15: Timeline of Eurex outages in 2020 and same-day news.

14 April 2020	
9:25	outage start
10:32 @Eurex	Due to technical problems, the Eurex T7 system is not available at the moment. We are investigating and will keep you informed.
11:47 DowJones	Trading on Deutsche Boerse AG's Xetra is currently suspended due to a technical fault, a spokesman for the German stock exchange operator said on Tuesday. The spokesman said to Dow Jones Newswires that he couldn't yet comment on when the trading would resume and communication related to trading resumption would be released on the Xetra Newsboard.
13:19 @Eurex	Trading will resume according to the following schedule: 13:15 CEST Pre-Trading Instrument State BOOK, 13:45 CEST Trading Instrument State OPENING AUCTION, 13:50 CEST Trading Instrument State CONTINUOUS.
14:00	outage end
17:51 Reuters	The outage was caused by a malfunction in the internal communication of the trading system, Deutsche Boerse said in a statement, adding that trading was operating smoothly again on Tuesday afternoon. A Deutsche Boerse spokesman said the outage was not due to a hacker attack.
1 July 2020	
8:49	outage start
9:39 Reuters	Frankfurt-based electronic trading system Xetra was experiencing a 'technical issue', affecting all securities traded on the platform, a Deutsche Boerse spokesman said on Wednesday. 'I just can confirm that there is a technical issue on Xetra... we're currently investigating the failure,' Patrick Kalbhenn, a spokesman for the German stock exchange, told Reuters.
10:11 @Eurex	Due to technical problems the trading system T7 is not currently available.
11:54 @Eurex	Update on the disruption: continuous trading on Eurex resumed at 1130.
11:31	outage end
17:25 Bloomberg	The disruption in the T7 system in April and today's failure had the same origin. They were due to faulty third-party software that is part of the trading system. We now understand the exact cause and have eliminated the issue. The system is now running stably and we expect it to remain so. External causes can be ruled out.

An [ESMA report](#) contains the following paragraphs about the Eurex outages in 2020: *'The first two incidents reported related to an issue with the Deutsche Börse T7 trading system. The first was reported on 14 April and trading was interrupted in the trading venue due to a software issue. This issue required the trading venue to stop and restart manually the system which was a heavy and time-consuming process. The second incident was reported on 1 July and due to a human error. Two failures of the trading venues' central network occurred which caused trading to be halted in Deutsche Börse. In both circumstances the incident affected a significant number of trading venues given that the T7 trading system is widely used across the EU.'*

What follows are newspaper reports for all previous outages on Eurex, in chronological order:

[6 February 2004](#): Computer glitch caused Xetra failure

Deutsche Boerse AG said on Friday a computer failure led to a nearly hour-long outage on its Xetra electronic trading platform. The outage was the fourth on Xetra, which accounts for 90 percent of trade in German stocks, since the electronic trading platform was introduced six years ago. Pre-trading resumed at 1045 GMT after a 45-minute trading halt while full trading resumed at 1130 GMT, Deutsche Boerse said in a statement. The stock market operator said trade in some stocks had been delayed even longer with dealing halted from 0900 GMT, caused by a failure in a host database. Traders said the Xetra failure caused some concern. 'It was a little bit hectic when it first began but luckily there was not that much important company news during the trading halt,' said one trader.

19 November 2007: Rare glitch hits Frankfurt stock exchange

Trading in German stock exchange operator Deutsche Boerse's electronic order-matching system Xetra was interrupted early on Monday by technical problems, which traders and the bourse said were rare. The impact of the disruption of just over one hour was minor because early volume had been 'very low,' said one Frankfurt-based stock broker, who could not recall more than two or three short Xetra disruptions in the past two years. 'Normally they (Deutsche Boerse) say the system is running nearly 100 percent,' he said, adding from his own experience. 'If it's down two times in one year, it's a lot.' Xetra trading was halted around half an hour after the start at 0800 GMT and resumed at 0940 GMT. During the interruption, Deutsche Boerse said it was investigating 'technical problems' in the system, which handled 16 million transactions in October, up 74 percent year-on-year. A Deutsche Boerse spokesman said after the restart of trading that he did not yet have information about the exact nature of the problem.

4 February 2009: Derivatives exchange Eurex resumes trading

Derivatives exchange Eurex resumed trading on Wednesday after a one-hour shutdown that traders took with a shrug. Eurex said a technical glitch halted trading between 1218 GMT and 1315 GMT on what is the main market for Europe's most traded fixed income futures and stock index options and futures. Traders said such an event was rare. A Eurex spokesman said the last time something similar had occurred was in 2006. '(For us) there was no effect, no losses as a result of this Eurex downtime,' said Sebastian Qureshi, head of German hedge fund manager Varengold, which specialises in managed futures strategies and trades also on Eurex. 'One hour is too short, it's not really affecting our business,' he said. 'We just went out to get a coffee,' a Frankfurt-based Eurex market-maker added.

18 November 2009: Eurex trade suspended until further notice

Trade on the Eurex derivatives exchange has been suspended since 1100 GMT due to technical problems, a Eurex spokesman said on Wednesday, adding it was not foreseeable when trade would resume. 'We are working hard to solve the problem,' the spokesman said. Eurex is jointly operated by Deutsche Boerse and SIX Swiss Exchange.

23 December 2009: Eurex Exchange says Wednesday opening delayed.

11 October 2011: Eurex trade suspended until further notice

Deutsche Boerse AG interrupted trading on its electronic derivatives platform Eurex on Tuesday to investigate technical problems. Trading on Eurex had been interrupted 'to avoid any threat to the functioning of Exchange trading', the stock exchange operator said in a statement. Earlier, Deutsche Boerse said it had been experiencing technical problems. 'We are investigating and will keep you informed,' Deutsche Boerse said in an e-mailed statement.

26 August 2013: Eurex Restarts Trading After Market Halt

German exchange operator Deutsche Boerse AG's (DB1.XE, DBOEF) Eurex Exchange arm re-

opened trading early Monday after a brief halt to certain trading earlier in the session. At roughly 0642 GMT, Eurex posted a note on its website saying that it had halted trading on its New Trading Architecture platform—a new electronic trading platform that was introduced by the firm in December—in order to avoid a threat to the functioning of exchange trading. The halted trading announcement came roughly 12 minutes after an earlier posting said there were technical problems at the exchange. Trading in all Eurex products was restarted at roughly 0720 GMT, according to the company’s website.

26 May 2014 (partial outage): Btp and Oat futures, delayed opening due to technical problems

Italian, French bond futures trading delayed, Eurex recorded message citing technical problems at exchange

17 February 2015: Eurex Restarts Trading After Market Halt

There is no trading currently on any Eurex futures or option products due to technical issues, according to one London-based trader. According to one London-based trader, the earliest start of opening Auctions on Eurex will be 0915 CET or 0815 GMT, while netting will start as of 0920 CET or 0820 GMT. Adds these times ‘not set in stone though.’ Eurex spokesman not immediately available to comment.

20 July 2015: Eurex Restarts Trading After Market Halt

Europe’s largest derivatives market, Eurex, suffered technical issues on Monday that delayed trading of all futures and options contracts and took two hours to fully resolve. The outage caused little disruption to broader cash equity and bond markets, however, unlike April’s Bloomberg terminal outage, which delayed debt sales and exacerbated a spike in volatility. Traders said the outage had effectively choked off liquidity in the derivatives market, with only over-the-counter deals available. But receding fears over Greece and recent declines in volatility meant it had less impact than it might have. ‘This type of outage is usually significant, but because of the broader environment things were much calmer,’ said a London-based equity derivatives trader. ‘It was a minor event in the end, but it could have been a major one had it hit a few weeks back.’ Frankfurt-based Eurex said complete trading had resumed at 0810 GMT. Index futures trading usually begins around 0600 GMT. A bond trader based in Frankfurt said the mood was ‘quite relaxed’. Another London-based trader said there had not been a big impact.

22 February 2016: Eurex says continuous trading to resume at 1150 GMT

Deutsche Boerse’s Eurex says pre-trading to start at 1125 GMT, continuous trading from 1150 GMT.

21 November 2016 (partial outage):

«For this outage, the only real-time confirmation we could find is on [Twitter](#).»

16 March 2018: Eurex hit by technical issues, bond and stock futures trading delayed

Many key European bond and stocks futures, including German Bund futures and DAX futures, did not open for trading on Friday as the Eurex trading system was hit by technical issues. German Bund futures, which allow investors to hedge against German government bonds, Italian BTP futures and French OAT futures were all down. Many stock futures were also down, including were Eurostoxx futures and Dax futures ‘There are some technical problems for the T7 system which has caused some delays. It’s under investigation currently and we will have updates on our production newsboard. As of now I have no further details on when it will be resolved,’ said a Eurex representative.

As a result, trading in government bonds is extremely thin, most likely because investors are

unable to hedge their investments, DZ Bank strategist Daniel Lenz said.

B.3 News Reports on Other Outages

Bloomberg Outage on 17 April 2015

[Yahoo](#): In the City of London [...] and financial centres around the globe traders and analysts were left gnashing their teeth as their Bloomberg computer terminals crashed on Friday morning, with fixed income markets suffering an especially large impact. Bloomberg [...] confirmed that its terminal system was unavailable worldwide, with the outage first cited at around 08:20 in London. [...] by 09:10 London time the company said that some customers had reported the terminal was back online.

[The Globe and Mail](#): Bloomberg's trading terminals, which are used by most of the world's biggest financial firms, went down for two-and-a-half hours on Friday due to apparent technical problems.

[UK Debt Management Office](#): Due to ongoing technical issues with the third party platform supplier [...] this morning's UK Treasury Bill tenders will be postponed.³⁸

Brokertec Outage on 11 January 2019

[Reuters](#): BrokerTec, an interdealer broker of U.S. government debt, planned to reopen its U.S. market platform at 3:35 p.m. (2035 GMT) following an outage earlier Friday, a company spokeswoman said. 'All trading participants should be able to reconnect to the system,' the spokeswoman said in an email statement.

[Bloomberg](#): Treasuries Hit by One-Hour Outage on Biggest Electronic Platform. BrokerTec, the biggest electronic trading platform for Treasury securities, shut down for more than an hour on Friday because of a technical malfunction, an outage that several traders said caused a market-wide slowdown in one of the world's biggest assets.

[Fed New York Treasury Market Practices Group meeting minutes](#): Members [...] noted that the timing and nature of this particular outage had a relatively limited impact given that [...] the outage occurred during a period of low trading activity.

Frankfurt stock exchange (FWB) Outage on 27 May 2015

[Reuters](#): Deutsche Boerse said floor trading on the Frankfurt Stock Exchange on Wednesday would not resume before 11 Central European Time, or 0900 GMT. Trading on the Xetra electronic exchange remained intact.

[Die Welt](#): On Wednesday a technical error brought the floor trading at the Frankfurt Stock Exchange to a halt all morning. Only around 1300 CET the trading began running smoothly again. Frankfurt Stock Exchange's electronic securities trading system Xetra, on which more than 90% of trade is handled, was not affected by the breakdown. Overnight there were problems in the data processing, a spokesman for stock exchange operator Deutsche Boerse said. Such a long breakdown is the absolute exception, according to the spokesman.

³⁸The UK Debt Management Office runs its auctions on the Bloomberg Auction System (see [Guide to DMO Primary Dealers](#)). In this sense, the fact that the auction was postponed is no prima facie that UK officials necessarily feared an impaired price discovery process (as we argue for the cancelled auction of Dutch bonds during the Eurex outage on 14 April 2020).

MTS Outage on 12 January 2010

Dow Jones: 'MTS Group can confirm that the cash market facility experienced an outage at the beginning of the trading day,' the company said in a statement. 'The issue was resolved and normal market conditions resumed. The outage caused no unverified transactions.' The problem resulted in users of MTS' system being unable to log in. It took place roughly between 0830 GMT and 0930 GMT after which the system was up a running again, Patrick Humphris, a spokesman for LSE, said.

Appendix C Previous Eurex Outages as Robustness Check

Section 3 in the main text suggests that market functioning for euro area sovereign bonds depends heavily on the futures market, but that evidence is based on just two Eurex outages in 2020. We put particular emphasis on these two most recent outages because we have the best data for this period, in particular regarding EGB transactions on the cash market, see Appendix A.5. The two Eurex outages in 2020 have not been without precedent, however. Hence, this section studies twelve previous outages on Eurex between 2009 and 2018 as a robustness check.

Section C.1 confirms that these previous outages also cause trading volumes on the cash market to decline and liquidity on MTS to evaporate.

Section C.2 exploits the fact that two outages did not affect the entire Eurex exchange. Instead, trading continued in 5-year and 10-year German bond futures. We find that these partial outages reduce liquidity on the cash market roughly half as much as system-wide outages, in line with the benchmark status of German bond futures.

Section C.3, lastly, studies a particular trading day (25 May 2015) on which – as an exception – Eurex was closed while MTS remained open.

C.1 System-wide Outages

The two Eurex outages in 2020 have not been without precedent. At least ten other times since 2008, the Eurex platform already experienced similar outages, see Appendix B.1.³⁹

Did these previous outages also cause trading on the cash market to decline? To find out, we run essentially the same regression as in Equation 1 for this older set of outages. The main difference is that we have to restrict this analysis to Germany, since the regulatory transaction-level dataset we use (the ‘Bafin’ dataset mentioned in Section 2.1) mainly captures trades in German bonds and we were not able to obtain intraday transaction data directly from trading platforms like we did for the 2020 outages. Thus, we regress the total trading volume in German bonds in a given maturity bucket and 30-minute time interval onto an outage dummy and fixed effects. The sample covers nine outage days plus the same day in the preceding and subsequent week, i.e. 27 days in total. Each day, we have 15 intraday observations (from 08:00 a.m. to 3:00 p.m.). Table A16 reports the results. In line with our previous finding, trading volumes on the cash market drop significantly when the futures market suffers an outage and compared to short-term bonds, longer-term bonds are particularly affected (though there is no differential effect for bonds with more than 10.5 years to maturity).

How about MTS? Did Eurex outages always lead to an evaporation of liquidity on MTS, as we have shown for the two 2020 outages in Section 3.2? Yes, as shown in Table A17. We run the same regression as in Equation 2 for the older outages and get basically the same results. For most countries and maturity buckets, the entire liquidity on MTS evaporates, see column (1), and this dry-up is particularly pronounced for bonds with longer maturity, see column (2). The only major difference is in the country-level results, see column (3). In particular, the nine Eurex outages between 2008-2018 had equally dramatic effects on German, French and Spanish bonds. Only the liquidity of Italian bonds was more robust. During the 2020 outages, also the liquidity in French and particularly Spanish bonds was more robust than in German bonds, cf. Table 2.

³⁹In the following analysis, we omit the outage on 23 December 2009, when the start of futures trading was delayed from 8:00 a.m. to 8:20 a.m., because this outage is too short and too early in the day to observe any effect on the cash market.

	(1) Aggregate	(2) Maturities
Outage	-1.63*** [0.46]	-1.14 [1.11]
Outage $\times$ 2.5-5.5y		-1.57* [0.77]
Outage $\times$ 5.5-10.5y		-2.49** [1.21]
Outage $\times$ >10.5y		2.10 [2.06]
FE Day	✓	✓
FE Time	✓	✓
FE Maturity Bucket	✓	
Observations	1620	1620
Adjusted R^2	0.500	0.502

Table A16: Effect of Previous Eurex Outages on Cash Trading Volumes. Each column shows results of a different regression, as in Equation 1. For brevity, the table shows estimates only for the outage dummy and interaction terms. Throughout, the dependent variable is the log of the transaction volume of German bonds of a given maturity bucket in 30-minute intervals. In column (2), bonds with less than 2.5 years to maturity serve as the baseline.

	(1) Aggregate	(2) Maturities	(3) Countries
Outage	-10.73*** [0.93]	-4.08*** [1.11]	-13.23*** [0.61]
Outage $\times$ 2.5-5.5y		-8.33*** [1.09]	
Outage $\times$ 5.5-10.5y		-8.90*** [1.28]	
Outage $\times$ >10.5y		-9.36*** [1.30]	
Outage $\times$ ES			2.07 [1.43]
Outage $\times$ FR			0.62 [0.47]
Outage $\times$ IT			7.29*** [1.01]
FE Day	✓	✓	✓
FE Time	✓	✓	✓
FE Country	✓	✓	
FE Maturity Bucket	✓		✓
Observations	47088	47088	47088
Adjusted R^2	0.601	0.631	0.617

Table A17: Effect of Previous Eurex Outages on MTS Order Book Depth. Each column shows results of a different regression, see Equation 2. Throughout, the dependent variable is the log of the quoted bid and ask volume of bonds of a given country and/or maturity bucket, at 5-minute snapshots. All explanatory variables are dummies, either for the maturity bucket (bonds with less than 2.5 years to maturity serve as the baseline) or for different countries (Germany serves as the baseline).

C.2 Partial Outages

Besides the system-wide outages discussed so far, there have been two outages on Eurex that affected all futures except those on 5-year and 10-year German bonds, see Appendix B.1. These events can shed additional light on the interaction between MTS and Eurex. Since two German bond futures were still available, we would expect that these partial Eurex outages i) lead to a smaller drop in the overall liquidity on MTS, ii) particularly for 5-10 year bonds and iii) particularly for German bonds.

To test these predictions, we repeat the dummy regressions from Equation 2 for this new type of outage. In particular, we regress the order book depth of all bonds of a given country and maturity bucket onto an outage dummy that equals one during the partial Eurex outages. We again include six days (the two outage days plus the preceding and subsequent week), this time using 109 intraday observations per day to cover all outage times (5-minute snapshots from 08:00 a.m. to 5:00 p.m.). All regressions control for day and time-of-day fixed effects. We can then compare the estimated effects with those from the system-wide Eurex outages studied in Table 2. Table A18 shows the results.

	Maturity Buckets		Countries	
	(1) System-Wide	(2) Partial	(3) System-Wide	(4) Partial
Outage	-4.95*** [0.21]	-2.89* [1.18]	-18.03*** [0.43]	-7.12*** [0.60]
Outage $\times$ 2.5-5.5y	-8.50*** [0.32]	-3.20 [1.88]		
Outage $\times$ 5.5-10.5y	-7.73*** [0.61]	-3.82* [1.52]		
Outage $\times$ >10.5y	-7.40*** [0.72]	-4.80*** [0.48]		
Outage $\times$ ES			8.57** [2.46]	2.53* [1.19]
Outage $\times$ FR			3.87*** [0.07]	-1.16 [0.75]
Outage $\times$ IT			16.24*** [0.91]	3.76*** [0.47]
FE Day	✓	✓	✓	✓
FE Time	✓	✓	✓	✓
FE Country	✓	✓		
FE Maturity Bucket			✓	✓
Observations	8736	10464	8736	10464
Adjusted R^2	0.558	0.614	0.644	0.616

Table A18: Comparison of the Effect of Partial and System-Wide Eurex Outages on MTS Order Book Depth. The ‘system-wide’ effects refer to columns (2) and (3) from Table 2 and are based on the two Eurex outages in 2020. The ‘partial’ outages estimate the same regressions for the two Eurex outages on 26 May 2014 and 21 November 2016, which affected all bond futures on Eurex except those for 5-year and 10-year German bonds. See Equation 2 and Table 2 for details.

Prediction i) is clearly confirmed by the data: partial outages reduce the liquidity on MTS only roughly half as much as system-wide outages. Prediction ii) on the maturity-specific effect is also confirmed. While the liquidity of 2.5-10.5 year bonds drops significantly more than shorter-term bonds during system-wide outages, this differential effect is barely significant during partial outages. Both of these results are in line with the benchmark status of German bond futures.

Prediction iii), that other countries should be more affected by the partial Eurex outages than Germany, however, is not true. In fact, we see that the liquidity drops similarly for French bonds, slightly *less* for Spanish bonds and much less for Italian bonds. Recall that this is despite the fact that two German bond futures were still available for trading on Eurex while all French, Italian and Spanish futures were unavailable.

C.3 Non-Trading Days

One last piece of evidence on the interaction between Eurex and MTS comes from different non-trading days between the two platforms. While MTS usually has the same holiday schedule as Eurex, 25 May 2015 was an exception. On that day, Eurex was closed while MTS remained open.⁴⁰ Figure A6 shows that MTS was basically inactive that day. There were very few quotes for any bonds and barely any transactions. Of course, this non-trading day does not constitute a true shock, since it was not exogenous and since it could be well anticipated by market participants. But in a sense, this makes our point stronger: even when market participants have enough time to prepare, they are not willing to trade or quote EGBs without an active futures market.

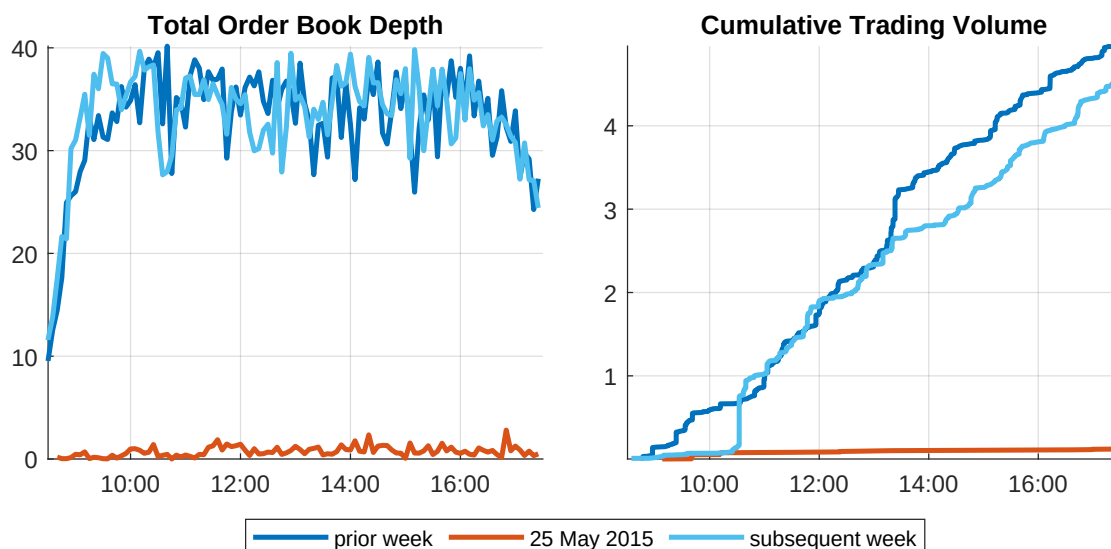


Figure A6: MTS Inactivity on Eurex Holiday. The left panel shows the total order book depth on MTS (in billion €), across all German, French, Italian and Spanish bonds and all market segments, in 5-minute snapshots. The right panel shows the total cumulative trading volume, in billion €. Red lines refer to 25 May 2015 (when Eurex was closed due to a holiday), dark and light blue lines refer to the previous and subsequent week.

⁴⁰Our results suggest an obvious explanation for why the two trading calendars coincide: since MTS is dependent on Eurex, it adopts their trading calendar. Regarding the 25 May 2015 exception, see [MTS trading calendar 2015](#) and [Eurex press release](#). The latter explains that ‘Eurex [...] decided, as an exception, not to offer any trading [...] on 25 May 2015. On that day there will be national public holidays in the U.S. (‘Memorial Day’), Great Britain (‘Late May bank holiday’), Germany, Austria and Switzerland (‘Whit Monday’), as well as in South Korea (‘Buddha’s Birthday’). Therefore, essential markets are not available.’

Appendix D Further Evidence on Eurex Outages

This section provides further evidence on the two Eurex outages in 2020 studied in [Section 3](#) of the main text.

Appendix [D.1](#) confirms that trading volumes on the EGB cash market catch up after the outages on the futures market are over.

Appendix [D.2](#) shows that the mispricing on the Bund market is restricted to the intraday periods of the two Eurex outages.

Appendix [D.3](#) provides further micro-level evidence on the trading behavior of different investor types during Eurex outages.

Appendix [D.4](#) shows that long-maturity interest rate swaps are severely affected by outages on Eurex. Short-term swaps, in contrast, continue to provide a reliable indicator of ‘fair’ short-term rates during the outages.

Appendix [D.5](#) documents that the liquidity dry-up on MTS was widespread and affected virtually all bonds.

Appendix [D.6](#) shows that the separately run local market segments on MTS were somewhat less affected by the outages on Eurex, potentially due to the ‘market making obligations’ that are enforced on these segments.

Appendix [D.7](#) provides robustness checks regarding the effect of Eurex outages on indicative EGB quotes on the cash market.

D.1 Trading Volumes

[Figure A7](#) shows cumulative trading volumes in 10-year government bond futures, comparing the two outage days with the same day in the previous and subsequent week. The figure confirms that during both outages, trading in bond futures came to a complete halt.

[Figure A8](#) reproduces the equivalent figure for the cash market, based on our extensive transaction-level dataset. Trading volumes drop while Eurex is offline and surge once it comes back online.

[Table A19](#) confirms the catch-up. Trading volumes are significantly higher roughly two hours after Eurex goes back online, suggesting that some investors are either forced or choose to delay their trades.

	(1) All Countries	(2) Germany
Outage Window	-1.46** [0.41]	-1.14* [0.51]
+ [0,30) min	-0.38 [0.37]	0.06 [0.74]
+ [30,60) min	0.30 [0.51]	0.83 [0.76]
+ [60,90) min	-0.00 [0.26]	0.73 [0.81]
+ [90,120) min	0.21 [0.37]	0.87** [0.24]
+ [120,150) min	0.81*** [0.20]	1.99** [0.54]
FE Day	✓	✓
FE Time	✓	✓
FE Country	✓	
Observations	360	90
Adjusted R^2	0.457	0.483

Table A19: Catch-up in Cash Trading Volumes after Eurex Outages. The dependent variable is the log of the transaction volume in German, French, Italian and Spanish bonds in 30-minute intervals, analogous to [Equation 1](#).

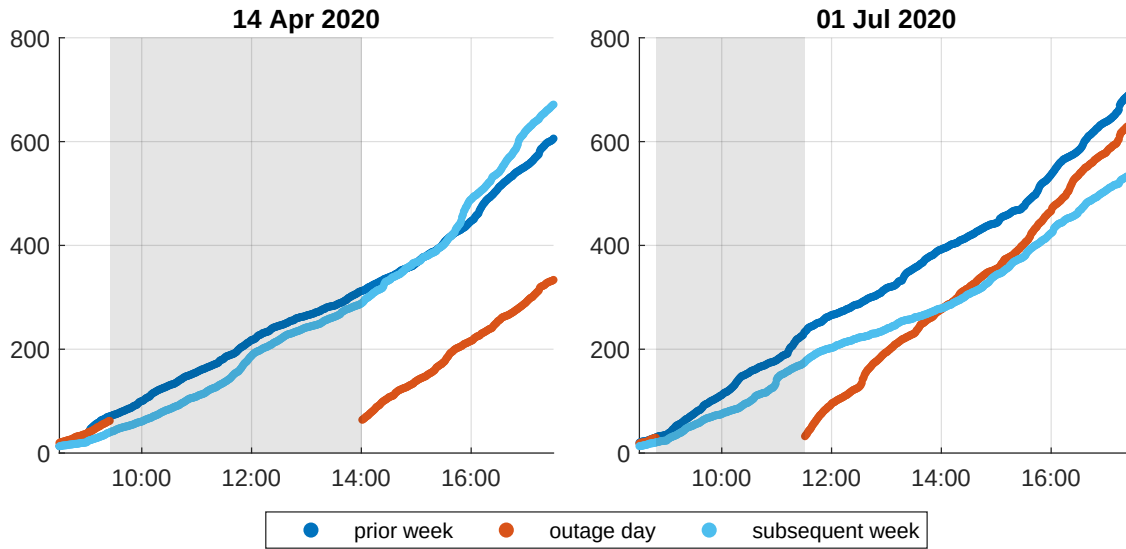


Figure A7: Trading Volume in 10-year Bond Futures. This figure shows the cumulative number of traded contracts (in thousands). Red dots refer to the outage day, dark and light blue dots refer to the same day in the previous and subsequent week.

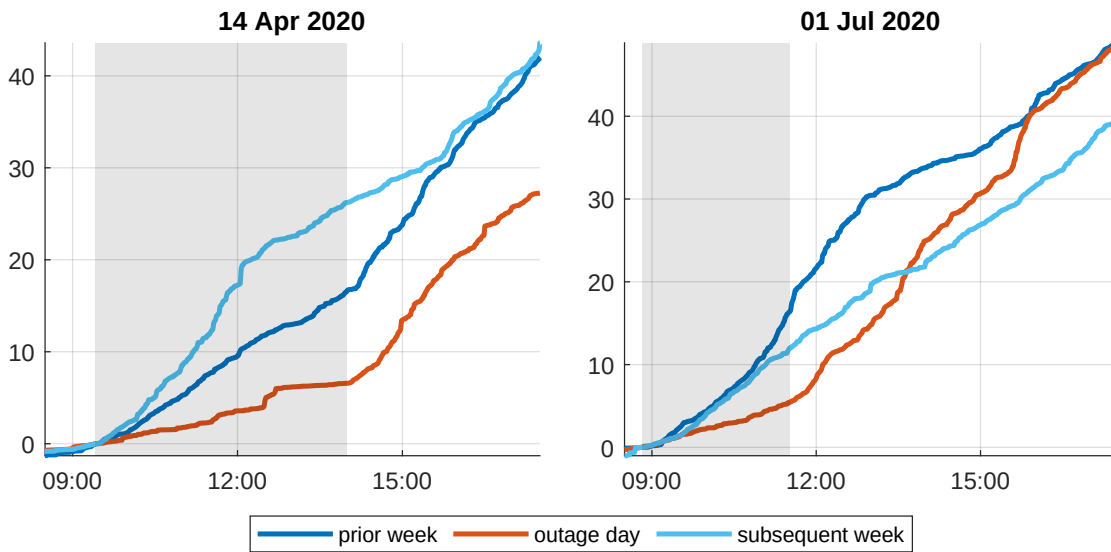


Figure A8: Trading Volume in EGBs on the Cash Market. This figure shows the cumulative trading volume in all German, French, Italian and Spanish sovereign bonds (in billions of Euro, normalized to zero at the intraday time of the outage). Red dots refer to outage days, dark and light blue dots to the previous and subsequent week.

D.2 Mispricing Before and After Eurex Outages

Figure A9 zooms in on the two outage days to rule out the possibility that pricing errors in German bonds are higher throughout the two outage days, for reasons other than the Eurex outages. We again fit yield curves and compute root mean squared pricing errors, but this time for hourly windows from one hour before till four hours after the outages. These hourly windows allow a sharp identification, while simultaneously ensuring that we have sufficiently many observations along the yield curve for the Svensson (1994) methodology to work. The figure shows that the huge spike in pricing errors is indeed restricted to the Eurex outage periods. The noise measure jumps up immediately during the first hour of the outage and then quickly recedes, reverting back

to normal once Eurex is back online.

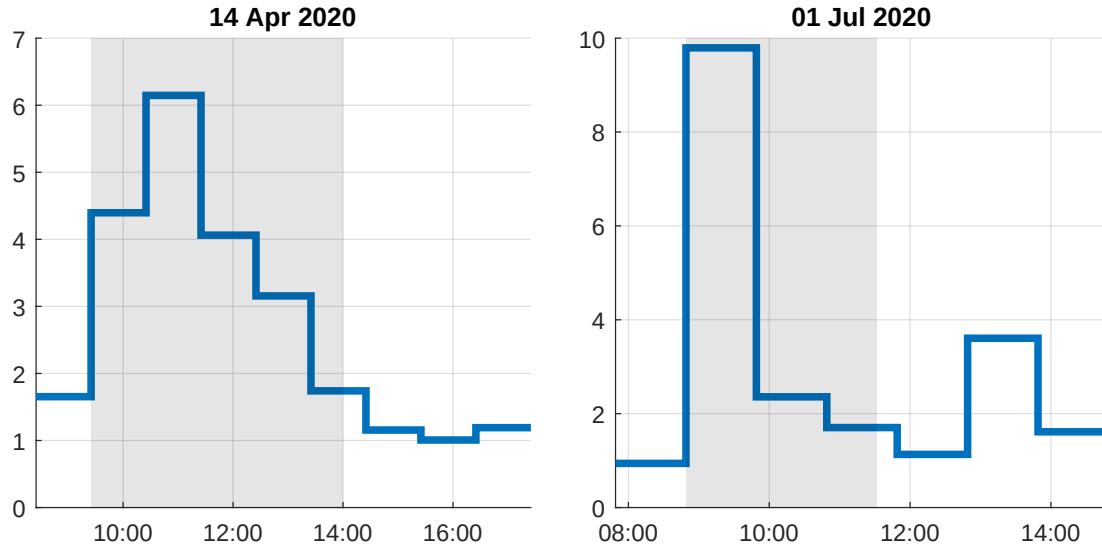


Figure A9: German Yield Curve Noise on Eurex Outage Days. This figure shows the noise measure proposed by [Hu et al. \(2013\)](#) on the two Eurex outage days, from one hour before the outage till four hours after the outage. For each one hour window, we fit a [Svensson \(1994\)](#) curve to the observed market yields and compute the root mean squared pricing errors. Grey areas mark the outage periods on Eurex.

D.3 Micro-Level Mechanisms

This section expands on the results presented in [Section 4.1-4.2](#) of the main text.

[Table A20](#) shows that the different hedging strategies across bond market dealers are barely correlated with other important characteristics. Neither the connectedness, the centrality, nor the size of the dealer reliably predicts how much of its cash market duration risk she offsets on the futures market. In a kitchen sink regression, the only statistically significant relationship is that more connected dealers use bond futures slightly more to hedge their inventory risk.

	(1)	(2)	(3)	(4)	(5)
# Connections	0.38 [1.02]				3.68** [1.59]
ln(Centrality)		-2.81 [2.16]			6.91 [17.77]
ln(Yearly # Trades)			-1.80 [2.38]		2.48 [5.29]
ln(Yearly Trading Volume)				-2.84 [2.07]	-17.50 [17.43]
Observations	32	32	32	32	32
Adjusted R^2	-0.029	0.022	-0.014	0.027	0.139

Table A20: Relation between Dealers' Hedging Intensity and other Characteristics. The dependent variable is the dealer-level β coefficient from [Equation 3](#), which measures how much of the cash market duration risk the dealer offsets via bond futures. See [Table 7](#) for details. # Connections refers to the number of different dealers each dealer has traded with and Centrality is the volume-weighted eigenvector centrality, based on all trades in 2020. As a proxy for size, we use the number of trades and total trading volume of each dealer in 2020. See [Section 4.2](#) for details.

Table A21 confirms that the results from Table 8 are virtually unchanged when we include dealer inventory as an additional explanatory variable. For each dealer, we compute the net trading volume in 1-10y German Bunds from the start of the prior trading day until the intraday time of the respective outage. Since the first outage on 14 April 2020 occurred on the Tuesday following the Easter holidays, we compute the net trading volume from the prior Thursday till 9:25 a.m. on Tuesday. For all other outage and control days, the previous day was a trading day. Since the inventory can be positive or negative, we use its inverse hyperbolic sine as a regressor. The results show that dealers with larger inventories cause have lower markups and trade less profitably.

	(1) mispricing	(2) markup	(3) profit
Outage $\times$ Dealer not hedging	0.36 [0.29]	-0.28 [0.63]	-0.43 [4.47]
Outage $\times$ Dealer moderately hedging	0.85*** [0.15]	0.25** [0.10]	3.13* [1.30]
Outage $\times$ Dealer intensely hedging	0.35* [0.17]	0.25 [0.45]	1.70 [2.75]
Outage $\times$ asinh(Inventory)	0.01 [0.01]	-0.02*** [0.00]	-0.11** [0.04]
FE Minute	✓	✓	✓
FE ISIN	✓	✓	✓
FE Client Sector	✓	✓	✓
Observations	1915	1915	1915
Adjusted R^2	0.069	0.033	0.037

Table A21: Effect of Eurex Outages on D2C Trades across Dealer Groups. This table replicates Table 8 from the main text, adding dealer inventory as an additional explanatory variable.

Table A22 replicates Table 10 using two additional explanatory variables. Firstly, trades are more beneficial for a client if it is the first trade with that particular dealer. This appears to contradict the value of established dealer-client relationships, but it is driven by a selection effect: during outages, clients are more likely to accept advantageous quotes by new dealers on electronic trading platforms (in line with Table 14). Another important characteristic is whether a client is active on the futures market at all. Trades by these clients are much more profitable (less unprofitable). This suggests that Eurex activity can serve as another proxy for informational advantages. In our particular application, clients that are usually active on Eurex are much more likely to be aware of the Eurex outages, and adjust their trading on the cash market accordingly.

	(1) mispricing	(2) markup	(3) profit
Outage	-0.87*** [0.20]	-1.89*** [0.20]	-4.66 [9.49]
Outage $\times$ # Connections	-0.04 [0.03]	0.00 [0.02]	0.18** [0.05]
Outage $\times$ Relationship w/ Dealer	-1.79** [0.63]	0.29 [0.86]	4.70** [1.65]
Outage $\times$ ln(Dealer Centrality)	-0.37*** [0.06]	-0.26 [0.20]	0.58 [0.87]
Outage $\times$ moderate hedging	0.90*** [0.18]	0.14 [0.11]	-3.09 [4.79]
Outage $\times$ intense hedging	0.98** [0.27]	0.21 [0.18]	-2.02 [5.03]
Outage $\times$ New Relationship	0.01 [0.20]	-0.08 [0.20]	2.88** [0.97]
Outage $\times$ Eurex-Active	0.62 [0.45]	1.06 [0.61]	5.00** [1.80]
FE Minute	✓	✓	✓
FE ISIN	✓	✓	✓
FE Sector	✓	✓	✓
Observations	1700	1700	1700
Adjusted R^2	0.132	0.043	0.036

Table A22: Role of Dealer-Client Network Characteristics during Outages. # Connections refers to the number of different dealers a client has traded with. Dealer Relationship measures the fraction of buying/selling volume of each client-dealer pair as a share of the total volume of each client. Dealer Centrality is the volume-weighted eigenvector centrality of the dealer involved in the transaction. The dependent variables are calculated from the client perspective. The sample includes all D2C transactions during the two outage and four non-outage control windows. We drop trades by households, since the various network measures cannot be computed, and trades on MTS and regular exchanges, as these occur anonymously in a CLOB.

D.4 Effect on Swap Market

Interest rate swaps are a key segment of the fixed-income market. These swaps exchange fixed-rate interest payments for floating-rate interest payments over a specified period.⁴¹ Figure A10 shows that quotes on €STR interest rate swaps become stale the moment bond futures become unavailable, particularly at longer maturities.⁴²

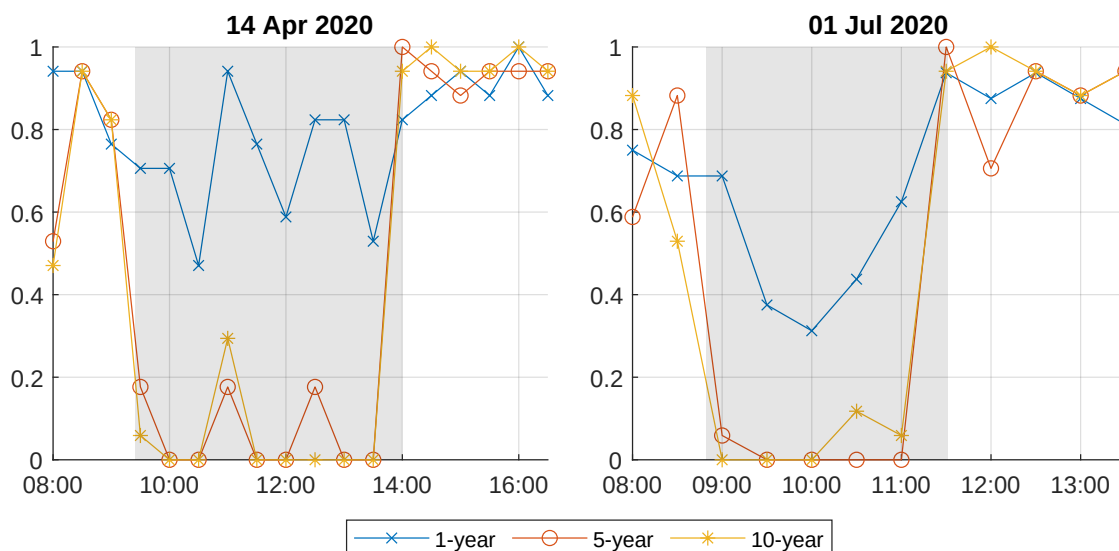


Figure A10: Quote Update Frequency for €STR Swap Rates on Bloomberg. We approximate the number of new quotes as the number of tick-by-tick price changes. To show all series on a single scale, we aggregate the number of quote updates in 30-minute windows and normalize them to a 0-1 range for each maturity. All series refer to overnight index swaps of different maturity based on the euro short-term rate (€STR)

Figure A11 confirms that bid-ask spreads for longer-dated swaps spike during outages, if quotes are available at all.

Figure A12 lastly, confirms these results with executable quotes from an interdealer broker. In particular, Compagnie Financière Tradition, a listed company on the Swiss stock exchange, runs the Trad-X platform. This platform is based on a central limit order book, i.e. immediately executable quotes, in contrast to the indicative quotes on Bloomberg. According to their own statements, Trad-X is the market-leading platform for interest rate derivatives and is used by market participants from over 29 countries. For ease of exposition, we focus on the most liquid instrument on Trad-X, namely 10-year/6-month Euro Interbank Offered Rate (Euribor) swaps, which exchange 10-year fixed-rate interest payments for six-month floating-rate interest payments. Eurex outages cause a complete evaporation of the order book for these swaps. The number of available order book levels declines and spreads widen immediately. After Eurex goes back online, the order book recovers within half an hour.

⁴¹See Dalla Fontana, Holz auf der Heide, Pelizzon, and Scheicher (2019) and Boudiaf, Frieden, and Scheicher (2024) for a detailed anatomy of the euro area interest rate swap market.

⁴²Recall that the €STR is closely linked to the ECB's policy rates, and that the ECB provided extensive forward guidance at the time. Between April and July 2020, the €STR rate fluctuated between -.53 and -.56 percent.

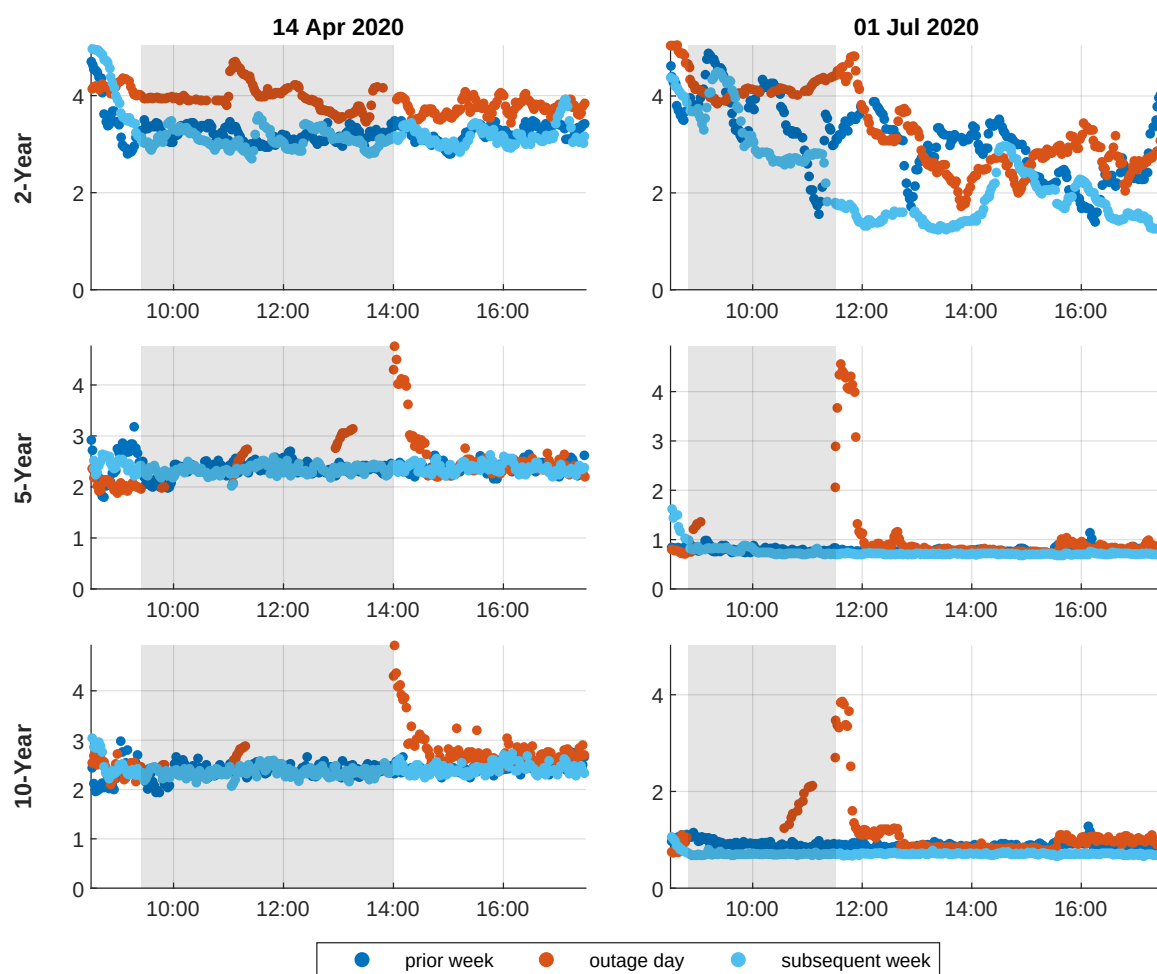


Figure A11: Bid-Ask Spread of €STR Swap Rates on Bloomberg. This figure shows the bid-ask spread (in basis points) for overnight index swaps of different tenors based on the euro short-term rate (€STR). Red lines refer to outage days, dark and light blue lines refer to the previous and subsequent week.

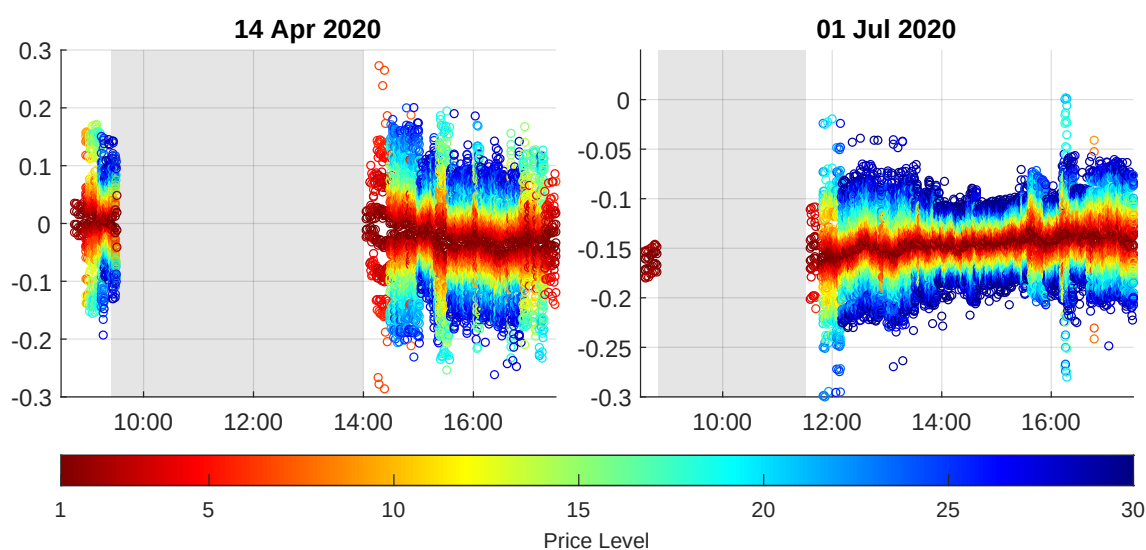


Figure A12: Bid and Ask Quotes for 10y/6m Euribor Swaps on Trad-X. The figure shows bid and ask quotes across different order book levels.

D.5 MTS Liquidity

This section provides robustness checks for the results in [Section 3.2.1](#) in the main text.

Recall that [Figure 1](#) in the main text shows that liquidity in the cheapest-to-deliver bonds underlying 10-year bond future contracts evaporates on MTS when Eurex goes down. [Figure A13](#) replicates this figure at the country-level.

The above results are based on MTS’s euro area wide ‘EBM’ platform (for ‘European Bond Market’). In parallel, MTS also runs local bond market platforms (labelled ‘GEM’ for Germany, ‘FRF’ for France, ‘MTS’ for Italy, and ‘ESP’ for Spain). [Figure A14](#) confirms that liquidity also evaporates on these local market segments.

Next, we confirm that our results are not confined to 10-year CTD bonds either. Our results hold for the entire universe of bonds. [Figure A15](#) documents a sharp decline in liquidity across countries.

In fact, [Figure A16](#) shows that the liquidity evaporates also for other euro area countries, which do not have an own futures market. [Figure A17](#), lastly, looks at non-euro countries whose bonds are traded on MTS. Except for Denmark, whose currency is pegged to the Euro, market liquidity seems unaffected by the outage on Eurex. In a sense, these results serve as a placebo test: the breakdown of MTS was indeed caused by the outage on Eurex, there was no simultaneous independent incident on MTS.

Another insightful exercise is to connect our order book data with the transaction data on MTS. When we focus on 10-year deliverable bonds for instance, i.e. bonds that were deliverable into the 10-year bond future, we only observe a single trade during each outage. [Figure A18](#) zooms in on these two trades. We see that in both cases, quotes are quite stable at first, but then a single trade causes massive quote adjustments. In the case of the French bond on the first outage, spreads widen massively before quotes disappear entirely. This is consistent with some stale quotes getting ‘picked off’.

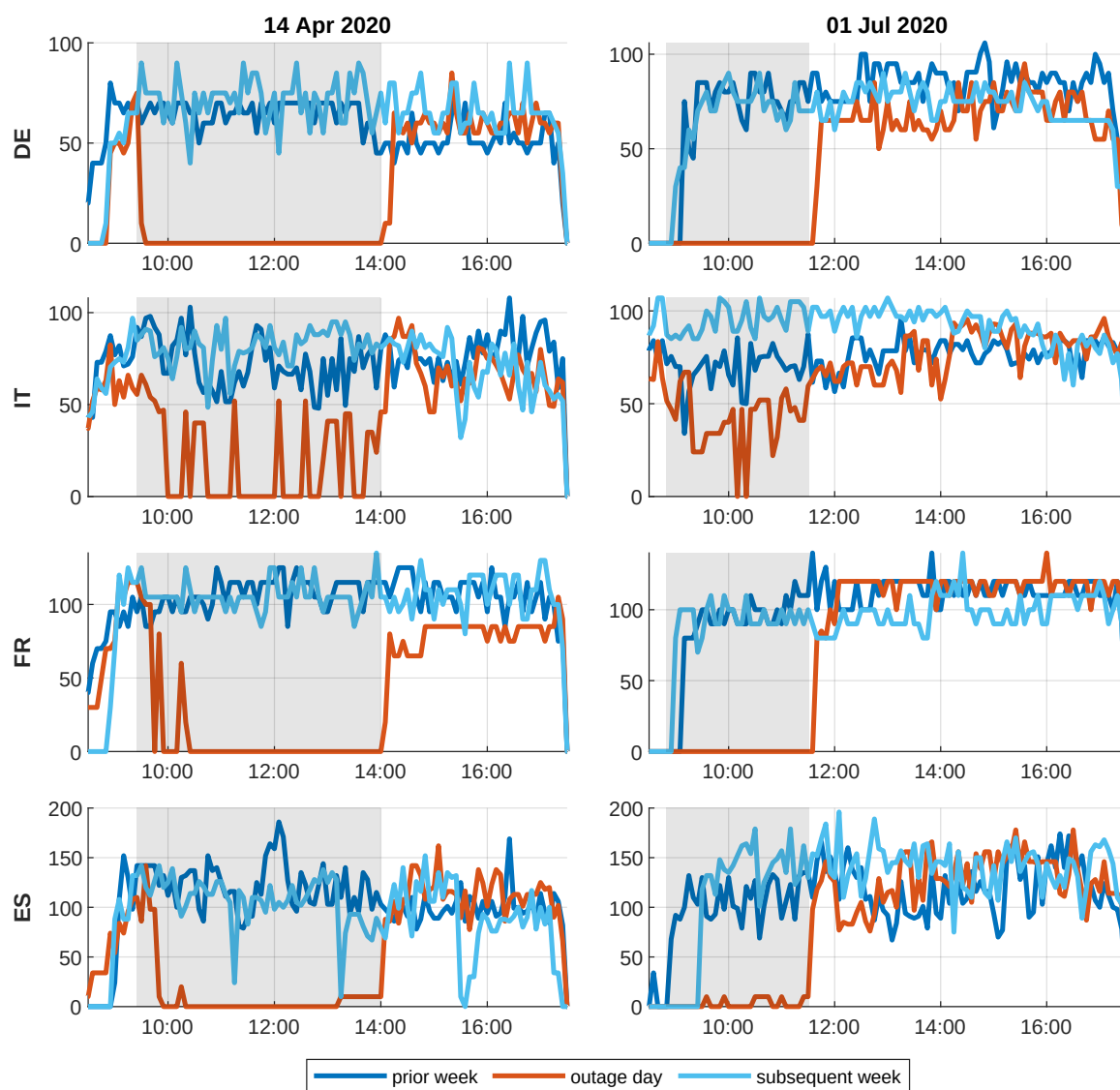


Figure A13: Order Book Depth on MTS. This figure shows the total quoted volume for 10-year CTD bonds (in million €) across all three levels and both sides of the order book, at 5-minute snapshots.

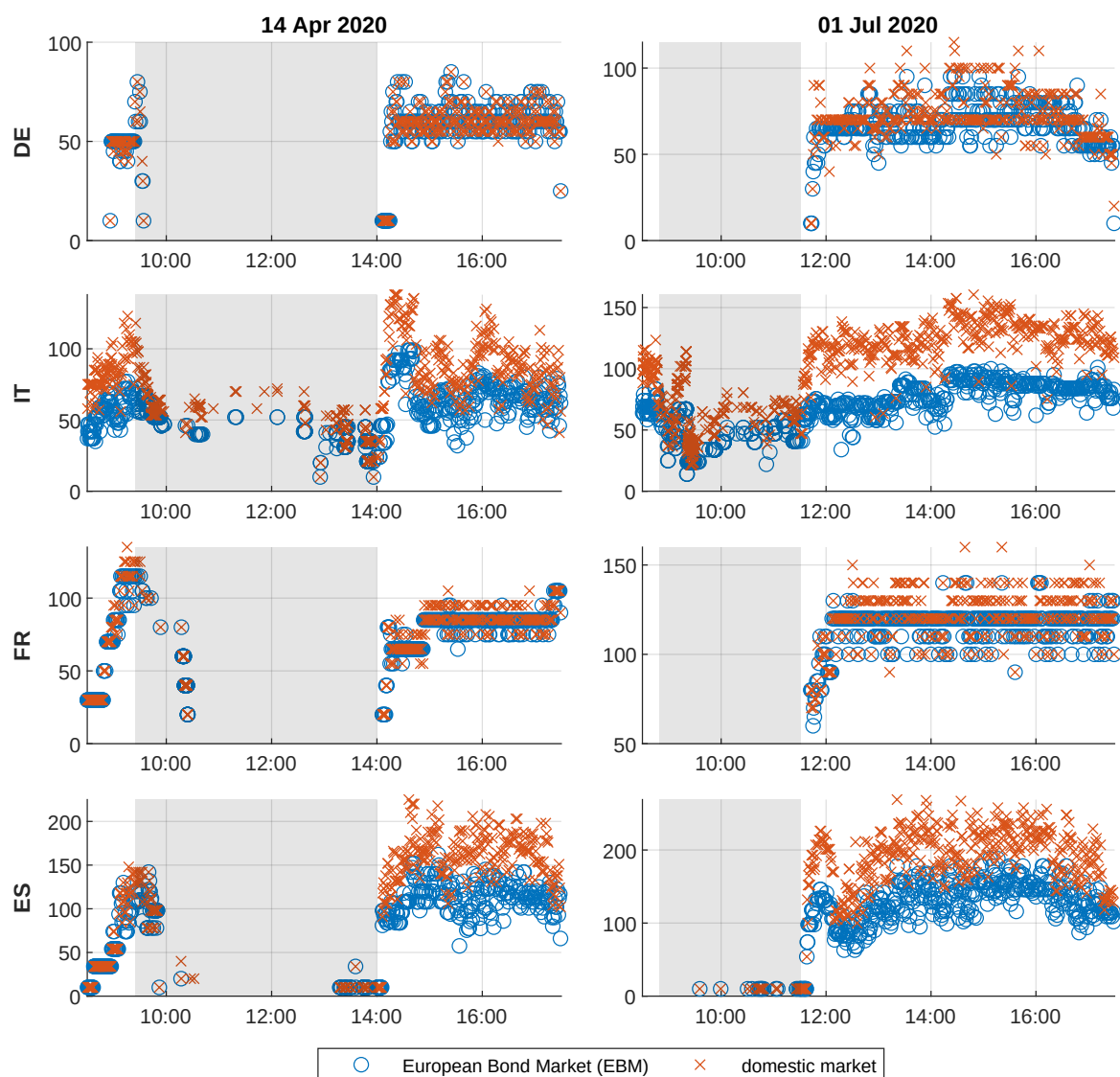


Figure A14: Order Book Depth of CTD bond on different MTS market segments. This figure shows the total quoted volume (in million €) across all three levels and both sides of the order book, for the cheapest-to-deliver bond underlying the 10-year future. Blue circles refer to the ‘European Bond Market’ segment (as in the main text), red crosses refers to the domestic market segment.

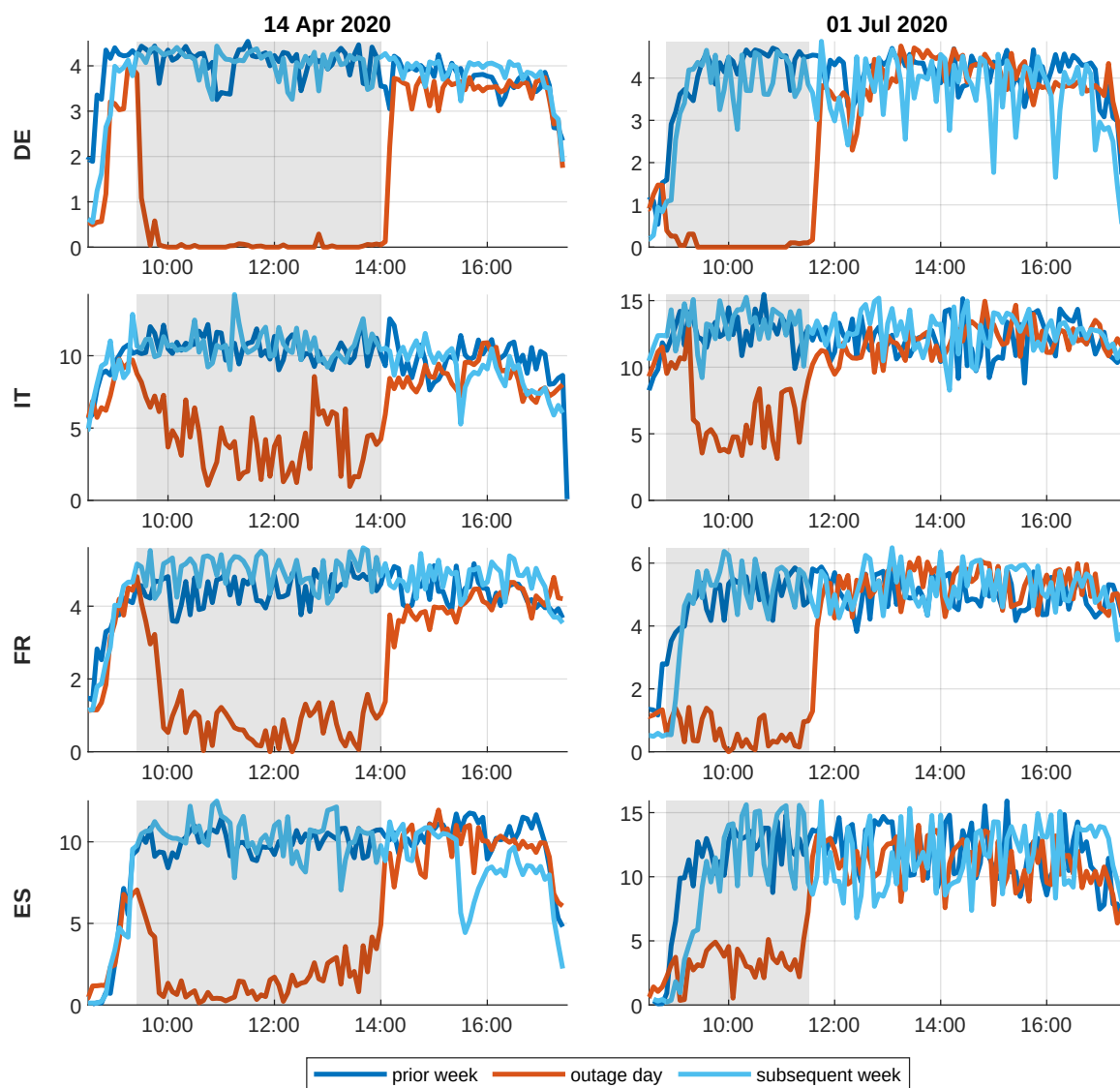


Figure A15: Total Order Book Depth on MTS across Countries. This figure shows the total quoted volume (in billion €) for German, French, Italian and Spanish bonds across all three levels and both sides of the order book, at 5-minute snapshots.

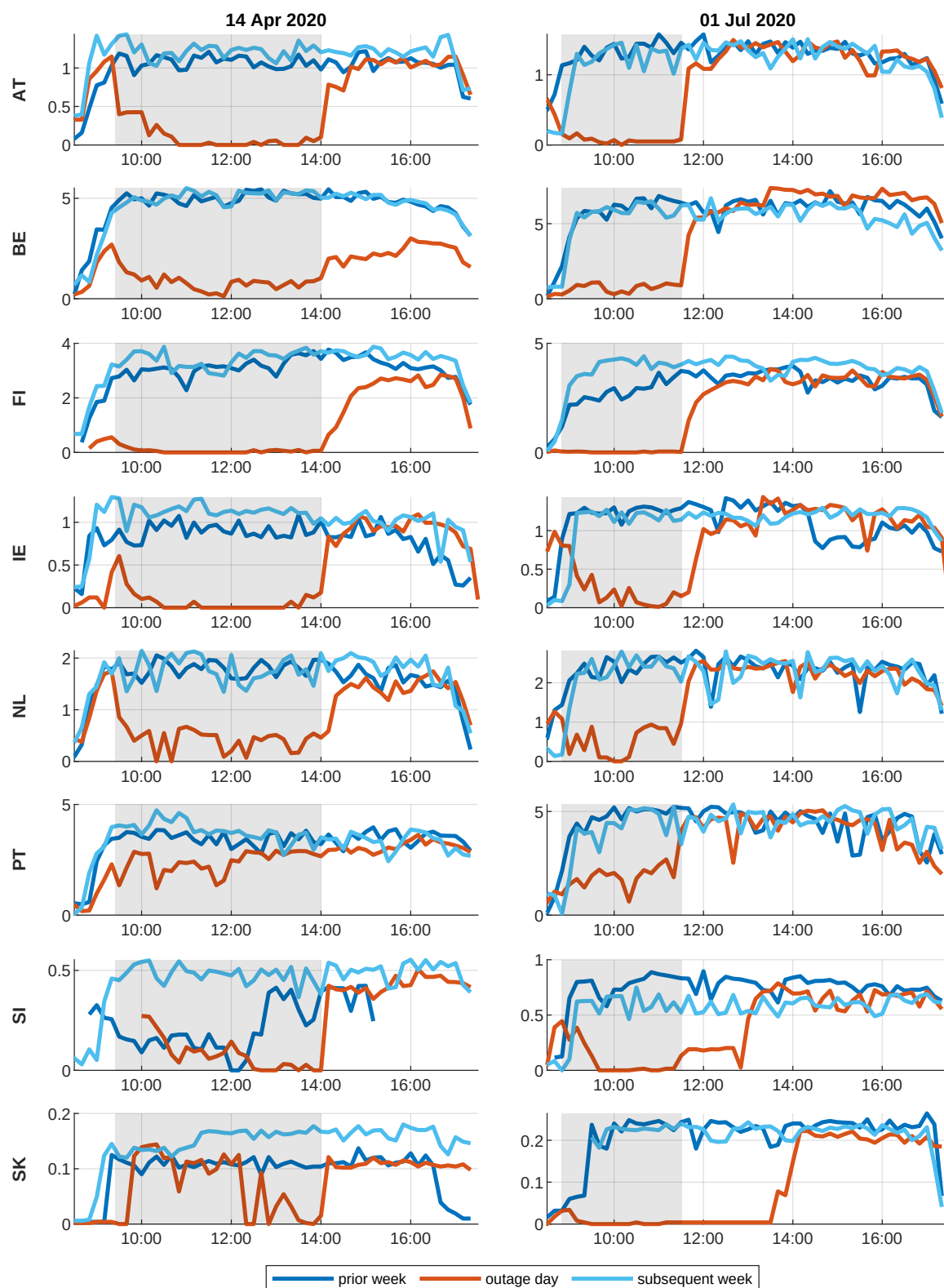


Figure A16: Total Order Book Depth on MTS for Other Euro Area Countries. This figure shows the total quoted volume (in billion €), across all three levels and both sides of the order book, at 10-minute snapshots, for other euro area member states whose bonds are traded on MTS.

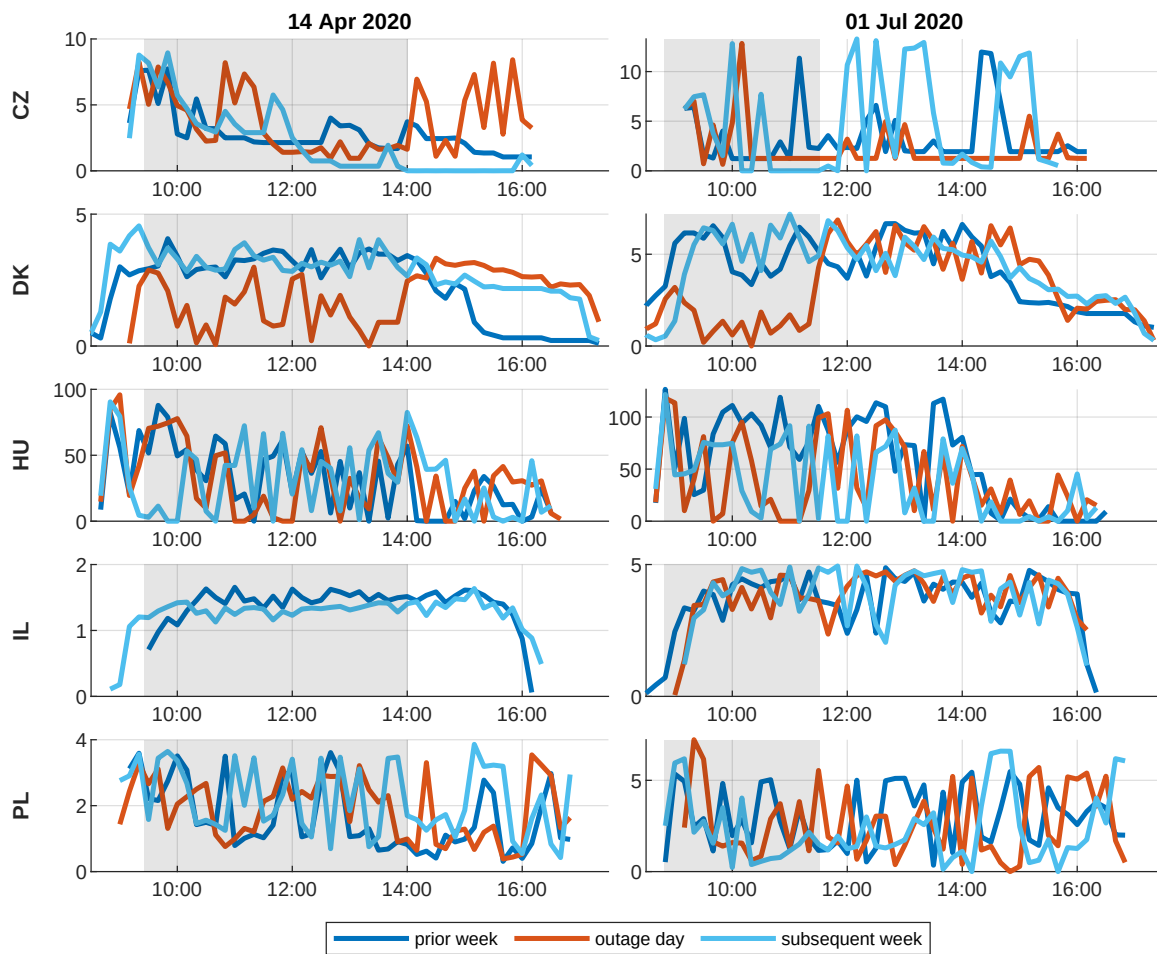


Figure A17: Total Order Book Depth on MTS for Non-Euro Area Countries. This figure shows the total quoted volume (in billion of local currency), across all three levels and both sides of the order book, at 10-minute snapshots, for non-euro countries (CZ for Czech Republic, DK for Denmark, HU for Hungary, IL for Israel, PL for Poland). The lack of liquidity on 14 April 2020 for Israeli bonds was due to a public holiday, not the outage on Eurex.

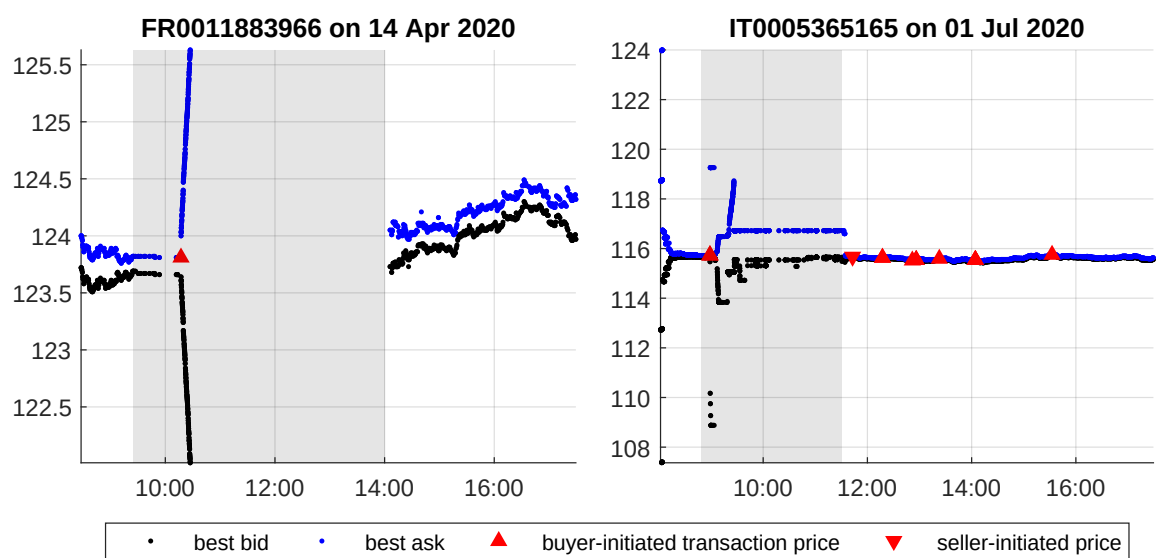


Figure A18: Quotes and Transactions Prices on MTS during Eurex Outage. This figure shows the only two trades in 10-year bonds that were deliverable into a 10-year futures contract.

D.6 MTS Quoting Obligations

A peculiarity on MTS are so-called ‘market making commitments’. While MTS’s euro area wide ‘EBM’ market has no such commitments, MTS also runs local bond markets, see previous section. On these local markets, each local issuer, i.e. usually the debt management office, can set obligations and requirements for their own country. For instance, market makers might have to provide bid and ask quotes for specific bonds, sometimes at pre-defined maximum spreads and minimum volumes and for a specified duration, e.g. at least five hours a day or 50% of trading time. See [Cheung, Rindi, and De Jong \(2005\)](#) for a detailed overview of the microstructure of the MTS trading platforms and the [MTS website](#) for current rules.

Do these quoting obligations matter? [Figure A19](#) compares the order book depth on the EBM and local markets for each country. We can clearly see liquidity drops on both segments for all countries, but it is hard to eyeball whether the relative decline differed across countries.

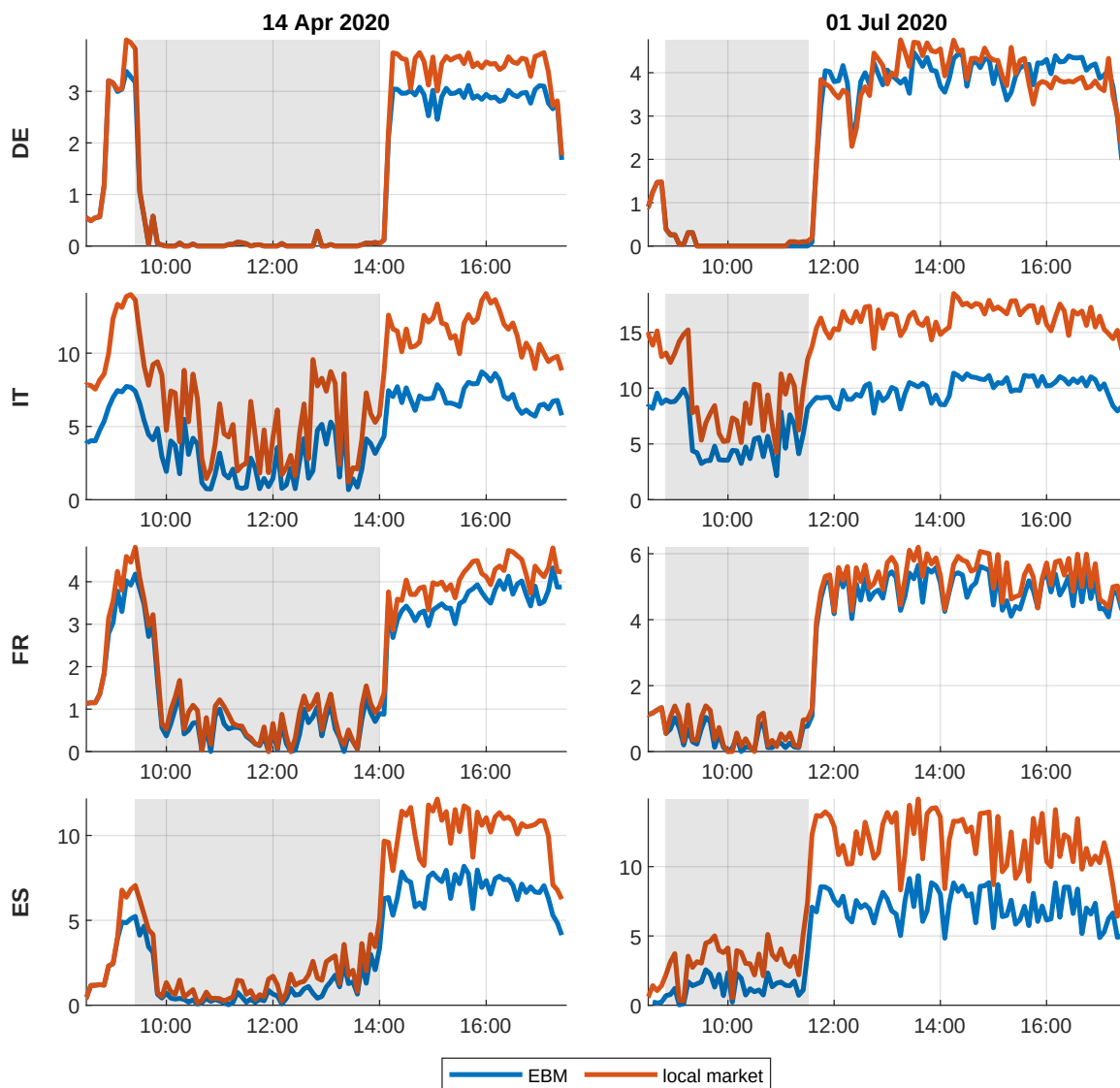


Figure A19: Order Book Depth across different Market Segments on MTS. This figure shows the total quoted volume (in million €) across all three levels and both sides of the order book, at 5-minute snapshots, for all bonds of a given country. Blue lines refer to the ‘European Bond Market’ segment (as in the main text), red lines refers to the respective domestic market segment.

Hence, [Table A23](#) reports results from another dummy regression, this time run separately for each country, namely:

$$\log(1 + OBdepth_{smt}) = \alpha + \beta_1 \times D_t + \beta_2 \times D_t \times local + \gamma \times FE + \epsilon_t \quad (A1)$$

where $OBdepth_{smt}$ is the order book depth (in €) of all bonds in maturity bucket m at time t , measured at 5-minute snapshots, quoted on market segment s . D_t is a dummy that equals one during the Eurex outages and $local$ is a dummy variable that equals one for MTS's local market segment and is zero for the EBM segment. We include six days (the two outage days plus the preceding and subsequent week) and 91 intraday observations per day (5-minute snapshots from 08:30 a.m. to 4:00 p.m.).

	(1) DE	(2) ES	(3) FR	(4) IT
Outage	-17.68*** [0.67]	-10.42** [2.71]	-14.18*** [0.56]	-1.36*** [0.14]
local	0.21** [0.08]	0.55*** [0.05]	0.18* [0.07]	0.48*** [0.01]
Outage $\times$ local	0.11 [0.24]	0.47*** [0.09]	-0.03 [0.08]	0.18*** [0.02]
FE Day	✓	✓	✓	✓
FE Time	✓	✓	✓	✓
FE Maturity Bucket	✓	✓	✓	✓
Observations	4368	4368	4368	4368
Adjusted R^2	0.805	0.538	0.647	0.567

Table A23: Differential Effect of Eurex Outages on MTS Market Segments. Each column shows results of a separate regression run at the country-level, see [Equation A1](#). Throughout, the dependent variable is the log of the order book depth of all bonds in four different maturity buckets, for the country mentioned in the column header. We differentiate between the EBM and local market segment.

We see that the order book depth is higher on the local market for all countries, especially so for Italy and Spain. And precisely for these two countries, we see a differential effect during the Eurex outage. The liquidity of Italian and Spanish bonds drops markedly less on the local market segment compared to the EBM market. This is in line with the stricter quoting obligations on the local Italian and Spanish market compared to the German and French market.

Note that these results are analogous to [Clark-Joseph, Ye, and Zi \(2017\)](#), who show that even mild quoting obligations significantly improve liquidity on US stock exchanges.

D.7 Indicative Quotes

This section reproduces the results presented in [Section 3.2.2](#) of the main text at the country-level. Recall that [Figure 2](#) shows across three different data sources that indicative quotes for EGB cash bonds become stale during the Eurex outage.

[Figure A20](#) confirms that the result holds for Germany, Italy, France, and Spain: the number of quote updates drops dramatically during Eurex outages. [Figure A21](#) plots the bid yield of 10-year bonds quoted for each country. Yields stay virtually constant while Eurex is offline. By all appearances, these prices are stale.

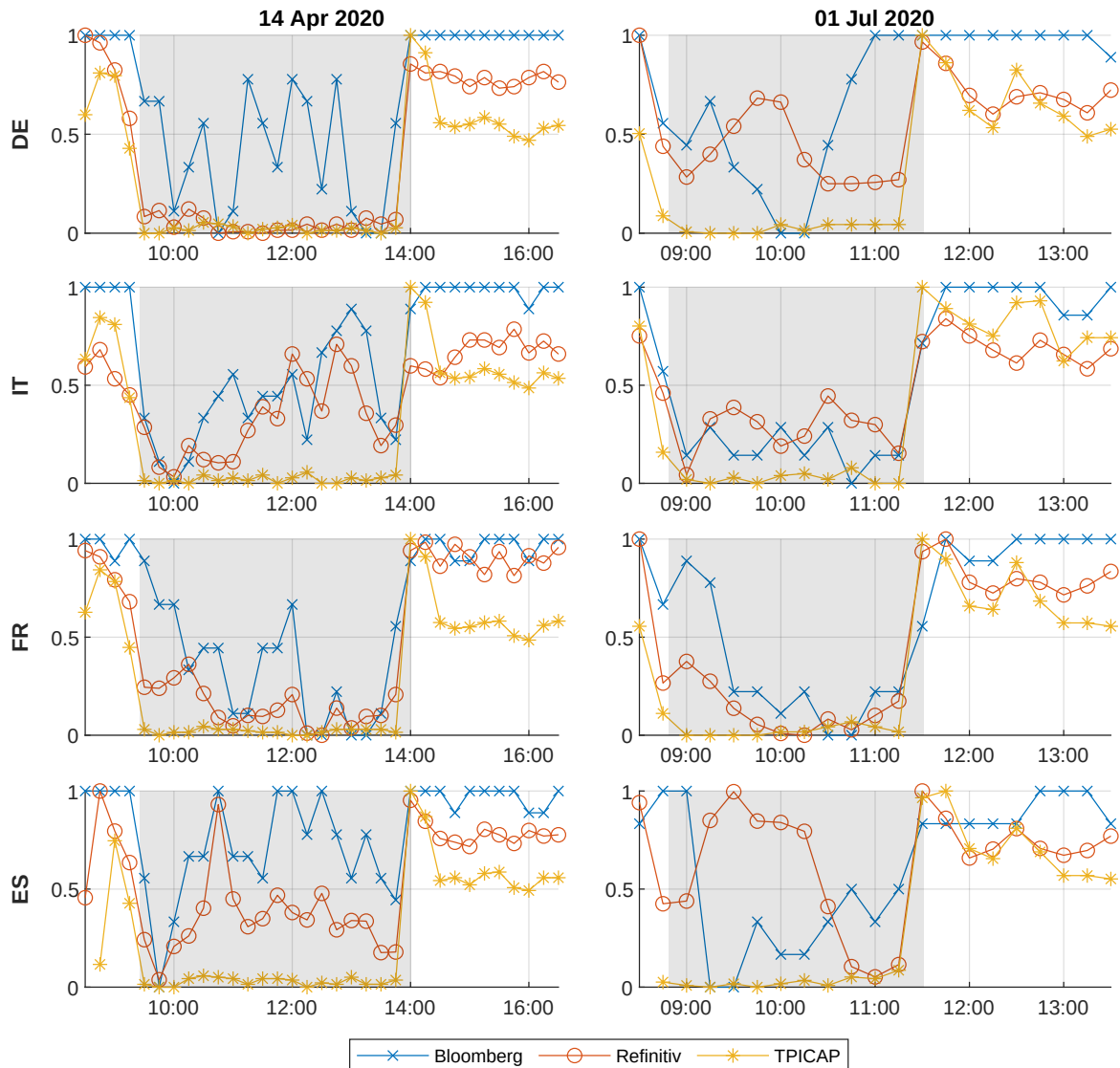


Figure A20: Quote Update Frequency for 10-Year Government Bonds on Different Platforms. Bloomberg and TPICAP data refer to quote updates in the cheapest-to-deliver bond, Refinitiv data to the on-the-run bond. For Bloomberg, we approximate the number of new quotes as the number of tick-by-tick price changes. To show all series on a single scale, we sum the number of quotes updates in 15-minute windows and normalize them to a 0-1 range for each data provider.

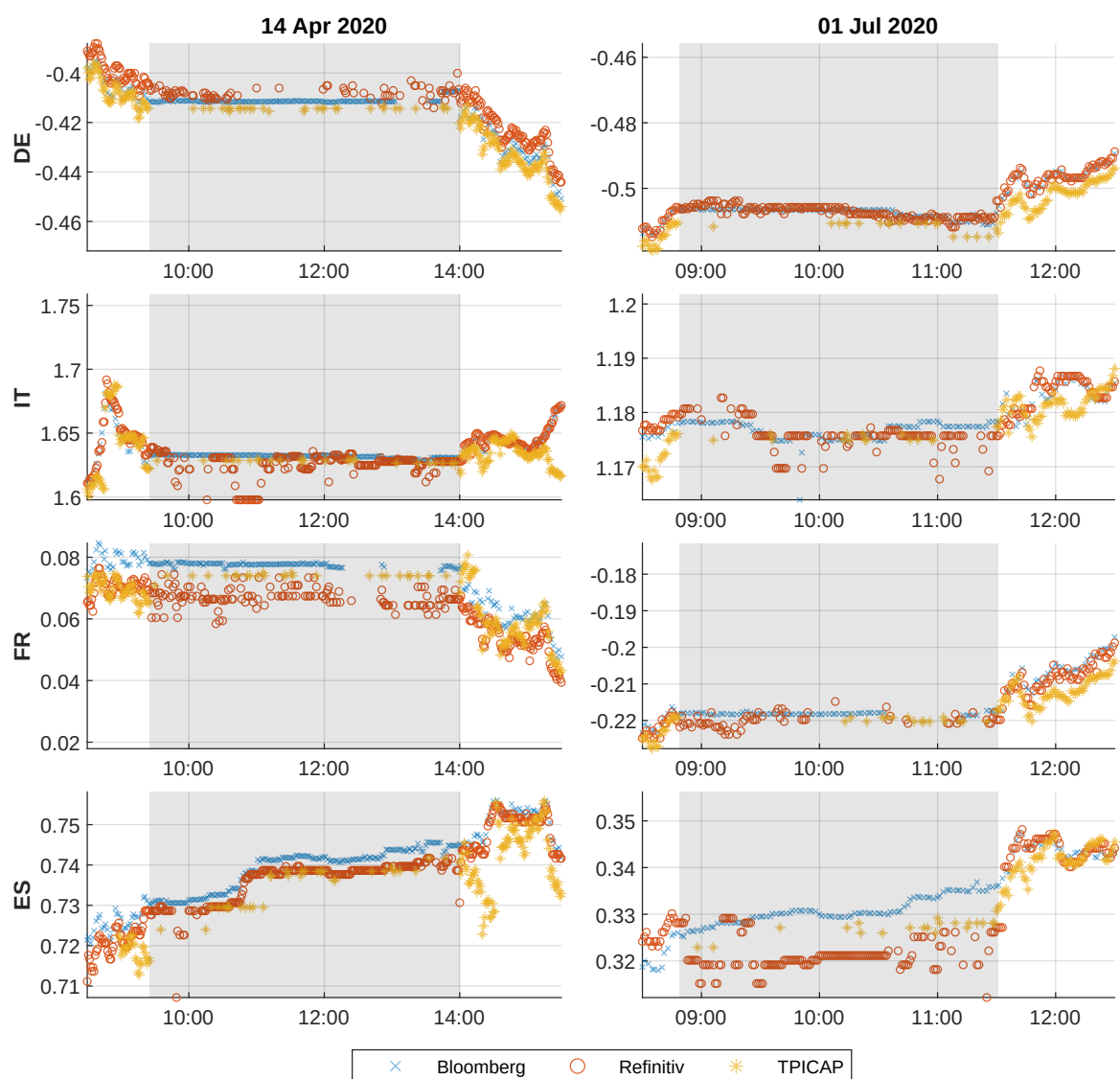


Figure A21: Quoted Bid Yield of 10-Year Government Bonds on Different Platforms. Bloomberg and TPICAP data refer to the cheapest-to-deliver bond, at minutely snapshots. Refinitiv data refers to the on-the-run ('benchmark') bond, at minutely snapshots and as the median value across all available 'pricing contributors'. To show all series on a single scale, we apply a level adjustment to the Refinitiv yield, such that the daily median yield matches the daily median yield of Bloomberg and TPICAP.

Appendix E Further Evidence from Cash Venue Outages

This section provides further evidence on the outages studied in [Section 5](#) of the main text. Recall that [Table 15](#) shows how cash venue outages affect various metrics of market functioning, like trading volumes and return volatility. The results refer to average effects across all four euro area countries we study (for the cash market) or all eight bond futures (for the futures market).

[Table A24](#) instead looks at the effect on aggregate and country-level cash trading volumes. At the aggregate level, only the trading volume decline during the Bloomberg outage is statistically significant. Looking at individual countries, we observe the largest effects for German and French bonds.

Similarly, [Table A25](#) looks at country-level trading volumes on Eurex. We see that the drop in trading volumes is entirely driven by German bond futures. In fact, the reduction in Italian, French and Spanish bond future trading volume is insignificant.

[Table A26](#) zooms in on the most liquid instrument on Eurex, namely 10-year German bond futures. In line with the average effects across all futures, trading activity tends to decline during cash venue outages but most other measures of market functioning are unaffected.

	MTS 12 Jan 2010 8:00-10:35	Bloomberg 17 Apr 2015 9:20-10:10	FWB 27 May 2015 8:00-11:00	MTS 26 Jul 2019 12:30-13:20	Pooled
All Bonds	-0.47 [0.27]	-2.09** [0.24]	-0.39 [0.44]	-1.05 [0.62]	-0.70** [0.25]
Obs.	96	33	111	33	273
DE Bonds	-0.35 [0.90]	-9.13*** [0.67]	0.49 [1.30]	-0.58 [0.30]	-1.10 [1.09]
IT Bonds	-2.34 [1.04]	-0.64 [0.73]	1.39 [1.15]	-0.97 [0.64]	-0.45 [0.94]
FR Bonds	-2.00 [0.95]	-8.31* [2.52]	-1.72 [1.12]	-3.42 [1.31]	-2.82*** [0.86]
ES Bonds	-0.63 [0.29]	0.30 [1.74]	-1.88 [1.46]	-4.45** [0.98]	-1.49** [0.67]

Table A24: Effect of Cash Venue Outages on Cash Market Trading Volumes. Each column refers to a different outage and regression. Each sample includes the outage period and the same intraday window on the preceding and subsequent week (as controls), at a 5-minute frequency. We omit the Brokertec outage because the EGB cash market was effectively idle during this time of day (7:43-9:35 p.m. local time). The last column refers to a pooled regression of all outages. The number of observations is equal across columns. The dependent variable differs across rows and refers to the log of the transaction volume in € plus one, in all EGBs (first row) or all bonds of the stated country. All regressions include time-fixed effects. For brevity, the table shows results only for the outage dummy.

Sections [E.1-E.2](#) below provide further evidence on MTS and Bloomberg outages. To put the size of these outages in perspective, note that the [2019 European Commission Report](#) estimates that each platform has a 20-30% market share in electronic EGB trading, compared to 0-5% for Brokertec and FWB (see Table 3 on p.168 of the report).

	MTS 12 Jan 2010 8:00-10:35	Bloomberg 17 Apr 2015 9:20-10:10	FWB 27 May 2015 8:00-11:00	Brokertec 11 Jan 2019 19:43-21:35	MTS 26 Jul 2019 12:30-13:20	Pooled
All Futures	0.13 [0.05]	-0.31 [0.15]	-0.77 [0.62]	-1.16** [0.12]	-0.41 [0.24]	-0.52* [0.26]
Obs.	465	153	540	339	153	1650
DE Futures	0.13 [0.05]	-0.56* [0.15]	-0.81 [0.62]	-1.16** [0.12]	-0.49 [0.29]	-0.57** [0.26]
IT Futures	0.13 [0.07]	0.83** [0.09]	-0.56 [0.70]		-0.41 [0.34]	-0.11 [0.24]
FR Futures		-0.94 [0.33]	-0.74 [0.54]		-0.46 [0.50]	-0.37* [0.21]
ES Futures					-0.11 [0.12]	-0.01 [0.01]

Table A25: Effect of Cash Venue Outages on Bond Future Trading Volumes. Each column refers to a different outage and regression. Each sample includes the outage period and the same intraday window on the preceding and subsequent week (as controls), at a minutely frequency. The last column refers to a pooled regression of all outages. The number of observations is equal across columns and corresponds to three times the duration of the outage in minutes. The dependent variable differs across rows and refers to the log of the number of traded contracts plus one, in all bond futures (first row) or all future of the stated country. Empty cells mean there were no futures available at the time or the trading volume was zero during the outage and control windows. All regressions include time-fixed effects. For brevity, the table shows results only for the outage dummy.

E.1 MTS Outage

While we did not find any newspaper reports about an outage on 26 July 2019, we noticed a prolonged and unexplained period without any activity on MTS using our intraday dataset of transactions and order book updates. Trading and quoting activity on MTS was lower than usual from about 10:00 that day, before breaking down entirely from 12:30 till 13:20. There were no quotes and no transactions in any bond (see [Figure A5](#)). MTS representatives could not provide an explanation for the incident, but the similarity with the confirmed outage from 2010 is striking in our view. We use this suspected outage as a robustness check.

Since this event took place quite recently, we can zoom in on the futures market using granular order book data. In particular, [Figure A22](#) confirms that there is no discernible drop in market liquidity for the 10-year bond future for any country when MTS is inactive. Similarly, indicative quotes on Bloomberg do not show any obvious reaction to the suspected MTS outage, see [Figure A23](#).

E.2 Bloomberg Outage

During the Bloomberg outage, aggregate trading volumes decrease on the cash and to a lesser extent also on the futures market (see [Table A24](#) and [Table 15](#)). Looking at individual countries, [Table A25](#) shows that trading volumes do considerably decrease in German and French bond futures. But this is offset by a substantial increase in trading volumes in Italian bond futures. This evidence is in line with [Bouveret, Haferkorn, Gaetano, and Panzarino \(2022\)](#). They document

	MTS 12 Jan 2010 8:00-10:35	Bloomberg 17 Apr 2015 9:20-10:10	FWB 27 May 2015 8:00-11:00	Brokertec 11 Jan 2019 19:43-21:35	MTS 26 Jul 2019 12:30-13:20	Pooled
Volume	0.42*** [0.00]	-0.68** [0.12]	-0.79 [0.53]	-1.25*** [0.08]	-0.48 [0.32]	-0.50 [0.29]
Observations	465	153	540	339	153	1650
#Trades	0.24 [0.14]	-0.51** [0.09]	-0.49 [0.32]	-0.89* [0.27]	-0.54 [0.35]	-0.37* [0.20]
Observations	465	153	540	339	153	1650
Volatility	0.15 [0.11]	-0.25** [0.03]	-0.78 [0.61]	-0.06 [0.04]	-0.12 [0.19]	-0.28 [0.26]
Observations	465	153	540	281	152	1591
Amihud	-0.06* [0.02]	0.05* [0.01]	0.05 [0.02]	0.13 [0.33]	-0.00 [0.00]	0.02 [0.04]
Observations	465	153	540	281	152	1591
Roll	-0.15 [0.17]	-0.08 [0.14]	0.19* [0.05]	-0.18 [0.20]	0.38*** [0.03]	0.05 [0.09]
Observations	464	153	540	132	152	1441

Table A26: Effect of Cash Venue Outages on 10y German FGBL Future. See [Table 15](#) for a description of the different dependent variables (rows). All regressions include time-fixed effects.

a similar divergence for flash crash episodes: trading activity in Italian bonds surges on the futures market and plummets on the cash market. [Figure A24](#) provides results for the MTS platform. The liquidity provision was slightly lower than normal during the Bloomberg outage, while the executed trading volumes were barely affected. [Figure A25](#) shows that the lower liquidity was restricted to Italian bonds.

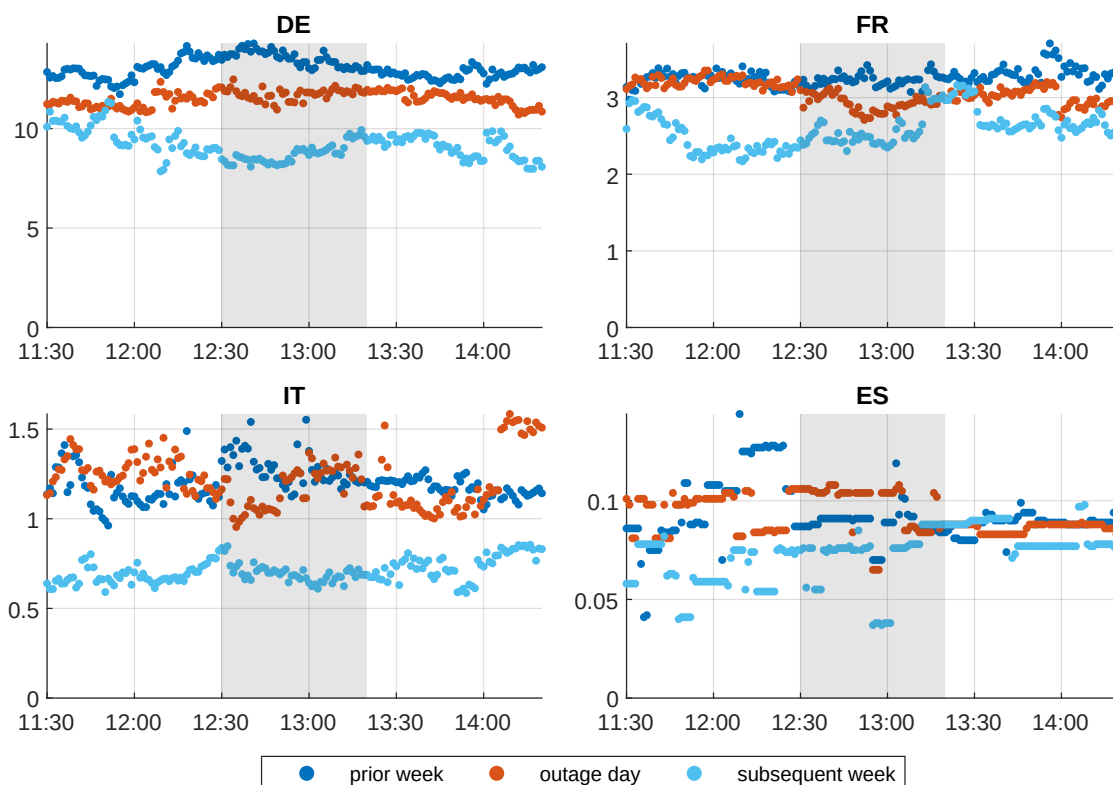


Figure A22: Order Book Depth of 10-year Bond Futures during suspected MTS Outage on 26 July 2019. This figure shows the total number of contracts (in thousands) quoted at the first fifteen levels on both sides of the order book, at minutely snapshots. Red dots refer to the suspected MTS outage day on 26 July 2019, dark and light blue dots refer to the previous and subsequent week.

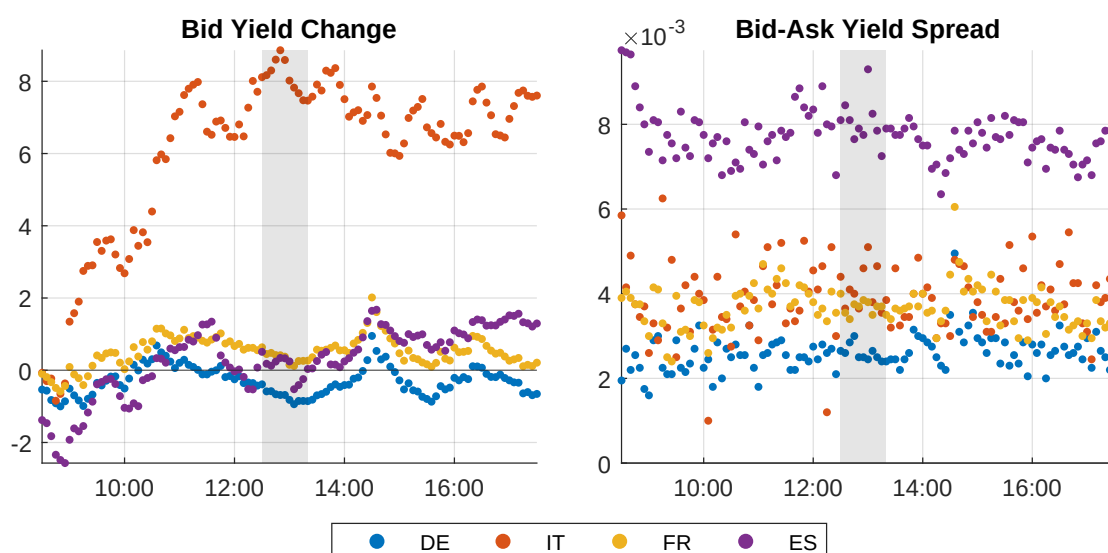


Figure A23: Indicative Quotes on Bloomberg on 26 July 2019. The left panel shows bid yield changes (in basis points since 8:00 a.m.), the right panel shows bid-ask yield spreads (in basis points). All data refers to 10-year cheapest-to-deliver bonds. The grey areas mark the times of the suspected MTS outage.

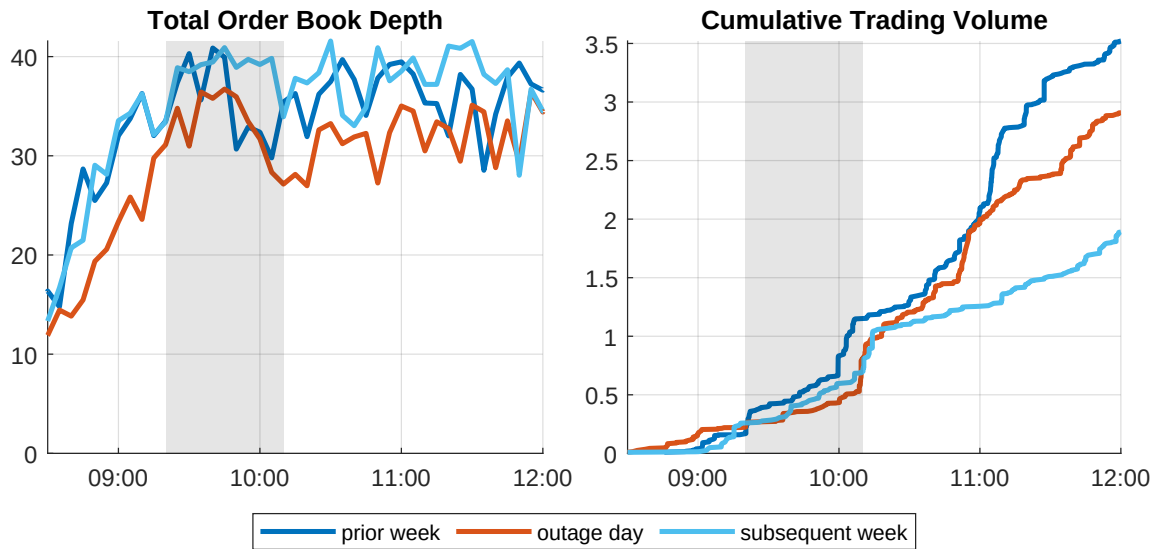


Figure A24: MTS Order Book Depth and Trading Volume during Bloomberg Outage. The left panel shows the total order book depth at 5-minute snapshots (in billion €), the right panel shows the cumulative trading volume (in billion €) in all German, French, Italian and Spanish sovereign bonds on MTS.

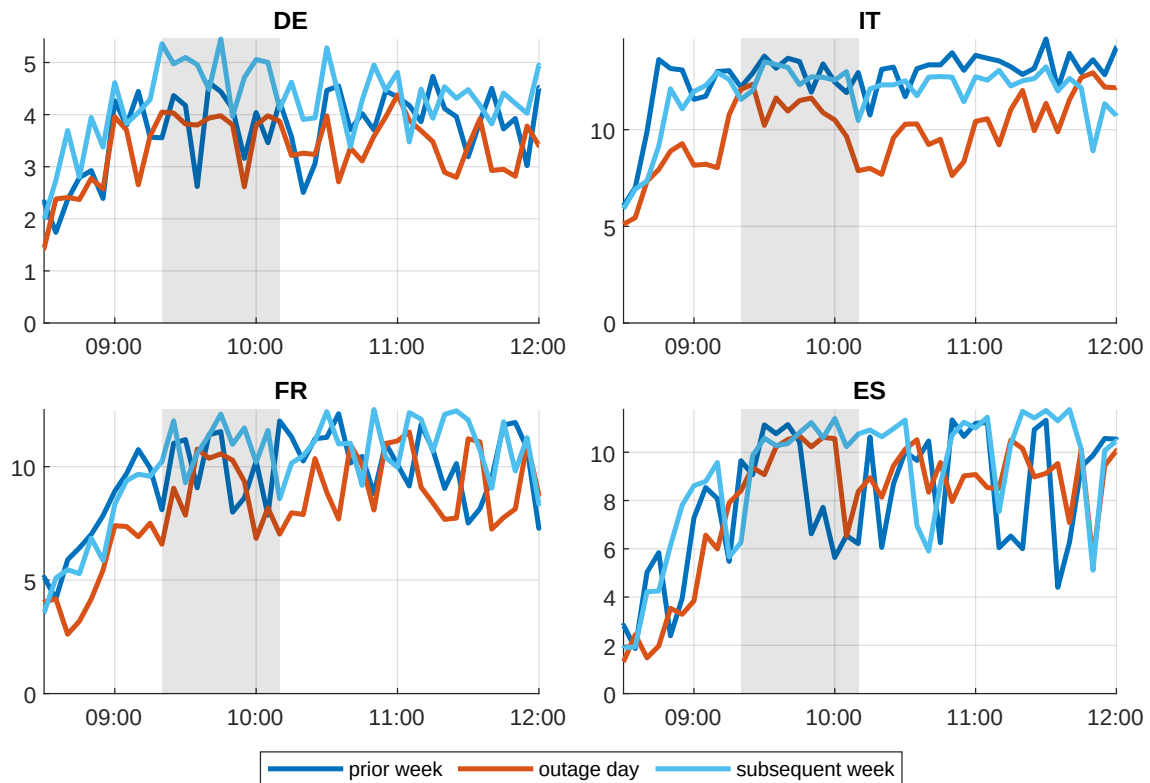


Figure A25: MTS Order Book Depth during Bloomberg Outage across Countries. This figure shows the total order book depth for German, French, Italian and Spanish sovereign bonds on MTS, at 5-minute snapshots (in billion €).

Appendix F Narrative Evidence

Appendix F.1 contains selected quotes from market participants regarding the euro area government bond market and Appendix F.2 cites from an internal memo of one of the major dealers in this market. Overall, this narrative evidence is in line with our empirical result that proper market functioning for EGBs vitally depends on an active futures market.

F.1 ESMA Consultations

The European Securities and Markets Authority (ESMA) has published two consultation papers that contain valuable information about euro area bond market functioning. The below sections summarize the most informative responses by market participants.

F.1.1 Market Outages

The below quotes refer to investor responses to the [ESMA Consultation Paper on Market Outages](#). The final report was published on 24 May 2023. The report’s focus is on stocks and the substitutability among stock exchanges, but some questions also concern the fixed-income market, e.g. question 12: *‘Is there any particular issue relating to trading of non-equity instruments that should be taken into account in the case of an outage? Where possible please differentiate between bonds and derivatives.’* The below quotes refer to investor responses to this question. We emphasize particularly informative bits in **bold font**.

Electronic Debt Markets Association (EDMA, 6 members are: BGC Fenics, Bloomberg, BrokerTec, MarketAxess, MTS and Tradeweb): ‘non-equity trading on EU trading venues is **less affected by outages, including fixed income markets, with trading more naturally moving to alternative platforms** [...] Given that end users have a plethora of alternative trading venues available to them trading the same financial securities, there is already appropriate (commercial) pressure on trading venues to reopen as quickly as it is safe to do so.’

Euronext: ‘in the fixed income market, it is **quite fragmented and trading generally moves to other venues when there are outages** in any case so there is less of an impact.’

Federation of European Securities Exchanges: ‘In particular for fixed income market, it is important to consider the different structure of the market, where there is **not the same reliance on the primary market as is the case for equity, and trading is distributed more widely across several trading venues and systematic internalisers**, so there is less of an impact in case of an outage of a single trading venue’

The Investment Association: ‘From a fixed income, bonds perspective, we echo the observation outlined by ESMA [...] that the trading of bonds is less affected by an outage regardless of the type of trading venue the outage occurs on, as trading does more naturally gravitate towards an alternative platform. Partly this is due to the vast differences in the trading landscape of equity and bond instruments.’ ‘Furthermore, we recommend that ESMA consider the **impact of market outages on futures exchange markets [...] and the trading of bond futures as an outage would most likely have significant implications on the liquidity portfolio of many government securities**’

F.1.2 Algorithmic Trading

The below quotes refer to investor responses to the [ESMA Consultation Paper on Algorithmic Trading](#). While the report’s focus is on stocks, some questions are informative also for the fixed-

income market, e.g. question 36 on market outages: *‘Do you believe any initiative should be put forward to ensure there is more continuity on trading in case of an outage on the main market, e.g. by requiring algo traders to use more than one reference data point?’* The below quotes refer to investor responses to this question. We emphasize particularly informative bits in **bold font**.

Deutsche Börse: ‘Regarding, initiatives aiming at continuity of trading in case of an IT incident/outage, DBG does not believe that such an initiative should be put forward given a close to 100% system performance of main markets. We would caution against forcing algorithms to include different sources of information. **The underlying assumption seems to be that regulated markets, MTFs (multilateral trading facilities) and potentially SIs are set on the same level in terms of price formation and information, with easy switch from one to the other, putting aside respective market shares and the notion of reference market.** The explored initiative would hence introduce an artificial change to the current market structure which is at odd with MiFID. To the contrary, the flight to execution quality at the height of volatility in the COVID-19 crisis proved once more that there was a need by investors to trade on transparent regulated markets when looking at the migration of volumes from dark, SI, and OTC trading to regulated markets. Last but not least, **it should be up to the trading participants to decide if they see merit in connecting to more than one reference data point or not**, but they should not be forced upon by regulation.’

Federation of European Securities Exchanges: ‘FESE considers that **declines in trading following outages are linked to the importance of price formation.** Despite the ability to trade on alternative venues, the **low confidence of traders in the price formation on alternative venues may deter them from trading on those markets** during the outage period.’

German Banking Industry Committee: ‘**we strongly oppose the notion that outages might be compensated by obliging intermediaries, i.e. investment firms, to connect to more than one trading venue so that in the event of an outage, trading can continue seamlessly on another venue.** This would mean market participants would have to maintain double memberships in all relevant main markets. This proposal is completely unproportionate, it might also lead to liquidity reduction in those main markets and will increase costs for intermediaries and clients alike. Neither do we see how the notion to require algo traders to use more than one reference data point might solve the real problem, nor do alternative markets exist for all financial instruments.’

Virtu: ‘Virtu believes that it is in the best interests of the market and orderly trading for the appropriate amount of time to be taken to properly resolve an incident and to restart afterward, instead of forcing trading venues to adhere to an arbitrary two hours of restart time. [...] **The market would be better served by improved resilience across the system as a whole, with true alternatives to primary markets (or any individual venue),** rather than a specific and arbitrary focus on restart times.’

ACI Financial Markets Association: ‘If an outage suspends trading on the main market, it is important to be able to migrate to another trading venue for the use of reference data points to ensure that market liquidity is not affected, since the **simultaneous suspension of Algorithmic trading by numerous market participants could result in high volatility and a drastic reduction in market liquidity.**’

World Federation of Exchanges (WFE): ‘The question as to whether to require use of more than one reference data point will, however, be a function of how well the alternative data points reflect

the market in question. **Fragmentation of markets can bring choice (and competition in terms of execution costs) but not all alternative venues generate (or are capable of generating) meaningful price discovery of their own, instead relying on ‘main’ venues to do so.**

Chicago Board Options Exchange (CBOE): ‘Technical outages by European trading venues are a reasonably regular occurrence and largely inevitable. When they do happen, they are highly disruptive, particularly when experienced by national stock exchanges (‘primary outages’), which facilitate trading in the stocks they list and are the sole operators of official opening and closing auctions for those stocks. **Primary outages in recent years have seen market-wide trading in instruments listed on those exchanges dry up to almost nothing.**

All Options International: ‘**The added liquidity and/or other fail-over benefits that the fragmented market structure and its supporters claim to have in practice is just false.** This is shown by the drop of liquidity of secondary venues the moment the primary venue has an outage. **We believe that the best way to maintain liquid markets is to concentrate all liquidity on one, single venue. Real price forming always happens on the primary market.** In order to generate a better, more liquid market in all scenarios the fragmentation of markets needs to be stopped and all off-book and SI trading needs to be prohibited.’

F.2 Citigroup incident

In 2005, the UK’s Financial Services Authority (FSA) ordered Citigroup to pay almost £14 million in fines for a trading strategy the bank had executed in August 2004, see e.g. the [FSA’s final notice](#) for a detailed description of the trading strategy and a timeline of events and this [Guardian article](#) for further background information.

What makes this incident particularly informative for our purposes is an internal memo cited by the FSA and in news articles by the [Financial Times](#), [New York Times](#) and [Euroweek](#) (now GlobalCapital). This memo was titled ‘*Challenging the dominance of Eurex futures*’ and was written by a member of Citigroup’s European government bond trading desk in London.

According to the FSA, the memo summarized the goal of Citigroup’s trading strategy as ‘*imposing a cost on competitors for offering liquidity on MTS, reducing the attractiveness of using the Bund future to hedge, widening cash market bid-offer spreads, [...] and helping remove some of the smaller primary dealers from the cash market.*’ The news articles quote the memo as saying ‘*Overall, these trades may help to reduce the market’s reliance on the Bund future and turn the European Government bond market into one that more closely resembles the US government bond market.*’

These quotes are in line with our evidence that proper EGB market functioning vitally depends on an active futures market.